The Holland Atom

Author: John Edward Holland **ORCID:** 0009-0001-5120-8712
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A Resonance-Locked Physical Reconstruction and Predictive Extension of Atomic Structure within General Expanse Tension Theory

Abstract

This paper introduces the Holland Atom: a standing-wave resonance reconstruction of atomic structure derived within the General Expanse Tension Theory (GETT) framework and the outcome of the scalar Holland Field Φ. The study investigates whether the experimentally observed architecture of atomic matter can be reconstructed from a physically real resonance-lock ontology rather than from fundamentally mathematical probabilistic orbital interpretation.

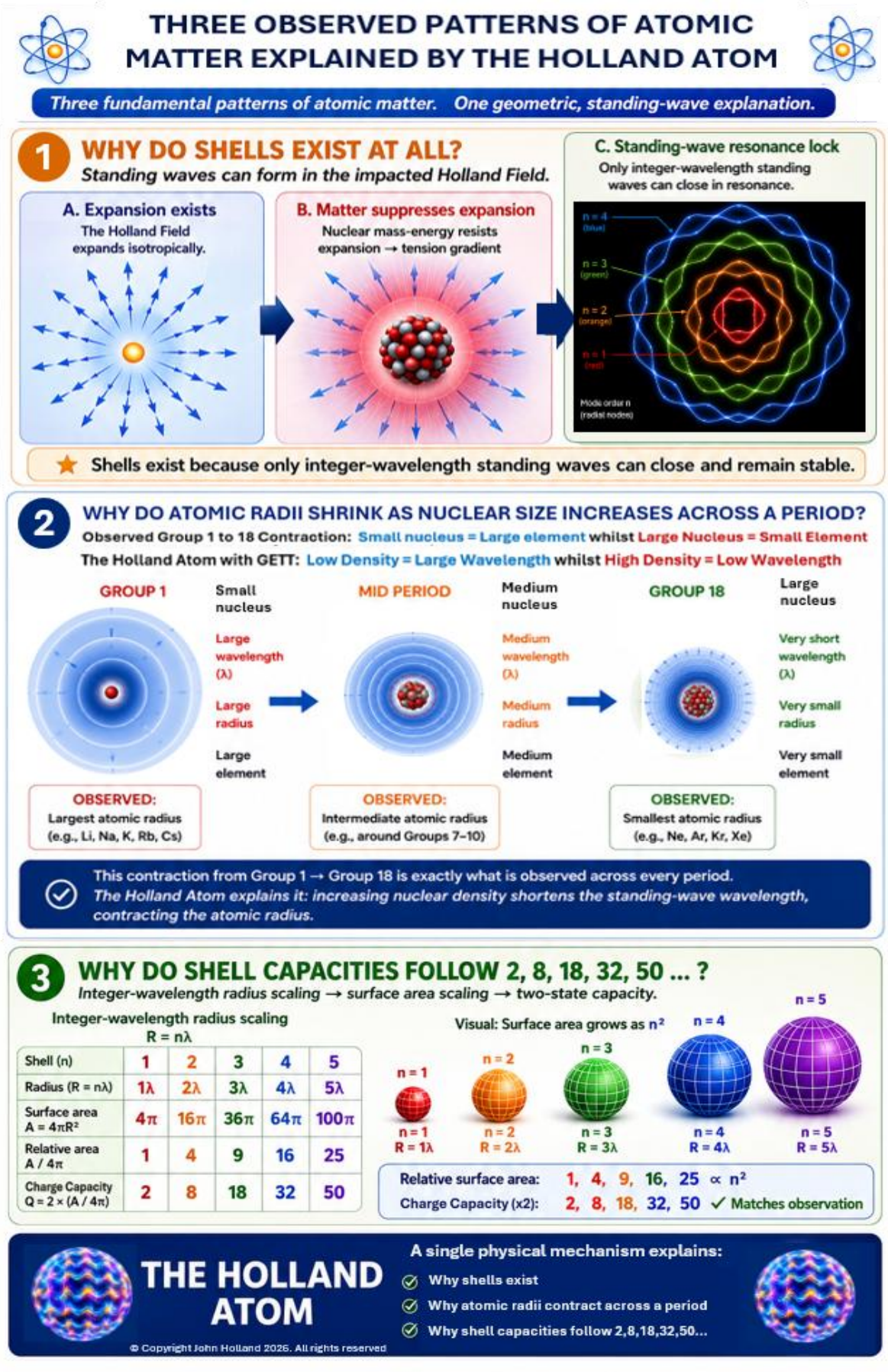
A formal postulate framework (H1–H9) is developed in which atomic shells emerge as spherical standing-wave resonance boundaries supported by the Holland Field. Shell capacities arise from surface-area scaling and angular resonance-family degeneracy, while sequential shell filling and anomalous atomic configurations emerge through dynamic resonance accessibility and global resonance optimisation respectively. The frozen Holland Atom framework was then subjected to a full reconstruction audit against the experimentally established periodic-table baseline. **The reconstruction programme successfully reproduced shell-capacity structure, angular resonance-family structure, shell-opening order, sequential shell filling, radius contraction behaviour, lanthanide and actinide compression, and historically anomalous transition-metal configurations** without anomaly-specific correction rules or post-calibration structural modification.

The study additionally identified preliminary evidence of resonance-scaling behaviour across atomic shell systems, including near-linear shell-radius scaling and repeated shell-opening and contraction scaling signatures, suggesting the possibility of a deeper Holland-field resonance-length structure underlying atomic geometry.

The Holland Atom therefore demonstrates that **a unified standing-wave resonance ontology can reconstruct a broad range of experimentally established atomic phenomena** within a single internally consistent geometric framework. Whether this ontology represents a deeper causal physical description beneath the successful mathematical formalism of quantum mechanics remains an open scientific question; however, the reconstruction programme presented here establishes the Holland Atom as a coherent, falsifiable, and empirically non-trivial candidate framework warranting serious further investigation.

Keywords:

Atom, standing-wave resonance, atomic structure, shell geometry, shell capacities, angular resonance families, periodic table reconstruction, resonance-lock mechanism, atomic radius scaling, shell contraction, anomalous atomic configurations, resonance optimisation, quantum ontology, atomic shell scaling, resonance geometry, unified atomic framework, General Expanse Tension Theory (GETT), Holland Field, Holland Atom.



Paper Roadmap: Three Empirical Patterns Addressed by the Holland Atom

1. Introduction, Scope, Objectives, and Study Boundaries

1.1 Introduction

Ever since Dalton proposed the first modern atomic model [1] and Thomson identified the electron [2], the atom remains one of the most successful predictive constructs in the history of science. From the earliest chemical models of matter, through the scattering experiments of Ernest Rutherford [3], through the quantised shell proposals of Niels Bohr [4], the matter-wave proposals of Louis de Broglie [5], and the orbital formalism of Erwin Schrödinger [6], atomic theory has evolved into one of the most experimentally successful frameworks in all of physics. Modern quantum mechanics accurately predicts atomic spectra, shell capacities, chemical periodicity, bonding behaviour, magnetic structure, and a wide range of condensed-matter phenomena with extraordinary precision. Yet despite this predictive success, the physical ontology of the atom remains open to interpretation. Quantum mechanics describes what is observed with remarkable mathematical precision, but questions concerning the physical nature of atomic charge distribution, shell geometry, orbital structure, block architecture, and the deeper causal origin of the periodic table remain encoded largely through mathematical formalism rather than through an explicitly identified physical substrate.

The present investigation emerges from **General Expanse Tension Theory (GETT)**, a broader physical framework built upon the existence of a **real, all-pervading, gauge-invariant, Lorentz-consistent singlet scalar field, Φ (the Holland Field), coupled to standard-model matter through a Higgs-portal interaction with density-conditioned modulation** [21]**.** Within GETT, Φ is treated not as an effective mathematical convenience, but as a physically real scalar substrate permeating all of spacetime, capable of supporting tension gradients, propagating ripple states, standing-wave resonances, and domain-dependent emergent phenomena across scales.

The underlying field-theoretic mechanism of GETT has undergone formal expert peer review within the context of quantum field theory and effective field theory, where its scalar-field construction, Higgs-portal coupling formalism, gauge consistency, Lorentz invariance, renormalisability, and vacuum stability were independently assessed and found fully consistent with established quantum field theoretic principles within its stated domain of applicability [21].

Since its original formulation, GETT has been progressively developed across multiple physical domains, including inertial reconstruction, Newtonian correspondence, relativistic correspondence, effective metric emergence, Einstein-domain gravitational recovery, gravitational lensing, compact-object structure, galaxy-scale dynamics, and large-scale cosmological applications. Across each of these domains, the role of Φ has remained physically consistent as a real scalar substrate whose interaction with mass-energy produces measurable tension gradients, standing modes, and domain-dependent emergent behaviour.

Most significantly, GETT proposes a direct **physical causal mechanism for gravitation itself.** Within this framework, mass locally resists the natural expansion of the surrounding Holland Field, producing a net tension-gradient imbalance whose equilibrium restoring response manifests observationally as gravitational attraction. Gravity therefore emerges not as a purely geometric abstraction, nor as an action-at-a-distance force, but as the measurable reaction of a real physical field to the presence of mass-energy [24-29].

It is within this broader, independently scrutinised, and increasingly cross-domain-tested physical framework that the present study introduces the **Holland Atom.** Rather than attempting to replace the extraordinary predictive mathematics of quantum mechanics, the Holland Atom asks a narrower and experimentally testable question: whether the architecture of atomic matter – including shell capacities, block structure, shell transitions, contraction behaviour, and historically anomalous configurations – may emerge naturally from the resonance-lock standing-mode geometry of the Holland Field Φ.

1.2 Paper Scope

This paper investigates whether atomic structure can be derived from the resonance-lock standing-mode geometry of the Holland Field. Specifically, this study addresses:

In Scope

✓ Experimental atomic baseline establishment
✓ Distinction between experimentally established observables and interpretive atomic ontology
✓ Physical derivation of the Holland Atom within GETT
✓ Definition and freezing of Holland Atom postulates H1–H9
✓ Derivation of spherical standing-wave shell geometry
✓ Derivation of shell-capacity structure $Q_n = 2n^2$
✓ Derivation of angular resonance-family structure (s, p, d, f, g, h, i)
✓ Derivation of sequential shell-opening architecture
✓ Dynamic resonance accessibility framework for shell population
✓ Global resonance optimisation framework for anomalous configurations
✓ Reconstruction of observed shell capacities
✓ Reconstruction of observed shell architecture
✓ Reconstruction of observed shell-utilisation behaviour
✓ Reconstruction of periodic-table block structure
✓ Reconstruction of shell-opening radius resets
✓ Reconstruction of period-contraction behaviour
✓ Reconstruction of lanthanide contraction
✓ Reconstruction of actinide contraction
✓ Reconstruction of historically anomalous atomic configurations
✓ Reconstruction audit against the frozen periodic-table baseline
✓ Preliminary investigation of resonance-scaling behaviour
✓ Discussion of implications for atomic ontology within GETT

1.3 Paper Goal

The primary goal of this study is to determine whether the physical architecture of atomic matter can emerge naturally from nested resonance-lock standing modes of the Holland Field Φ, and whether this ontology can predict the structure and behaviour of the periodic table using a transparent calibration-to-prediction methodology.

1.4 Study Objectives, Boundaries, and Acceptance Criteria for the Holland Atom

Ref	Category	Objective / Requirement	Measurement / Evidence	Success Criterion
O1	Experimental Baseline	Establish the experimentally observed baseline of atomic structure, including nucleus localisation, shell occupancy, shell capacities, spectral transitions, atomic radii, block architecture, and historical anomalies.	Literature audit, spectroscopy, ionisation data, atomic radii databases, periodic-table observations.	Complete baseline established with no unresolved observational conflicts.
O2	Physical Ontology	Derive the Holland Atom from first principles within GETT, including Φ boundary formation, resonance-lock shell geometry, spherical standing modes, harmonic sub-bands, and nested shell boundaries.	Mathematical derivation, geometric construction, physical consistency review.	A self-consistent causal ontology is established without internal contradiction.
O3	Formal Model Definition	Define and freeze the Holland Atom postulates H1–H9 and associated mathematical architecture prior to reconstruction audit.	Postulate table, mathematical architecture table, symbol definitions, appendices D–I.	A fixed, internally consistent model is established before application to the periodic-table baseline.
O4	Reconstruction Audit	Apply the frozen Holland Atom framework to reconstruct the experimentally observed atomic architecture across the known periodic table.	Shell, angular-family, radius, contraction, accessibility, and optimisation audits.	Successful reconstruction achieved without postulate modification or anomaly-specific correction rules.
O5	Falsification	Test the Holland Atom against historically exceptional atomic configurations.	Dedicated anomaly case studies and configuration comparisons.	Exceptional configurations structurally reproduced using the frozen framework.
B1	Scope Boundary	Restrict the study to ground-state neutral atoms under standard laboratory conditions.	Model assumptions, dataset filters.	No scope violations.
B2	Out of Scope	Exclude molecular bonding, reaction kinetics, solid-state physics, nuclear decay, QED reconstruction, fine-structure corrections, and multi-body wavefunction derivations.	Scope audit.	All excluded topics remain outside analysis.
S1	Acceptance Criterion	Shell capacities emerge naturally.	Shell derivation.	(2,8,18,32...) reproduced.
S2	Acceptance Criterion	s, p, d, f block structures emerge without orbital postulates.	Harmonic-mode derivation.	All blocks reproduced from standing-mode geometry.
S3	Acceptance Criterion	Shell radius reset behaviour is predicted at shell transitions.	Atomic-radius trend analysis.	Group 1 radius jumps reproduced.
S4	Acceptance Criterion	Progressive contraction within periods is predicted.	Radius trend charts across periods.	Contraction behaviour reproduced.
S5	Acceptance Criterion	Historical anomalies are predicted.	Case studies: Chromium, Copper, Palladium, Platinum, Gold.	Anomaly configurations structurally reproduced.
S6	Acceptance Criterion	Prediction robustness maintained across full application range.	End-to-end audit from H to Og.	No post-calibration tuning required.

Table 1 Study Objectives, Boundaries, and Acceptance Criteria for the Holland Atom

Table 1 defines the formal objectives, study boundaries, methodology, reconstruction audit, and acceptance criteria for the Holland Atom investigation. Each objective is assigned a unique reference identifier (O = Objective, B = Boundary, S = Success Criterion) to enable full traceability between initial study definition, model development, blind application, and final audit closure. These identifiers are intentionally preserved throughout the paper and revisited in Section 8, where each

objective is independently assessed against observational evidence, predictive performance, and model completeness.

1.5 Study Boundaries

To maintain scientific discipline, reproducibility, and clear falsifiability, explicit study boundaries are defined prior to model construction. The present investigation is intentionally restricted to the physical structure of neutral ground-state atoms under standard laboratory conditions, with specific focus on experimentally established shell occupancy, shell capacities, atomic radii, block architecture, spectroscopic shell transitions, and historically exceptional atomic configurations across the periodic table. The purpose of these boundaries is to ensure that the Holland Atom is first evaluated against the most stable, well-characterised, and experimentally reproducible manifestations of atomic structure before extension into more complex physical domains. Particular emphasis is placed on shell-capacity formation, spherical harmonic band emergence, shell-transition radius resets, period contraction behaviour, transition-metal structure, lanthanide behaviour, and historically anomalous atomic configurations, as these provide high-value falsification tests of the model.

To preserve clarity of scope, several related but substantially more complex physical domains are intentionally excluded from the present study. These exclusions include molecular bonding, chemical reaction kinetics, condensed-matter band structure, crystal-field effects, nuclear structure modelling, radioactive decay processes, particle-scattering amplitudes, quantum electrodynamic corrections, relativistic fine-structure derivations, spin-statistics formalism, and full many-body wavefunction reconstruction. Such phenomena undoubtedly represent important future applications of GETT, however they introduce additional layers of physical complexity that could obscure the primary objective of this work. By explicitly defining these boundaries at the outset, the present study establishes a controlled and transparent framework in which success, failure, limitations, and predictive capability can be objectively audited against the criteria established in Table 1 and subsequently re-evaluated during final model closure in Section 8.

1.6 Scientific Positioning

This paper does **not** seek to replace the predictive mathematics of quantum mechanics. Rather, it addresses a different scientific question: **What is the atom physically?**

The Holland Atom is therefore proposed not as a replacement for quantum prediction, but as a testable physical ontology that may explain why the periodic table exists in the form observed. If successful, the Holland Atom would represent a deeper causal reconstruction of atomic matter within General Expanse Tension Theory, extending beyond correspondence into direct physical application.

1.7 Paper Structure and Contents

The paper is organised as a structured reconstruction programme progressing from the experimentally established atomic baseline toward a full standing-wave resonance reconstruction of atomic structure within the Holland Atom framework.

Sections 2 and 3 establish the accepted empirical and quantitative baseline of atomic structure. Section 4 introduces the Holland Atom postulates (H1–H9) together with the associated physical ontology and geometric resonance interpretation. Section 5 develops the complete mathematical architecture of the Holland Atom. Section 6 then performs the full reconstruction audit against the experimentally observed periodic-table structure through a series of dedicated appendices and validation studies. Section 7 investigates preliminary resonance-scaling behaviour emerging from the reconstruction programme, including shell-radius scaling, shell-opening expansion scaling, and contraction scaling signatures. Sections 8–10 provide discussion, final reconstruction evaluation, and

conclusions regarding the Holland Atom as a candidate standing-wave resonance interpretation of atomic structure within GETT.

2. Historical Baseline of Atomic Development and Experimental Evidence

2.1 Introduction

The modern concept of the atom did not emerge from a single theoretical breakthrough, but through a sequential series of increasingly precise experimental observations, each exposing limitations in the preceding model and motivating progressively more sophisticated descriptions of atomic structure. From the indivisible solid spheres of classical chemistry, through the discovery of subatomic charge, nuclear localisation, quantised shell energies, and ultimately quantum mechanical orbital formalism, each successive atomic model was developed in direct response to experimental evidence that could not be explained within the preceding framework.

This historical progression is scientifically important for the present study for two reasons. First, it establishes which features of atomic structure are directly supported by experiment, independent of theoretical interpretation. Second, it allows the Holland Atom to be evaluated not against philosophical preference, but against the same observational evidence that historically motivated each successive development in atomic theory. Table 2 therefore provides a chronological baseline of atomic-model evolution, summarising the key experimental observations, the corresponding theoretical developments, and the unresolved causal or ontological assumptions that remained following each stage.

Era	Model	Principal contributors	Key observation / evidence	What the model successfully explained	Remaining causal or ontological gap
~1803	Solid sphere atom [1]	**John Dalton**	Chemical stoichiometry, fixed composition ratios	Matter composed of discrete atoms	No internal structure
~1897	Plum pudding model [2]	**J. J. Thomson**	Cathode-ray experiments	Negative charge exists within atoms	Positive charge distribution unknown
~1911	Nuclear atom [3]	**Ernest Rutherford**	Alpha-particle scattering	Compact central nucleus; atom mostly empty space	Why electrons do not collapse into nucleus
~1913	Quantised shell atom [4]	**Niels Bohr**	Hydrogen spectral lines	Discrete shell energies; stable atomic spectra	Physical origin of shell quantisation
~1924	Matter-wave atom [5]	**Louis de Broglie**	Quantised momentum relationships	Wave interpretation of bound states	What physically waves
~1926	Wave-mechanical atom [6]	**Erwin Schrödinger**	Atomic spectra, multi-electron structure	Orbital mathematics, shell capacities, block structure	Physical meaning of wavefunction
~1927	Probabilistic quantum atom [7]	**Werner Heisenberg**	Measurement statistics	Predictive quantum formalism	Ontology becomes observer-dependent
Modern	Quantum-mechanical atom [8,9]	**Dirac + collective development**	Spectroscopy, chemistry, condensed matter	Highly accurate prediction across physics	Physical mechanism for shell geometry, block structure, anomalies remain interpretive

Table 2 Historical Evolution of Atomic Models, Experimental Drivers, and Remaining Ontological Gaps

Table 2 summarises the historical evolution of atomic models, the experimental observations that motivated each successive development, the predictive successes achieved at each stage, and the causal or ontological questions that remained unresolved. This historical baseline presents a

structured evidence trail showing how atomic theory evolved in response to increasingly precise experimental constraints [1–9].

2.2 The Accepted Atomic Model in 2026

As of 2026, the internationally accepted standard model of atomic structure is the **quantum mechanical atom**, developed through the combined contributions of Erwin Schrödinger, Werner Heisenberg, Paul Dirac, and subsequent developments in quantum electrodynamics and computational chemistry [6-9]. This framework forms the foundation of modern atomic physics, chemistry, condensed-matter physics, spectroscopy, and materials science, and remains among the most experimentally successful predictive models in all of science [15,16]. Within this framework, the atom is described as a compact positively charged nucleus surrounded by quantised electronic states represented by solutions to the Schrödinger or Dirac equations. These states are commonly represented as orbitals, characterised by discrete quantum numbers, probability amplitudes, spin states, and exclusion rules that collectively reproduce the observed shell architecture of the periodic table. The modern quantum mechanical atom successfully predicts a wide range of experimentally observed phenomena, including atomic spectra, ionisation energies, chemical periodicity, bonding behaviour, magnetic structure, fine structure, and relativistic corrections in heavy elements.

However, while its predictive power is extraordinary, the physical ontology of the quantum mechanical atom remains interpretive. The wavefunction itself is not directly observable, the continuous physical existence of point-like electrons between interactions remains experimentally unresolved, and many structural features of the periodic table – while mathematically reproduced – are encoded through quantum-number formalism rather than through an explicitly causal physical substrate.

Feature	Current accepted interpretation (2026)	Experimental Support	Ontological Status
Nucleus	Compact positively charged nucleus containing protons and neutrons	Directly supported	Established
Electromagnetic shell charge totals [15,19,20]	Spectroscopy, ionisation energies, chemical periodicity, charge balance measurements	Directly supported	Empirically established
Stable orbiting electron particle trajectories within shells	Not directly observed - inferred through quantum-mechanical orbital formalism and indirect measurement interpretation	Inferred	Interpretative / model-dependent
Shells	Discrete energy shells	Strongly supported	Established
s, p, d, f blocks [10,11,19,20]	Orbital angular momentum states	Strongly supported mathematically	Interpretation of physical mechanism
Wavefunction	Probability amplitude describing electron state	Predictively successful	Not directly observable
Electron position	Undefined until interaction / measurement	Consistent with experiment	Interpretive
Electron trajectory	No classical trajectory	Consistent with experiment	Interpretive
Spin	Intrinsic quantum property	Strongly supported	Physical mechanism unresolved
Exclusion principle	One state per quantum occupancy	Strongly supported	Mechanism unresolved
Shell anomalies	Half-filled / filled orbital stability	Strongly supported	Descriptive rather than causal

Table 3 The Accepted Quantum Mechanical Atom (2026 Baseline)

Table 3 summarises the accepted quantum mechanical description of atomic structure as understood in 2026. The purpose of this baseline is to clearly distinguish between experimentally established observables and the mathematical or interpretive constructs used to model them. This distinction is essential for the present study, whose objective is not to replace established predictive mathematics, but to investigate whether a deeper causal ontology may reproduce the same observed atomic architecture through resonance-lock interactions within the Holland Field Φ.

For clarity and scientific fairness, the present study therefore first establishes the current accepted quantum mechanical description of the atom before distinguishing between experimentally established observations and ontological assumptions.

2.3 Experimental Facts vs Non-Proven Ontological Assumptions

Despite more than a century of extraordinary experimental success, atomic physics still contains an important distinction between what has been directly measured and how those measurements have been historically interpreted. Before proceeding further, it is therefore necessary to press this distinction to its formal limit, separating experimentally established observables from ontological assumptions that, while widely adopted, are not themselves directly proven by experiment. Table 4 defines this boundary explicitly, identifying the hard empirical constraints that any future atomic model must preserve, while clarifying the remaining interpretational space available for causal reconstruction.

Empirically Established Facts	**Not Proven by Experiment**	**Permitted GETT Interpretation Within the Scope of Facts**
Positive mass and charge concentrated in a compact nucleus **[Fact]**	Electrons orbit the nucleus as accelerating point particles **[?]**	Nucleus establishes a compact **Φ tension boundary and resonance seed**
Negative charge exists away from the nucleus in discrete radial regions **[Fact]**	Negative charge must always exist as localised particles between measurements **[?]**	Negative charge may exist as **stable matter–Φ charge clouds** confined to resonance shells
Stable occupancy occurs only at discrete shell energies **[Fact]**	Charge exists continuously between shells **[?]**	Charge occupies only **resonance-locked shell layers**; inter-shell states are sub-threshold and non-persistent
Emission and absorption occur at discrete energies **[Fact]**	Energy itself is fundamentally granular **[?]**	**Continuous energy** becomes observable only through **discrete resonance-lock transitions**
Localised electron-like detection events occur during ionisation, scattering, or extraction **[Fact]**	The electron is always a point particle prior to detection **[?]**	Localised **electron events** may represent **collapse/extraction** of stable shell charge into **observable particle states**
Shell radii increase systematically with shell number **[Fact]**	Shell structure must arise from orbital angular momentum quantisation **[?]**	**Shell radius** may be **set by integer Φ standing-mode** boundary conditions
Shell capacities follow (2,8,18,32...) **[Fact]**	Spin degeneracy is the only explanation for shell capacity **[?]**	**Shell capacity** may emerge **geometrically from resonance-shell surface scaling** and accessible charge-supporting surfaces
Spectral lines arise only from shell transitions **[Fact]**	Wavefunction collapse is required **[?]**	**Spectral emission** may arise from **shell-to-shell matter–Φ reconfiguration**

Table 4. Atomic observables, interpretational assumptions, and allowable GETT reinterpretation

Table 4 confirms that the central experimental facts of atomic physics – compact nuclear structure, discrete shell occupancy, forbidden inter-shell stability, quantised emission and absorption, systematic shell scaling, and localised electron-like detection events – do not uniquely require orbiting particles, wavefunction ontology, spin-based degeneracy, or collapse-based interpretations. Any future causal model remains formally permitted, provided it preserves the observed shell radii,

charge capacities, transition energies, spectral outputs, and all experimentally established emission, absorption, and ionisation observables.

3. Quantitative Atomic Baseline and Process Capability

Having defined the paper scope, historical context, accepted atomic baseline, and distinction between experimentally established facts and ontological assumptions, the study now moves from definition to measurement. The purpose of this section is to establish the quantitative baseline against which the Holland Atom will later be tested. This section measures the known structure of the periodic table as a controlled dataset: shell capacities, shell-opening sequences, atomic-radius resets, period contraction, block filling behaviour, transition-metal irregularities, lanthanide behaviour, and historically anomalous electron configurations. These features represent the process behaviour that any physical model of the atom must explain.

The objective is to convert the modern atomic record into an auditable baseline. Once measured, these same features will be revisited after the Holland Atom is derived, allowing the model to be tested against the original baseline conditions rather than against selectively chosen examples. This section defines the current process capability of atomic theory as observed in nature: what repeats reliably, what shifts periodically, what contracts, what resets, and which exceptional cases must be explained without post-hoc adjustment.

3.1 The Accepted Shell Architecture of the Modern Atom

Figure 1 summarises the accepted shell architecture of the modern quantum mechanical atom as understood in 2026. The atom is represented as a compact central nucleus containing positively charged protons and electrically neutral neutrons, surrounded by a sequence of discrete electronic shell regions conventionally labelled K through Q, corresponding to principal quantum numbers:

$$n = 1 \rightarrow 7$$

Within the standard quantum mechanical framework, each shell n possesses a maximum occupancy defined by:

$$N_{\max} = 2n^2$$

producing the well-established shell-capacity sequence:

$$2,\ 8,\ 18,\ 32,\ 50,\ 72,\ 98$$

These shell capacities are among the most robustly established structural features in atomic physics, supported independently by spectroscopy, ionisation energies, X-ray shell transitions, chemical periodicity, and the observed architecture of the periodic table. They are accepted here beyond doubt [15,17,19,20]. Within the standard model, these capacities are derived mathematically from the allowed combinations of quantum numbers associated with orbital angular momentum, magnetic orientation, and spin occupancy. While the predictive success of this formalism is unquestioned, the physical causal origin of why atomic charge organises into this precise geometric sequence remains encoded within quantum-number formalism rather than an explicitly identified physical substrate [19,20].

Figure 1 therefore serves as the quantitative baseline for the present investigation. Before proposing the Holland Atom, the following sections will measure how this shell architecture manifests throughout the periodic table, including shell-opening sequences, shell-capacity utilisation, radius resets, period contraction, block structure, and historically exceptional configurations that any physical atomic ontology must ultimately explain.

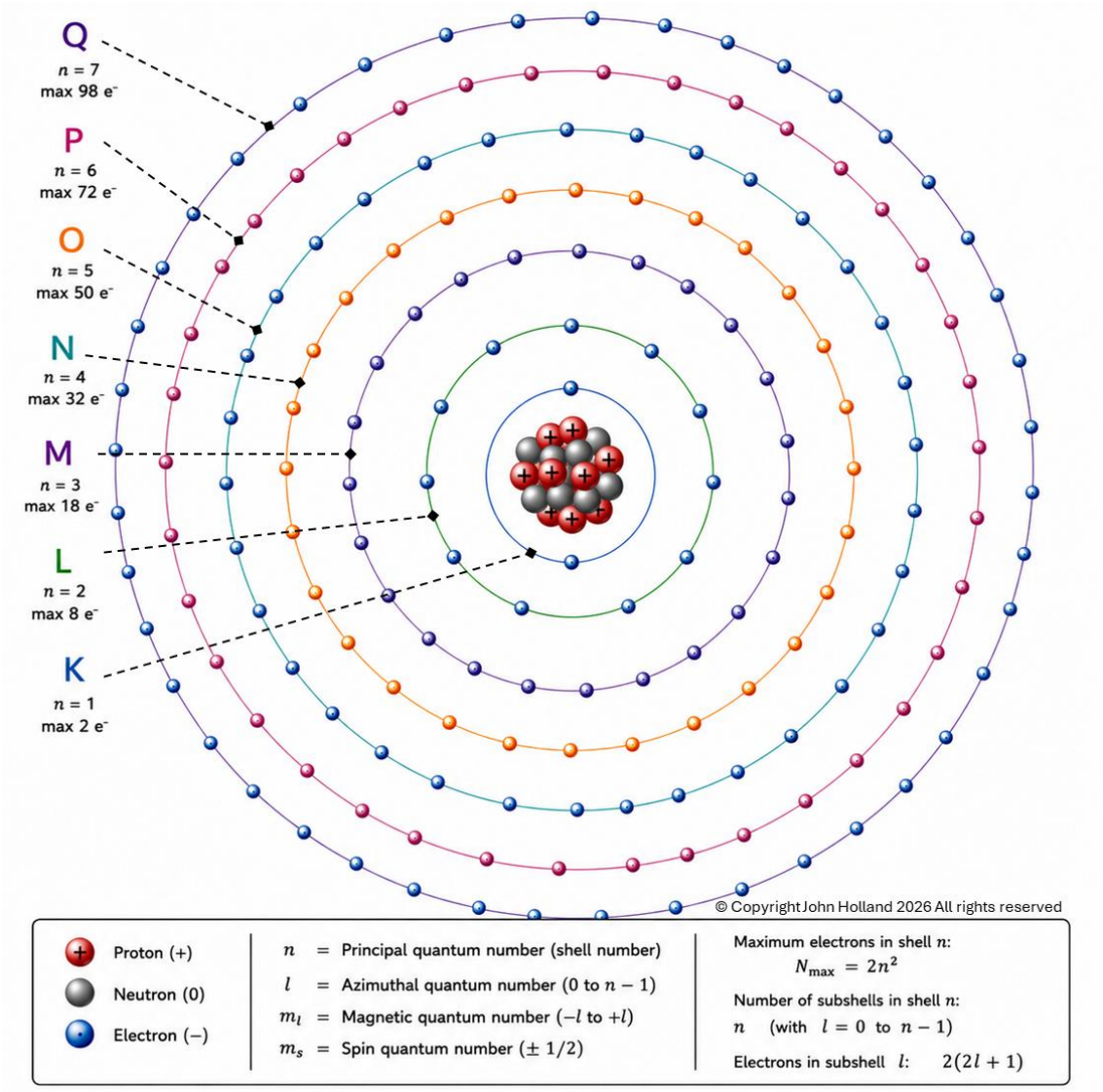


Figure 1. Shell architecture of the modern quantum mechanical atom

Figure 1. Accepted shell architecture of the modern quantum mechanical atom, showing the principal shell sequence K–Q ($n = 1 \rightarrow 7$), corresponding maximum shell occupancies $N_{max} = 2n^2$, and the quantum-number framework conventionally used to derive shell and subshell capacities. This figure establishes the accepted structural baseline of atomic shell organisation prior to quantitative measurement and subsequent evaluation against the Holland Atom proposed in this study.

3.2 Standard Quantum Mechanical Derivation of Shell Capacity

Before evaluating any alternative physical ontology, it is necessary to establish how the modern quantum mechanical framework derives the observed shell-capacity sequence of the periodic table. Within standard quantum mechanics, each principal shell n contains a set of allowed orbital angular momentum states:

$$l = 0 \rightarrow n - 1$$

Each orbital family contains:

$$2l + 1$$

allowed spatial orientations.

$$2l + 1$$

Summing across all allowed orbital families gives:

$$\sum_{l=0}^{n-1} (2l + 1) = n^2$$

Including the two allowed spin states yields:

$$N_{\max} = 2n^2$$

This derivation reproduces the observed shell-capacity sequence of the periodic table and forms one of the foundational structural results of modern atomic quantum mechanics [11,19]. Table 5 quantifies this derivation across the experimentally observed shell sequence.

Shell	Principal shell (n)	Allowed orbital states (l)	Number of orbital sub-states	Spin degeneracy	Total calculation	Maximum occupancy
K	1	0	$1^2 = 1$	2	1×2	2
L	2	0,1	$2^2 = 4$	2	4×2	8
M	3	0,1,2	$3^2 = 9$	2	9×2	18
N	4	0,1,2,3	$4^2 = 16$	2	16×2	32
O	5	0,1,2,3,4	$5^2 = 25$	2	25×2	50
P	6	0,1,2,3,4,5	$6^2 = 36$	2	36×2	72
Q	7	0,1,2,3,4,5,6	$7^2 = 49$	2	49×2	98

Table 5. Standard Quantum Mechanical Derivation of Shell Capacity

Table 5 summarises the standard quantum mechanical derivation of atomic shell capacity. By summing the allowed orbital angular momentum states within each principal shell and applying two-fold spin degeneracy, the modern quantum mechanical framework derives the shell-capacity sequence $2, 8, 18, 32, 50, 72, 98$. *While this formalism reproduces the observed architecture of the periodic table with extraordinary predictive success, the derivation remains fundamentally mathematical, leaving open the separate ontological question of what physical mechanism causes atomic charge to organise into this precise geometric sequence. This unresolved causal question forms one of the central motivations of the present investigation.*

3.3 The Quantum Numbers of the Modern Atom: Definition, Predictive Role, and Ontological Status

The shell-capacity derivation presented in Table 5 emerges from the quantum-number formalism of modern atomic physics. Within the accepted quantum mechanical framework, the electronic structure of the atom is described using four quantum numbers that collectively define the allowed electronic states surrounding the nucleus. These quantum numbers are among the most successful predictive constructs in modern science, underpinning atomic spectroscopy, shell capacities, chemical periodicity, magnetic behaviour, and condensed-matter physics. Their predictive power is not in dispute.

However, while these quantities are mathematically indispensable within modern quantum mechanics, their physical ontological interpretation remains fundamentally model-dependent. The quantum numbers themselves are not directly observed as physical objects or measured as independent geometric entities. Rather, they are inferred state descriptors derived from solutions to the Schrödinger or Dirac equations, whose validity is established through agreement with observed atomic behaviour [19]. For clarity and scientific precision, Table 6 summarises the accepted modern definitions of these quantum descriptors, together with their present experimental and ontological status.

Quantum descriptor	Symbol	Current accepted definition	Predictive role	Directly observed as physical object?	Ontological status
Principal quantum number	n	Defines allowed shell energy levels and approximate radial extent [4,6,19]	Shell capacities, shell energies	Indirectly through spectra	**Strongly supported**
Orbital angular momentum	l	Defines orbital family (s,p,d,f...) and angular node structure [6,19,20]	Block architecture, shell filling	Not directly observed	**Inferred** state descriptor
Magnetic quantum number	m_l	Defines allowed spatial orientation of orbital states [6,19]	Degeneracy, field splitting	Not directly observed	**Inferred** state descriptor
Spin quantum number	m_s	Defines two allowed spin states [8,9,19]	Magnetic structure, exclusion	Indirectly through magnetic measurements	**Inferred** intrinsic property
Orbital wavefunction	ψ **(psi)**	Probability amplitude of electron state [6,19,20]	Predicts electron distributions	Not directly observed	**Interpretive** construct
Electron trajectory	—	No classical trajectory assigned [7,19]	Not used	Not observed	**Explicitly undefined**
Orbital geometry	—	Probability-density representation [6,19,20]	Chemical bonding, shell structure	Not directly observed	**Interpretive** construct

Table 6. The Quantum Numbers of the Modern Atom: Accepted Definitions and Ontological Status (2026)

Table 6 summarises the accepted quantum descriptors used in modern atomic theory. These descriptors possess extraordinary predictive success and are strongly supported through indirect experimental agreement across spectroscopy, chemistry, and condensed-matter physics. However, the descriptors themselves are not directly observed as independent physical objects, geometric surfaces, or measurable trajectories. Rather, they function as inferred state descriptors derived from quantum mechanical formalism. This distinction is scientifically important for the present study, because it investigates whether the same observed atomic architecture may emerge from an alternative physical ontology.

3.4 Observed Shell Utilisation Across the Periodic Table

Having established the accepted shell-capacity derivation of the modern quantum mechanical atom, the next step is to examine how these capacities are populated in nature. While the shell-capacity sequence:

$$2,\ 8,\ 18,\ 32,\ 50,\ 72,\ 98$$

defines the maximum occupancy of each principal shell, experimental observation reveals that atomic

charge does not populate these shells as a simple sequential filling process. Instead, the periodic table exhibits a far richer architecture in which new outer shell boundaries may emerge before lower-energy internal shell capacities are fully occupied, producing the characteristic s, p, d, f, g, h, and i block structure of the modern periodic table.

3.4.1 Subshells and Blocks Definition

The shell-capacity sequence $2n^2$ describes the maximum number of electrons that may be assigned to a principal shell n. However, modern atomic theory does not treat each shell as a single undivided container. Instead, each shell is subdivided into **subshells**, with labels defining different orbital-state families within a principal shell and are essential for understanding the structure of the periodic table.

The letters **s, p, d,** and **f** used throughout modern atomic physics are historical labels originally introduced by early spectroscopists to classify distinctive families of spectral lines observed in atomic emission and absorption experiments. Specifically, **s** referred to *sharp*, **p** to *principal*, **d** to *diffuse*, and **f** to *fundamental*. Although their original spectroscopic meanings are no longer used operationally, these labels were retained as quantum mechanics developed and now denote distinct families of allowed electronic states within an atom. In the modern quantum mechanical framework, s, p, d, and f correspond to successive values of the orbital angular momentum quantum number $l = 0, 1, 2, 3$, respectively, and define different orbital-state families within a given principal shell. Each family possesses a characteristic number of allowed orientations and therefore a characteristic **maximum occupancy** – 2, 6, 10, and 14 electrons, respectively – forming the structural basis of the familiar s-block, p-block, d-block, and f-block architecture of the periodic table [11,19,20]. Importantly, while these labels are extraordinarily successful predictive descriptors, they do not represent directly observed physical objects, surfaces, or trajectories, but rather mathematically derived state classifications within the modern quantum mechanical formalism.

The notation **3s**, **3p**, or **3d** combines two pieces of information. The number refers to the **principal shell**, n, while the letter refers to the **subshell type**, determined by the orbital angular momentum quantum number l. Thus, **3s** means the s-type subshell within the third principal shell, **3p** means the p-type subshell within the third principal shell, and **3d** means the d-type subshell within the third principal shell.

In standard quantum mechanics, these subshells are not physical rings or mechanical orbits. They are families of allowed quantum states derived from the mathematical structure of the atomic wavefunction. Their shapes, orientations, and capacities arise from the allowed values of quantum numbers. The periodic table is divided into blocks because elements are grouped according to which subshell is being populated by the next electron.

Subshell	Quantum number l	Number of orientations $2l+1$	Capacity with spin $2(2l+1)$	Plain meaning
s	0	1	2	One spherical state family
p	1	3	6	Three directional state families
d	2	5	10	Five angular state families
f	3	7	14	Seven higher-order angular state families
g	4	9	18	Nine higher-order angular resonance families
h	5	11	22	Eleven higher-order angular resonance families
i	6	13	26	Thirteen higher-order angular resonance families

Table 7. Meaning and Capacity of s, p, d, f,…Subshells

Table 7 summarises the experimentally established and mathematically extended angular resonance-family structure of atomic subshells. The familiar s-, p-, d-, and f-families correspond to increasing angular standing-mode order l, producing $2l + 1$ independent angular orientations and total capacities of $2(2l + 1)$ giving the observed capacities 2, 6, 10, and 14. The higher-order g-, h-, and i-families are mathematically permitted continuations of the same angular resonance progression and will be shown to emerge naturally within the Holland Atom formalism without requiring additional occupancy rules or modified shell-capacity assumptions. These labels are predictive and operationally successful, but they should not be mistaken for directly observed physical objects; they are state descriptors within the quantum mechanical formalism.

3.4.2 Maximum Occupancy in Block and Subshell Notation

Notation	Meaning	Maximum occupancy	Example role
1s	s subshell in shell (n=1)	2	Hydrogen and helium
2s	s subshell in shell (n=2)	2	Lithium and beryllium
2p	p subshell in shell (n=2)	6	Boron to neon
3s	s subshell in shell (n=3)	2	Sodium and magnesium
3p	p subshell in shell (n=3)	6	Aluminium to argon
3d	d subshell in shell (n=3)	10	Transition metals Sc to Zn
4s	s subshell in shell (n=4)	2	Potassium and calcium
4p	p subshell in shell (n=4)	6	Gallium to krypton
4f	f subshell in shell (n=4)	14	Lanthanides

Table 8. Examples of Subshell Notation

Table 8 explains standard subshell notation. The leading number identifies the principal shell n, while the letter identifies the subshell family. For example, 3p denotes a p-type subshell within the third principal shell, while 4f denotes an f-type subshell within the fourth principal shell. This notation is central to the modern description of electron configuration and periodic-table block structure.

Experiment does not directly observe “orbitals,” “d-orbitals,” or “f-orbitals” as physical objects. What experiment directly establishes is that atomic charge does not populate principal shells as simple sequential $2n^2$ containers. Instead, new outer shell boundaries emerge before lower shells reach their nominal maximum capacity, while spectroscopy, magnetic behaviour, ionisation energies, and chemical periodicity independently confirm the existence of multiple internally distinct charge states within the same principal shell. The modern concepts of s, p, d, and f subshells therefore arise as mathematically inferred state classifications introduced to describe this experimentally observed non-sequential shell utilisation.

To consolidate the present scientific understanding of subshell architecture, spatial orbital orientations, occupancy capacities, and their relationship to shell population, **Appendix B** provides a complete visual baseline of the modern quantum-mechanical subshell model, including the experimentally inferred **s, p, d, and f** orbital families, their three-dimensional orientation states, electron occupancy rules, geometric progression in orientation number (1,3,5,7), and the resulting subshell capacities (2,6,10,14) that underpin the observed shell-capacity sequence of the periodic table.

3.4.3 Observed Shell Utilisation

This behaviour is among the most important structural observations in atomic physics. Any physical ontology of the atom must therefore explain not only shell capacities themselves, but also how those capacities are actually utilised across the periodic table, including early shell opening, internal band filling, transition-metal behaviour, lanthanide contraction, and historically exceptional atomic configurations. Table 9 establishes the measured shell utilisation pattern across the major regions of the periodic table, identifying the outer boundary shell, the active filling region, the observed occupancy behaviour, and the first appearance of non-sequential shell population.

Atomic range	Representative elements	Outer boundary shell	Active filling region	Observed structural behaviour
Z = 1–2	Hydrogen–Helium	1	1s	First shell fills completely
Z = 3–10	Lithium–Neon	2	2s, 2p	Second shell fills sequentially
Z = 11–18	Sodium–Argon	3	3s, 3p	Third shell begins
Z = 19–20	Potassium–Calcium	4	4s	New outer shell opens
Z = 21–30	Scandium–Zinc	4	3d	Inner shell fills before outer shell closure
Z = 31–36	Gallium–Krypton	4	4p	Outer shell resumes filling
Z = 37–38	Rubidium–Strontium	5	5s	New shell reset
Z = 39–48	Yttrium–Cadmium	5	4d	Internal band filling
Z = 49–54	Indium–Xenon	5	5p	Shell closure
Z = 57–71	Lanthanum–Lutetium	6	4f	Lanthanide contraction
Z = 89–103	Actinium–Lawrencium	7	5f	Actinide filling

Table 9. Observed Shell Utilisation Across the Periodic Table

Table 9 summarises the experimentally observed utilisation of shell capacity across the periodic table. While principal shell capacities follow the well-established $2n^2$ sequence, nature does not populate these capacities as a simple sequential process. Instead, new outer shell boundaries frequently emerge before lower-energy internal capacities are fully occupied, producing the characteristic s, p, d, and f block architecture of the periodic table. This non-sequential shell utilisation represents one of the most important structural constraints that any physical model of atomic matter must explain.

3.5 Predictive Success, Wavefunctions, and the Outstanding Ontological Gap in Modern Atomic Theory

The shell capacities, subshell structures, block architecture, magnetic behaviour, spectroscopic transitions, and chemical periodicity of the modern atom are reproduced with extraordinary precision by quantum mechanics. Few scientific frameworks in history have demonstrated comparable predictive success across such a broad range of physical phenomena. The modern quantum mechanical atom remains one of the most experimentally validated predictive models in all of science. At the mathematical core of this framework lies the atomic wavefunction, conventionally denoted:

$$\hat{H}\psi = E\psi$$

where $\hat{H}$ represents the Hamiltonian operator of the atomic system, ψ represents the wavefunction, and E represents the allowed quantised energy states. By solving this equation subject to the boundary conditions imposed by the atomic nucleus, quantum mechanics generates a discrete set of mathematically allowed solutions. These solutions produce the principal shells, subshell families, quantum numbers, orbital-state classifications, magnetic splitting, spin occupancy, and exclusion rules that collectively reproduce the observed structure of the periodic table.

The predictive success of this formalism is not in dispute. Spectroscopic line energies, ionisation potentials, shell capacities, chemical periodicity, magnetic moments, and a vast range of atomic observables are reproduced to remarkable accuracy through wavefunction-based quantum mechanics. In operational and predictive terms, the modern quantum mechanical atom is among the most successful models ever constructed.

However, the physical ontology underlying this success remains **fundamentally interpretive**. The wavefunction itself is not directly observed as a physical object, medium, surface, or measurable geometric structure. The orbital families denoted s, p, d, and f are not directly imaged as physical entities, nor are the associated quantum numbers measured as independent physical objects. Rather, these quantities emerge as mathematically inferred state descriptors whose validity is established through agreement between calculated observables and experimental measurement.

At the empirical level, experiment directly establishes that atomic negative electric charge occupies discrete energy states, that shell capacities follow highly repeatable numerical sequences, that new outer shell boundaries emerge before lower shell capacities are fully populated, and that atomic charge responds to excitation, ionisation, magnetic fields, and chemical interaction in highly structured and reproducible ways. What experiment does not directly establish is what physical entity is actually oscillating, what physical mechanism generates the standing-state solutions of the wavefunction, or what underlying causal substrate gives rise to the specific shell, subshell, and block architecture of atomic matter.

As such, while modern quantum mechanics provides an extraordinarily successful predictive formalism, the deeper physical question – **what is the atom physically?** – remains an open ontological question in present-day atomic physics. The distinction between predictive success and causal physical mechanism therefore represents one of the central unresolved knowledge gaps motivating the present investigation.

3.6 Group 1 Radius Reset: The Opening of New Atomic Boundaries

Having established the shell-capacity sequence, subshell architecture, and the quantum mechanical formalism used to describe atomic structure, the next quantitative baseline examines one of the most visually striking and experimentally reproducible features of the periodic table: the abrupt increase in atomic radius observed whenever a new Group 1 element begins a new period.

Unlike orbital-state descriptors, atomic radius is a directly measurable physical quantity inferred from X-ray crystallography, molecular bond lengths, electron scattering, and spectroscopic measurements [17]. Across the periodic table, atomic radii generally contract as shell populations increase within a given period. However, at the beginning of each new period, this contraction is interrupted by a pronounced and repeatable radius reset, most clearly observed in the alkali metals of Group 1.

This behaviour is of particular significance because it represents a direct geometric signature of atomic boundary formation. Each Group 1 element marks the first occupancy of a newly accessible outer shell, producing an immediate increase in the measured atomic radius before progressive contraction resumes across the remainder of the period.

3.6.1 Radius Convention Used in the Present Study

To maintain physical interpretability, consistency, and independence from model-dependent quantum mechanical calculations, all radius analyses in the present study use **empirical atomic radii**, derived from experimentally measured interatomic distances obtained through crystallographic, diffraction, and spectroscopic methods.

Empirical radii were selected in preference to covalent, van der Waals, metallic, or calculated radii because they represent the closest presently available operational measure of the physical spatial extent of atomic matter, and therefore provide the most appropriate geometric baseline for evaluating the resonance-lock shell boundaries proposed by the Holland Atom.

To preserve methodological integrity, this radius convention is frozen throughout the study and applied consistently across calibration, blind prediction, anomaly analysis, and final audit closure.

3.6.2 Radius Reset Empirical Evidence

Any physical ontology of the atom must therefore explain not only shell capacities and subshell architecture, but also why the opening of a new atomic period produces these abrupt and highly reproducible geometric boundary resets [12,13,14]. Table 10 establishes the measured Group 1 radius-reset sequence across the periodic table.

Element	Atomic Number	Period	Outer Shell	Empirical Atomic Radius (pm)	Previous Noble Gas	Previous Noble Gas Radius (pm)	Radius Reset Δr (pm)	Observed Behaviour
Lithium	3	2	2	**145**	Helium	120	**+25**	First shell reset
Sodium	11	3	3	**180**	Neon	160	**+20**	New outer boundary
Potassium	19	4	4	**220**	Argon	71	**+149**	Major radius jump
Rubidium	37	5	5	**235**	Krypton	88	**+147**	Boundary reset
Caesium	55	6	6	**260**	Xenon	108	**+152**	Largest stable reset
Francium	87	7	7	**~270***	Radon	120	**~+150**	Final predicted reset

Table 10. Group 1 Radius Reset Baseline (Empirical Atomic Radii)

Table 10 summarises the empirically derived atomic radii of the Group 1 elements, selected using a single frozen radius convention throughout the present study. Empirical radii were chosen in preference to covalent, van der Waals, metallic, or calculated radii because they represent the closest presently available operational measure of the physical spatial extent of atomic matter, being derived from experimentally measured interatomic distances obtained through crystallographic, diffraction, and spectroscopic methods. The radius-reset quantity Δr is defined as the difference between each Group 1 element and the immediately preceding noble-gas shell closure, providing a direct geometric measure of new shell-boundary formation. The large, repeatable, and highly structured radius resets observed across Group 1 represent one of the clearest experimentally measured signatures of atomic shell opening in nature.

Inspection of the empirical Group 1 radius-reset sequence reveals a non-trivial, unexpected and previously under-emphasised geometric regularity. From the fourth principal shell onward, the measured shell-opening increments appear to converge toward an approximately constant radius reset of ~150 pm, despite substantial increases in atomic number, nuclear charge, and internal shell complexity. While the individual radius measurements themselves are well established within atomic physics, this near-constant shell-opening increment does not appear to be explicitly recognised as an independent geometric scaling law within standard atomic theory, where such behaviour is typically described qualitatively in terms of shielding and effective nuclear charge.

3.7 Period Contraction: Progressive Compression of Atomic Boundaries

The abrupt radius resets observed in Group 1 establish the opening of new outer atomic boundaries at the beginning of each period. However, these boundary expansions do not remain fixed as additional atomic charge is introduced. Instead, one of the most reproducible geometric signatures of atomic structure is the progressive contraction of atomic radius observed across each period as shell population increases.

Unlike shell capacities or orbital-state classifications, atomic radius represents a directly measurable geometric quantity derived from crystallographic, diffraction, and spectroscopic measurements. The contraction of atomic radii across each period is therefore an experimentally observed physical phenomenon, independent of theoretical interpretation. This contraction occurs systematically as atomic number increases from Group 1 toward Group 18, producing a highly repeatable compression of the outer atomic boundary despite the continued addition of negative charge to the atom. The effect is observed across all major periods of the periodic table and represents one of the clearest geometric signatures of internal charge reorganisation in atomic matter [12,14,18].

Within standard atomic theory, this behaviour is typically described in terms of increasing effective nuclear charge, electron shielding, and orbital-state population. However, regardless of interpretation, the measured geometric trend itself is unambiguous: once a new shell boundary opens, that boundary progressively contracts as shell occupancy increases. Table 11 establishes the empirical period-contraction baseline across the major periods of the periodic table.

Period	Opening Element	Radius (pm)	Closing Element	Radius (pm)	Total Contraction (pm)	Relative Contraction
2	Lithium	145	Neon	38*	107	74%
3	Sodium	180	Argon	71	109	61%
4	Potassium	220	Krypton	88	132	60%
5	Rubidium	235	Xenon	108	127	54%
6	Caesium	260	Radon	120	140	54%

Table 11. Period Contraction Baseline (Empirical Atomic Radii)

Table 11 summarises the empirically observed contraction of atomic radius across successive periods of the periodic table using the frozen empirical radius convention established in the present study. Each period begins with a Group 1 shell-opening reset, followed by a systematic and measurable contraction of the outer atomic boundary as shell population increases toward noble-gas closure. The repeatability of this contraction across multiple periods represents one of the clearest experimentally measured geometric signatures of atomic structure. Any physical ontology of the atom must therefore explain not only the abrupt shell-opening resets identified in Table 10, but also the progressive compression of these boundaries as shell occupancy increases. The unusually small empirical radius assigned to Neon reflects a known radius-definition ambiguity associated with noble gases, which do not form stable covalent bonds under standard conditions. Accordingly, noble-gas radii – particularly Neon – carry greater definitional uncertainty than directly bonded covalent radii, and the corresponding Period 2 contraction magnitude should therefore be interpreted as approximate.

Inspection of the empirical radius-reset and period-contraction data reveals an intriguing and potentially non-trivial geometric regularity. From approximately the fourth principal shell onward, the opening of each new Group 1 boundary appears to produce a near-constant outward radius reset of approximately **+150 pm**, while the subsequent contraction toward noble-gas closure appears to converge toward a similarly repeatable inward compression of approximately **−130 to −140 pm**. At present, this observation is presented only as an empirical pattern within the measured atomic

baseline. However, if confirmed across a frozen dataset and independent radius conventions, it may suggest the presence of an underlying dual geometric scaling law governing the expansion and compression of atomic boundaries – an observation of potential significance for any causal physical ontology of atomic structure.

3.8 Complete Atomic Radius Baseline: Hydrogen to Oganesson (H → Og)

The shell-opening resets and period-contraction trends established in the preceding sections represent local geometric features of atomic structure. However, before any predictive physical model can be constructed, it is necessary to establish the complete radius baseline across the entire known periodic table. To maintain physical interpretability and independence from model-dependent quantum mechanical calculations, the present study adopts **empirical atomic radii** as the frozen geometric baseline for all subsequent calibration, prediction, and audit activities. Empirical radii are derived from experimentally measured interatomic distances obtained through crystallographic, diffraction, and spectroscopic methods, and therefore represent the closest presently available operational measure of the physical spatial extent of atomic matter.

The complete radius dataset from $Z = 1 \rightarrow 118$ establishes the geometric process behaviour of atomic matter across the full periodic table. This dataset captures not only shell-opening resets and period contraction, but also transition-metal compression, lanthanide contraction, heavy-element boundary behaviour, and historically exceptional radius anomalies. This frozen dataset serves as the master geometric baseline against which the Holland Atom will later be calibrated, applied, and independently audited.

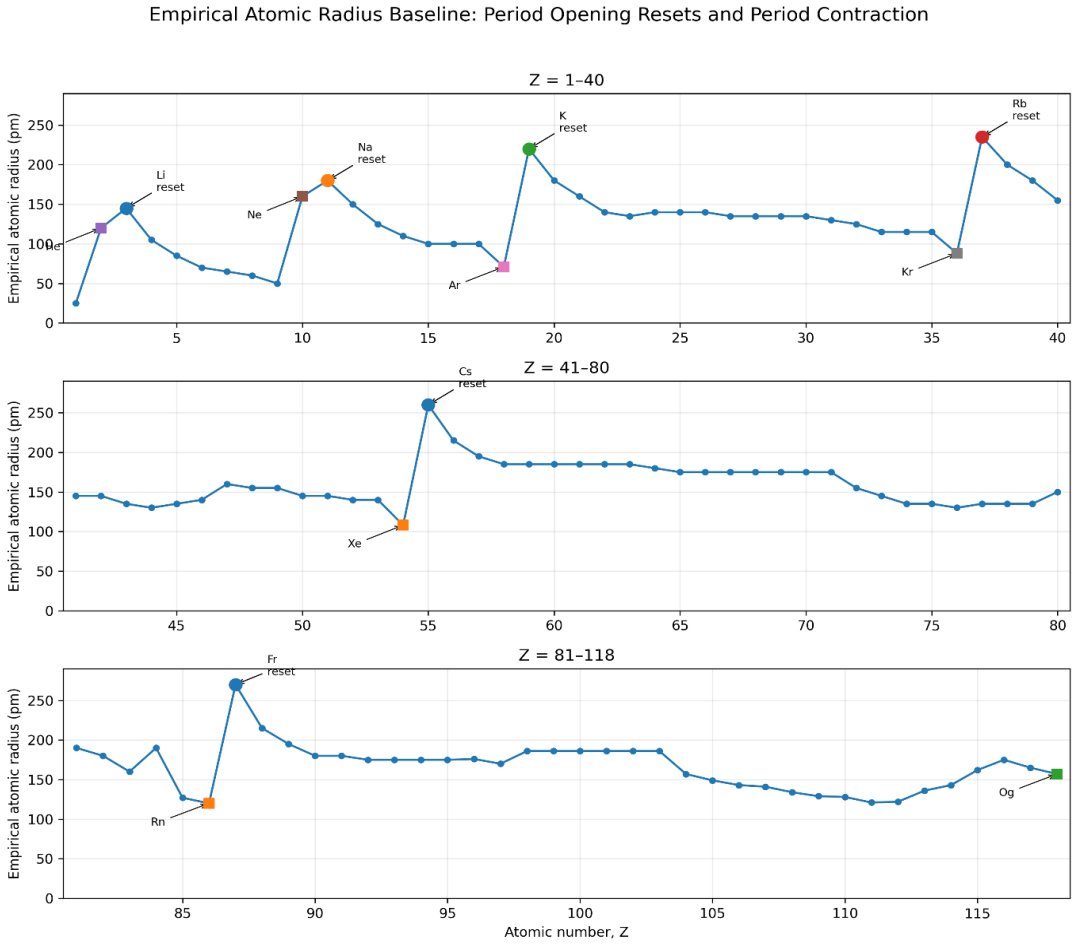


Figure 2. Empirical Atomic Radius Baseline: Hydrogen to Oganesson

Figure 2 presents the complete empirical atomic-radius baseline from Hydrogen ($Z = 1$) to Oganesson ($Z = 118$), displayed in three contiguous panels for visual clarity. Atomic radii are shown using the frozen empirical-radius convention adopted throughout the present study, derived from experimentally measured interatomic distances obtained through crystallographic, diffraction, and spectroscopic methods. The dataset reveals a highly structured sawtooth geometry characterised by abrupt radius expansion at the opening of each new period, followed by progressive intra-period contraction toward noble-gas closure, together with higher-order signatures including transition-metal compression, lanthanide contraction, and heavy-element boundary flattening. This complete geometric baseline forms the master empirical dataset used for all subsequent calibration, prediction, anomaly analysis, and audit of the Holland Atom.

Inspection of the complete empirical atomic-radius baseline shown in Figure 2 reveals that atomic geometry is neither random nor smoothly monotonic across the periodic table. Instead, the full radius sequence from Hydrogen ($Z = 1$) to Oganesson ($Z = 118$) exhibits a highly structured sawtooth architecture characterised by abrupt radius expansion at the opening of each new period, followed by progressive intra-period contraction toward noble-gas closure. Superimposed upon this global behaviour are additional higher-order geometric signatures, including transition-metal compression, lanthanide contraction, actinide compression, and heavy-element boundary flattening.

Taken collectively, the shell-capacity sequence, subshell architecture, filling order, shell-opening resets, period contraction, and full empirical radius baseline establish that atomic matter exhibits a deeply ordered geometric process behaviour extending across the entire periodic table. These experimentally established structural features now complete the measurement baseline required for the present investigation. For completeness, Appendix A consolidates the experimentally established atomic shell architecture, subshell structure, filling sequence, periodic-table block structure, and full H–Og geometric baseline into a single reference infographic, while Appendix C extends the present analysis through quantitative audit of shell-opening reset magnitudes, period-contraction magnitudes, and preliminary geometric scaling trends identified within the empirical dataset.

With the full atomic baseline now established, the investigation may proceed from **what atomic structure measurably does**, toward the deeper question of **what physical mechanism causes it to do so.**

3.9 Transition-Metal Anomalies: Deviations from Sequential Shell Population

While the shell-capacity sequence and subshell-filling architecture established in the preceding sections successfully describe the majority of atomic structure across the periodic table, a number of experimentally confirmed transition-metal elements are known to deviate from the simple sequential filling order predicted by naïve shell-population rules [15,19,20]. These deviations are not rare measurement artefacts, nor theoretical corrections introduced after the fact. Rather, they represent stable, reproducible, and experimentally confirmed ground-state electron configurations that have remained established through spectroscopy, magnetic measurements, ionisation studies, and chemical behaviour.

Of particular significance are a small number of transition-metal elements whose observed ground-state shell populations systematically depart from the expected Aufbau filling sequence [11,19,20]. These anomalies are among the most frequently cited examples of shell-population behaviour that requires additional quantum mechanical interpretation involving orbital-energy splitting, exchange stabilisation, electron correlation, and relativistic correction terms. Table 12 establishes four of the most historically significant transition-metal anomalies, then Table 13 explores these deeper.

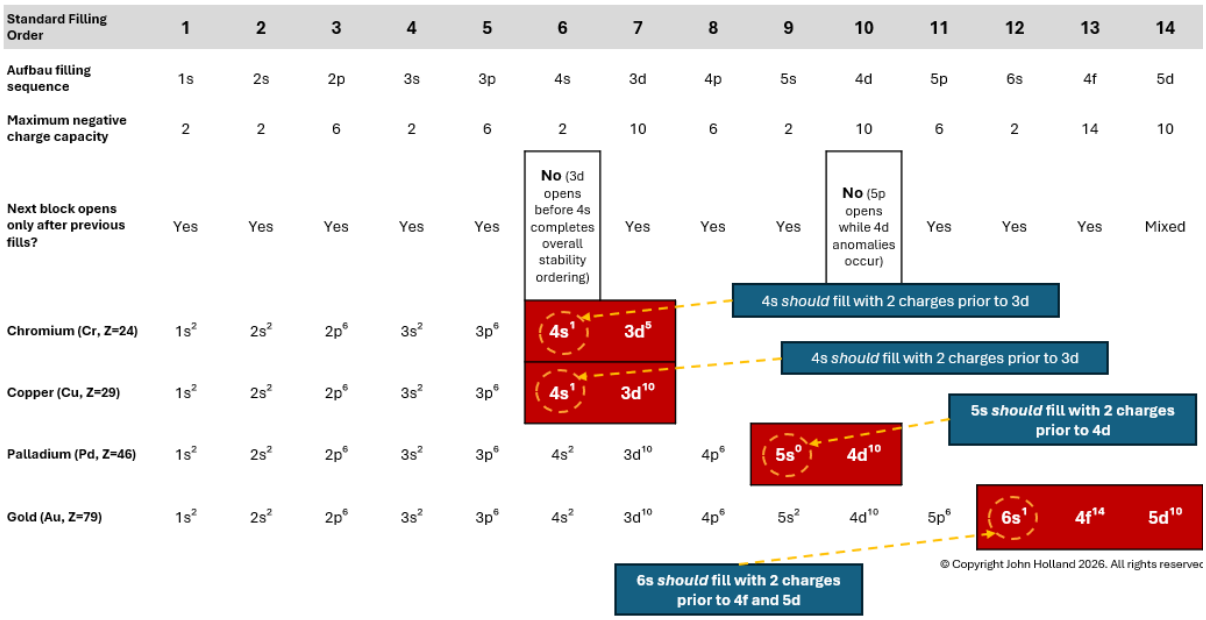


Table 12. Empirical Exceptions to the Aufbau Filling Sequence in Atomic Charge Structures

Table 12 shows the standard Aufbau filling sequence predicts sequential orbital occupation according to lowest total energy states. Highlighted examples show empirically observed deviations, where atomic negative charge reorganises into alternative stable configurations that depart from simple sequential filling expectations.

Element	Expected configuration	Observed configuration	Observed deviation	Experimental status
Chromium	$4s^2\ 3d^4$	$4s^1\ 3d^5$	One 4s electron promoted into 3d	Established
Copper	$4s^2\ 3d^9$	$4s^1\ 3d^{10}$	One 4s electron promoted into 3d	Established
Palladium	$5s^2\ 4d^8$	$5s^0\ 4d^{10}$	Complete 5s depopulation	Established
Gold	$6s^2\ 5d^9$	$6s^1\ 5d^{10}$	One 6s electron promoted into 5d	Established

Table 13. Transition-Metal Shell-Filling Anomalies

Table 13 summarises four experimentally established transition-metal shell-population anomalies in which the observed ground-state atomic configurations depart from the simple sequential shell-filling order established by the Aufbau principle. These deviations are directly supported by spectroscopic, magnetic, ionisation, and chemical measurements, and form part of the accepted atomic baseline of modern quantum mechanics. The existence of such stable anomalies demonstrates that atomic shell population does not always proceed through simple shell-capacity maximisation and therefore represents a critical baseline condition for any causal physical model of atomic structure.

Inspection of Table 13 reveals a striking common pattern: in each case, atomic charge reorganises toward **half-filled or fully filled subshell states**, even when this requires apparent violation of the simple sequential shell-filling order established elsewhere in the periodic table [15,19,20]. The physical significance of this repeatable behaviour forms one of the most important empirical baselines for the present investigation.

3.10 Lanthanide Contraction: Progressive Compression Within the 4f Series

Beyond the shell-opening resets, period contraction, and transition-metal anomalies established in the preceding sections, the periodic table exhibits an additional and highly structured geometric signature known as the **lanthanide contraction**. This phenomenon describes the progressive reduction in atomic radius observed across the lanthanide series as atomic number increases from Lanthanum ($Z = 57$) to Lutetium ($Z = 71$). Unlike abrupt shell-opening resets, the lanthanide contraction occurs as a smooth and highly repeatable inward compression of atomic boundary size, despite the continued addition of both nuclear charge and internal shell population [12,14]. This behaviour has been experimentally established through crystallographic measurements, spectroscopy, chemical bonding studies, and ionic-radius analysis, and remains one of the most important geometric signatures in atomic structure [15,16].

Within standard atomic theory, the lanthanide contraction is typically described as arising from incomplete shielding of increasing nuclear charge by the progressively populated 4f subshell [18-20]. Regardless of interpretation, the measured geometric behaviour itself is unambiguous: as the 4f series is populated, atomic boundaries progressively contract in a highly structured and reproducible manner.

Table 14 establishes the empirical lanthanide contraction baseline.

Element	Atomic Number	4f Occupancy Progression	Empirical Atomic Radius (pm)	Relative Behaviour
Lanthanum	57	Series opening	**195**	Maximum radius
Cerium	58	Early filling	**185**	Initial contraction
Neodymium	60	Progressive filling	**185**	Stable compression
Gadolinium	64	Mid-series	**180**	Continued contraction
Dysprosium	66	Mid-series	**175**	Continued contraction
Ytterbium	70	Late filling	**175**	Near minimum
Lutetium	71	Series closure	**175**	Minimum radius

Table 14. Lanthanide Contraction Baseline (Empirical Atomic Radii)

Table 14 summarises the empirically measured atomic radii across the lanthanide series, revealing a progressive and highly structured inward contraction as the 4f series is populated from Lanthanum to Lutetium. Unlike the abrupt shell-opening resets observed at the beginning of each period, lanthanide contraction represents a smooth intra-series compression of atomic boundary size. This behaviour is directly supported by crystallographic, spectroscopic, and chemical measurements, and forms one of the most important experimentally established geometric signatures of atomic structure.

Inspection of the empirical lanthanide baseline reveals an approximately monotonic inward contraction of ~20 pm across the complete 4f series, despite substantial increases in nuclear charge and internal shell complexity. The existence of such smooth, structured geometric compression within a partially populated internal shell represents a critical baseline condition for any causal physical model of atomic structure.

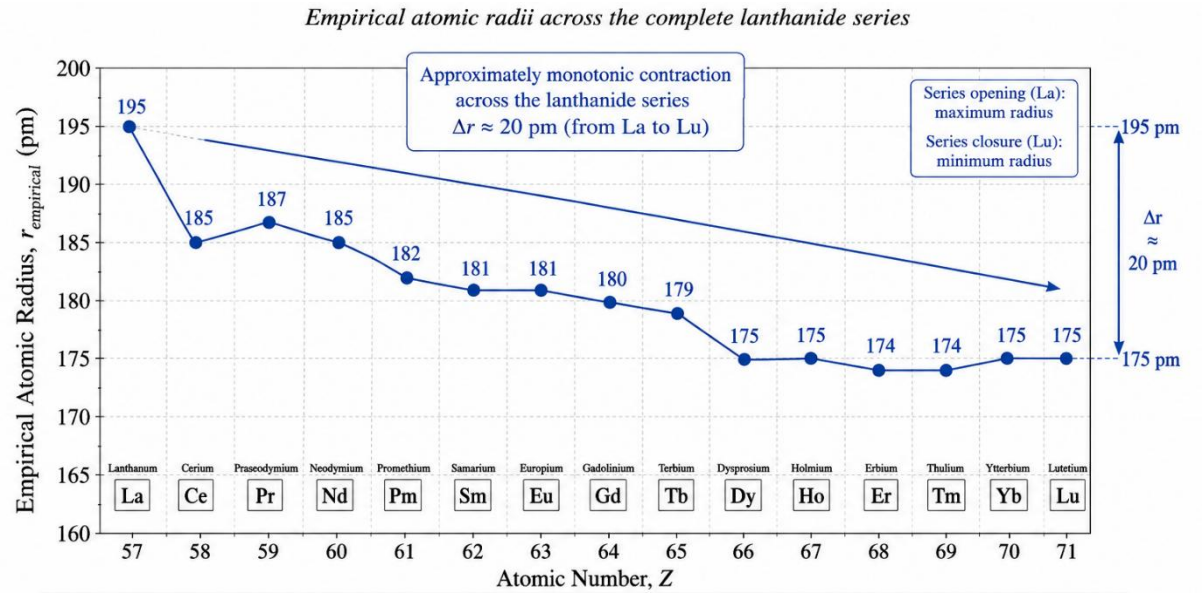


Figure 3. Empirical Lanthanide Contraction Baseline ($Z = 57 \rightarrow 71$)

Figure 3 shows the complete empirical atomic-radius baseline across the lanthanide series from Lanthanum ($Z = 57$) to Lutetium ($Z = 71$). Unlike the abrupt shell-opening resets and sawtooth period contraction observed elsewhere in the periodic table, the lanthanide series exhibits a smooth and highly structured inward compression of atomic boundary size as the 4f series is progressively populated. The approximately monotonic contraction of ~20 pm across the series represents one of the clearest experimentally established examples of internal shell-driven geometric compression in atomic structure.

3.11 Actinide Contraction: Independent Confirmation of Internal Shell Compression

A second series of internal-shell-driven geometric compression is observed within the actinide series, extending from Actinium ($Z = 89$) to Lawrencium ($Z = 103$). As with the lanthanides, empirical atomic radii exhibit a progressive inward contraction as the internal f-series is populated. Although experimental uncertainties are generally larger owing to radioactivity, limited sample availability, and shorter isotopic lifetimes, the overall geometric trend remains consistent with the lanthanide baseline: progressive internal shell population is accompanied by measurable compression of the outer atomic boundary [14,18]. This independent recurrence of internal-shell-driven contraction provides additional experimental support for the existence of structured geometric compression mechanisms operating within atomic matter.

Element	Z	Empirical radius (pm)	Behaviour
Actinium	89	195	Series opening
Thorium	90	180	Initial contraction
Uranium	92	175	Continued contraction
Curium	96	176	Stable compression
Californium	98	186*	Dataset variability
Lawrencium	103	~186*	Series closure

*Values are less directly constrained than lanthanides.

Table 15. Actinide Contraction Baseline

Table 15 summarises the empirically observed atomic-radius behaviour across the actinide series from Actinium ($Z = 89$) to Lawrencium ($Z = 103$). Unlike the lanthanides, several late-series actinide radii remain less directly constrained owing to radioactivity, short isotopic lifetimes, limited sample availability, and partial reliance on inferred or model-dependent radius estimates. Late-actinide atomic radii remain less directly constrained experimentally than corresponding lanthanide radii, and published values for elements such as Californium (Z = 98) and Lawrencium (Z = 103) exhibit greater inter-source variability than lighter periodic-table systems. The values adopted here follow the covalent-radius compilation of Pyykkö and Atsumi [14] and should therefore be interpreted as approximate first-order empirical estimates rather than uniquely established atomic radii. Importantly, the existence of a broad actinide contraction trend remains evident from the independently established early- and mid-series measurements and does not depend critically upon the precise values of these late-series entries.

3.12 Existing Quantum Mechanical Predictive Completeness Audit

Having established the complete experimental baseline of atomic structure – including shell capacities, subshell architecture, shell-filling behaviour, atomic-radius geometry, shell-opening resets, period contraction, transition-metal anomalies, lanthanide contraction, and actinide contraction – it is now appropriate to assess the predictive completeness of the modern quantum mechanical atom against these experimentally established baseline conditions.

Since its development in the early twentieth century, quantum mechanics has demonstrated extraordinary predictive success across atomic physics. Through the Schrödinger equation, angular-momentum operators, spin formalism, exclusion rules, relativistic corrections, and many-body approximations, the modern quantum mechanical framework successfully reproduces an exceptionally broad range of experimentally observed atomic behaviours with remarkable precision. These predictive successes include discrete shell capacities, subshell occupancy rules, orbital-state splitting, spectroscopic transition energies, ionisation potentials, magnetic moments, chemical periodicity, shell-filling architecture, transition-metal configuration anomalies, and the broad geometric trends observed across the periodic table. Few scientific frameworks in history have demonstrated comparable quantitative agreement with experiment across such a wide range of physical observables. To objectively establish the present predictive baseline, Table 16 audits the completeness of modern quantum mechanics against the experimentally established atomic conditions defined throughout Section 3.

3.12.1 Quantum Mechanical Predictive Completeness Audit

Baseline condition	Experimental status	QM predictive status	Audit
Nuclear centralisation	Established	Successfully reproduced	PASS
Discrete shell capacities	Established	Successfully reproduced	PASS
Subshell architecture (s,p,d,f)	Established	Successfully reproduced	PASS
Shell-filling sequence	Established	Successfully reproduced	PASS
Spectroscopic transitions	Established	Successfully reproduced	PASS
Magnetic splitting	Established	Successfully reproduced	PASS
Chemical periodicity	Established	Successfully reproduced	PASS
Group 1 shell-opening resets	Established	Successfully reproduced	PASS
Period contraction	Established	Successfully reproduced	PASS
Transition-metal anomalies	Established	Successfully reproduced	PASS
Lanthanide contraction	Established	Successfully reproduced	PASS
Actinide contraction	Established	Successfully reproduced	PASS

Table 16. Quantum Mechanical Predictive Completeness Audit

Table 16 audits the predictive completeness of the modern quantum mechanical atom against the experimentally established baseline conditions defined throughout Section 3. Across all major structural, spectroscopic, magnetic, geometric, and anomalous atomic behaviours examined in the present study, quantum mechanics successfully reproduces the observed experimental outcomes, demonstrating extraordinary predictive completeness across the full atomic baseline.

Critically, throughout the present section, "*quantum mechanical predictive status*" refers specifically to the demonstrated ability of **wavefunction-based mathematical formalisms** to reproduce experimentally observed atomic outcomes through probabilistic state evolution and measurement postulates and should not be interpreted as implying direct causal physical identification of what underlying entity, medium, or mechanism physically generates the observed atomic structure.

Importantly, the purpose of the present audit is not to challenge the predictive success of quantum mechanics, which remains among the most experimentally successful scientific frameworks ever constructed. On the contrary, the present audit confirms that the modern quantum mechanical atom successfully reproduces all major experimentally established baseline conditions examined in this study. However, predictive completeness alone does not necessarily constitute causal physical completeness. The distinction between *what a theory successfully predicts* and *what physical mechanism actually causes those predictions to emerge* therefore motivates the next stage of the present investigation.

3.13 Definitive Causal Gap Audit: What Remains Physically Unexplained

The preceding section confirms that modern quantum mechanics reproduces the experimentally established atomic baseline with extraordinary predictive completeness. This success is not disputed in the present study. However, the purpose of the Holland Atom investigation is not merely to ask whether atomic behaviour can be calculated, but whether the physical architecture of the atom can be causally understood.

A distinction must therefore be maintained between **predictive completeness** and **causal physical completeness**. Quantum mechanics provides a mathematical formalism that successfully predicts atomic observables through wavefunctions, quantum numbers, operators, degeneracy rules, exclusion principles, and probabilistic measurement outcomes. What remains unresolved is whether these constructs identify the physical mechanism that causes atomic matter to organise into the observed shell, subshell, block, radius, contraction, and anomaly structures.

The predictive audit presented in Table 16 and the ontological audit presented below in Table 17 evaluate fundamentally different categories of scientific completeness. Table 16 assesses the demonstrated predictive capability of standard quantum-mechanical formalisms. Within this **mathematical predictive domain**, quantum mechanics successfully reproduces the experimentally observed atomic spectra, shell occupancies, angular structures, periodic trends, and anomaly configurations of atomic matter, and therefore correctly receives a "PASS" classification throughout the predictive audit.

Table 17, by contrast, evaluates **unresolved ontological and causal** interpretation questions associated with the physical meaning of the underlying quantum-mechanical formalism itself. These questions do not necessarily represent failures of predictive accuracy but instead concern the absence of an explicitly specified physical mechanism underlying the successful mathematical formalism.

Importantly, several entries within Table 17 – particularly those associated with wavefunction ontology, spin interpretation, and electron localisation – represent longstanding foundational interpretation questions within the history of quantum theory itself and may ultimately remain experimentally underdetermined. Accordingly, the purpose of the ontological audit is not to diminish the extraordinary predictive success of quantum mechanics, but rather to identify areas in which the

Holland Atom proposes an alternative physically causal standing-wave interpretation of the same experimentally observed atomic phenomena.

Table 17 therefore summarises the principal **causal gaps** identified from the measured atomic baseline.

Ref	Baseline feature	What is experimentally established	What QM predicts successfully	Remaining causal / ontological gap
G1	Shell existence	Atomic charge occupies discrete stable energy regions	Shell energies and capacities	What physical mechanism creates stable shell regions?
G2	Shell capacity	Capacities follow (2,8,18,32...)	$2n^2$ derived mathematically	Why does physical charge organise according to this geometry?
G3	Subshells	Internal state families exist within shells	s,p,d,f formalism	What physical structure causes (2,6,10,14) capacity families?
G4	Orientation counts	Subshells follow (1,3,5,7) orientation sequence	$2l+1$ degeneracy	Why should nature physically generate odd-numbered orientation families?
G5	Spin occupancy	Two-state occupancy is observed	Spin and Pauli exclusion	What physical mechanism enforces two occupancy states per orientation?
G6	Wavefunction	Wavefunction predicts atomic observables	ψ evolves mathematically	What, if anything, is physically waving?
G7	Orbital shapes	Probability-density patterns match observations	Orbital solutions	Are orbitals physical structures or mathematical probability maps?
G8	Shell opening	New outer shells open at period starts	Aufbau sequence	What physical event creates a new atomic boundary?
G9	Non-sequential filling	4s opens before 3d completion, etc.	Energy-ordering rules	Why does nature open outer boundaries before inner capacities are full?
G10	Group 1 radius reset	Atomic radius jumps at new period openings	Shielding / effective nuclear charge	What physical mechanism produces repeatable geometric reset jumps?
G11	Period contraction	Radius contracts across periods	Increasing effective nuclear charge	What physically compresses the outer boundary as charge is added?
G12	Transition anomalies	Cr, Cu, Pd, Au deviate from naïve filling	Exchange/correlation/ relativistic corrections	Why do stable atoms reorganise toward half-filled or filled bands?
G13	Lanthanide contraction	4f filling compresses atomic radius	Incomplete shielding	What physical mechanism converts internal filling into outer-boundary contraction?
G14	Actinide contraction	A second f-series contraction occurs	Relativistic and shielding effects	Why does analogous internal compression recur at higher Z?
G15	Localised detection	Electron-like events appear during interaction	Measurement postulates	Does the electron exist as a point object between detections?
G16	Atomic ontology	Atomic observables are predicted	Formalism complete operationally	What is the atom physically?

Table 17. Causal Gap Audit of the Modern Quantum Mechanical Atom

Table 17 identifies the principal causal and ontological gaps that remain after the predictive completeness of modern quantum mechanics is acknowledged. The table does not claim that quantum mechanics fails experimentally; rather, it distinguishes successful mathematical prediction from physical causal explanation. The unresolved questions listed here define the explanatory target for the Holland Atom: a proposed physical ontology intended to account for the same atomic baseline through resonance-lock structure rather than through purely formal state descriptors.

The most important conclusion from Table 16 is that the unresolved issues are not isolated curiosities. They form a connected explanatory pattern. The residual causal gap and ontological gap point toward the same deeper question: *why does atomic charge organise into such a precise, repeatable, geometric architecture?*

Modern quantum mechanics answers this question operationally by assigning allowed wavefunction solutions and quantum numbers. It tells us which states are permitted, which transitions occur, and what probabilities should be expected. It does not uniquely identify a physical substrate, medium, or causal mechanism that generates those allowed states as real structures in space. So, we can predict with extraordinary precision, whilst still debating what is physically real underneath the formalism.

This distinction defines the central opportunity for the Holland Atom. If a physical field ontology can reproduce the same baseline facts while also explaining why the architecture takes the observed form, then the question shifts from whether quantum mechanics predicts the atom to whether its current ontology is physically complete.

The present study therefore proceeds from measurement into analysis with a clear target: to determine whether the observed architecture of atomic matter can be reconstructed as a causal geometric system rather than accepted only as a successful mathematical formalism.

The ambitious imagination here dares to consider that many of these gaps (G1 – G16) may not be independent – shell capacities, odd orientation counts, radius resets, contraction, half-filled stability, lanthanide compression, and localised detection – may be **one phenomenon, viewed from sixteen different angles.**

4. The General Expanse Tension Theory (GETT) Framework

The preceding sections establish that atomic matter exhibits a highly ordered and experimentally reproducible architecture. Modern quantum mechanics reproduces these observations with extraordinary predictive success, yet the deeper physical mechanism responsible for this geometric organisation remains unresolved.

4.1 The Holland Field Φ

The present investigation approaches this causal gap through the **General Expanse Tension Theory (GETT)**, a field-based physical ontology proposing that gravity, inertia, temporal behaviour, and structured matter emerge through interaction with a real, all-pervading scalar field, conventionally denoted:

$$\Phi$$

Within GETT, the Holland field Φ is proposed as a physically real, gauge-invariant, Lorentz-consistent, singlet scalar field permeating all space and all matter [21-23]. Unlike purely geometric or probabilistic formalisms, Φ is treated as a genuine physical substrate capable of supporting spatial gradients, restoring forces, standing modes, boundary conditions, and propagating tension disturbances. In its simplest physical interpretation, mass locally resists the natural expansive tendency of Φ, producing measurable field gradients whose restoring reaction manifests macroscopically as gravitation. Under this ontology, gravity is not treated as a standalone fundamental interaction, but as an emergent response of a deeper physical field.

4.2 Continuous Density-Dependent Modulation

A defining feature of GETT is the presence of a **continuous density-dependent modulation** governing the effective coupling between the Holland scalar field Φand the Higgs sector. In the full field framework, this interaction is regulated by the dimensionless coupling modulation function:

$$S(\Sigma)$$

where Σ represents the local coarse-grained mass-energy density, interpreted covariantly through local energy–momentum invariants. This modulation permits the scalar–Higgs interaction strength to vary

continuously across extreme physical environments, enabling GETT to recover standard Einstein-domain behaviour at intermediate densities while naturally permitting modified field behaviour in ultra-low-density cosmological voids and ultra-high-density compact-object interiors.

Extensive prior GETT analysis has identified four principal transition regimes: an ultra-low-, extreme-low-, ultra-high- and extreme-high-density activation as follows;

- The ***Extreme-low*** regime of outer galactic regions ("*dark matter*") $\Sigma \lesssim 10^{-21.6}\ \mathrm{kg\,m^{-3}}$
- The ***Ultra-low*** regime of late-time acceleration ("*dark energy*") at $\Sigma \lesssim 10^{-27}\ \mathrm{kg\,m^{-3}}$
- *The* ***Extreme-high*** regime producing glitchy pulsars at $\Sigma \gtrsim 10^{17}\ \mathrm{kg\,m^{-3}}$
- The ***Ultra-high*** regime at black hole cores with symmetry-restoration behaviour emerging beyond approximately $\Sigma \gtrsim 10^{30}\ \mathrm{kg\,m^{-3}}$.

Between these limits lies the **mid-density Einstein domain** [35,36], where the coupling modulation is effectively constant, spatially smooth, and adiabatic:

$$10^{-21.6} \lesssim \Sigma \lesssim 10^{17}\ \mathrm{kg\,m^{-3}},\ \frac{dS}{d\Sigma} \approx 0$$

The present atomic investigation is conducted entirely within this mid-density Einstein domain. Accordingly, while the full density-dependent coupling architecture remains an essential part of GETT, no threshold switching, non-adiabatic transition, or density-triggered coupling modulation is invoked within the Holland Atom. For the purposes of the present work, the scalar–Higgs coupling may therefore be treated as effectively constant across all atomic environments considered.

The Holland Atom is not introduced as an independent atomic model separate from GETT, but as an atomic-scale application of the same density-regulated scalar-field framework developed throughout the General Expanse Tension Theory programme. In the underlying singlet-scalar formulation, the effective scalar–matter interaction strength is modulated through the density-dependent portal function $S(\Sigma)$, where Σ represents the local effective mass-energy density environment.

Within the Holland Atom, the shell-forming regime is interpreted as occupying a comparatively narrow density domain relative to the many orders of magnitude spanned by the full GETT framework. Consequently, variations in $S(\Sigma)$ across neighbouring shell boundaries are assumed to remain sufficiently smooth that local standing-wave closure, shell accessibility, and resonance partitioning may be treated using first-order approximations. In this sense, the Holland Atom adopts an atomic-scale analogue of the smooth, locally stable regime used elsewhere in GETT to justify effective-domain approximations.

The postulates developed in the present work should therefore be interpreted as a first-order reconstruction of atomic structure within a locally stable density regime of the broader $S(\Sigma)$ framework, rather than as a complete derivation of atomic behaviour directly from the underlying scalar-field action.

4.3 GETT Framework Status

The foundational field mechanism underlying GETT has undergone independent expert peer review within the context of quantum field theory and effective field theory, where the proposed scalar-field–Higgs-portal interaction was explicitly assessed for consistency with established principles of gauge invariance, Lorentz symmetry, renormalisability, vacuum stability, and boundedness. Expert review

concluded that the mechanism is *fully consistent* with established quantum field theoretical principles, while identifying the need for further phenomenological application and experimental validation – precisely the purpose of the broader GETT research programme.

Since establishment of the foundational mechanism, GETT has been systematically applied across an expanding portfolio of physical domains, including gravitational correspondence, inertial mechanics, special relativity, general relativity, galaxy-scale dynamics, dark-matter-like phenomena, void dynamics, cosmological expansion, gravitational lensing, compact-object physics, gravitational-wave reinterpretation, and atomic-scale reconstruction. This programme includes the **Solving Science** series, focused on causal reinterpretation of historically unresolved physical phenomena [24-29], together with the **GETT Correspondence Series: A Domain-Limit Reconstruction of Physics (GCS-DLRP)**, which systematically reconstructs established physical laws as emergent domain limits of the Holland Field under experimentally defined conditions [30-38].

Across these investigations, the central working hypothesis of GETT has remained unchanged: where modern physics successfully predicts observed behaviour, those predictions may represent mathematically accurate domain limits of a deeper physical field ontology rather than complete causal descriptions in themselves. The present investigation extends that hypothesis to atomic matter, asking whether the experimentally established shell architecture of the periodic table may likewise emerge as a structured resonance phenomenon of the Holland Field. With this field framework now established, the investigation proceeds to define the specific atomic postulates of the **Holland Atom**.

4.4 First Physical Postulates of the Holland Atom

Here we introduce the **Holland Atom**: a proposed physical reconstruction of atomic structure based upon the interaction of nuclear matter with the all-pervading Holland Field:

	Φ	Eq. 4.4.1

The Holland Atom does not begin by assuming orbiting electrons, probability-density orbitals, wavefunction collapse, or pre-defined quantum-state descriptors. Instead, the model begins from a simpler physical premise:

Atomic structure emerges as a stable resonance-lock geometry formed through the interaction of nuclear mass-energy with the real scalar field Φ [38]**.**

To establish this ontology rigorously, the Holland Atom is defined through the following first physical Holland (H) postulates.

H1 — Nuclear Mass Generates Local Field Boundary Conditions

The atomic nucleus is proposed to act as a local mass-energy boundary condition within the Holland Field. Through its finite mass-energy density, the nucleus locally resists the expansive tendency of Φ, generating a stable three-dimensional field disturbance centred on the nuclear core. Under this postulate, the nucleus does not merely act as a Coulomb centre for orbiting charge, but as the primary geometric source of local field structuring.

H2 — Nuclear Density Defines a Characteristic Resonance Wavelength

Each atomic nucleus is proposed to establish a characteristic local resonance wavelength within the surrounding Holland Field. This wavelength is determined by the effective nuclear mass-energy density:

	Σ_N	Eq. 4.4.2

such that nuclei of differing density generate distinct local field wavelengths and therefore distinct atomic geometric scales.

Under this postulate, atomic size and shell structure are not arbitrary, but emerge from nuclear-density-conditioned field geometry and resonance behaviour. The precise scaling behaviour may depend on the local coupling regime of the Holland Field, including possible departures from the smooth Einstein-domain limit within inner-shell regions.

H3 — Stable Standing Modes Form Spherical Resonance Boundaries

The local field disturbance generated by the nucleus is proposed to form stable standing-wave solutions within Φ, producing discrete spherical resonance boundaries surrounding the nuclear core. These boundaries define the physically real shell architecture of the atom. Under this postulate:

Atomic shells are <u>not</u> probability distributions.

They are: **Stable standing-mode boundaries of the Holland Field.**

H4 — Shell Radii Correspond to Standing Nodes of the Field

Each allowed atomic shell is proposed to correspond to a discrete standing-node radius of the local field structure. The experimentally observed shell sequence:

$$K,\ L,\ M,\ N,\ O,\ P,\ Q$$

is therefore interpreted as the physical node sequence of the nucleus-driven Holland Field resonance. Under this postulate, shell radii are directly measurable geometric boundaries rather than mathematically inferred occupancy regions.

H5 — Stable Atomic Charge Exists as Distributed Surface Occupancy

Negative atomic charge is proposed <u>not</u> to exist as continuously orbiting point particles between interactions, but as stable distributed electromagnetic occupancy bound to the resonance boundaries of the Holland Field. Under this postulate:

Particles do <u>not</u> orbit shells. Charge occupies shells.

Localised electron-like events are interpreted as interaction-dependent extraction, excitation, absorption, or detection of this distributed shell-bound charge.

H6 — Spherical Surface Area Defines Shell Charge Capacity

Each stable Holland-field standing mode establishes a characteristic resonance radius. The corresponding spherical surface area determines the maximum negative charge capacity of that shell. Under the local constant-wavelength standing-wave approximation associated with the atomic resonance plateau regime, shell capacities follow geometrically from standing-wave shell closure and surface-area scaling. The standing-wave shell structure is treated under a local adiabatic approximation in which the Holland-field resonance wavelength varies sufficiently slowly across the shell-forming region. Under this postulate, the experimentally observed shell capacities:

$$2,\ 8,\ 18,\ 32,\ \ldots$$

are interpreted as geometric consequences of standing-mode surface architecture rather than abstract quantum occupancy rules. Therefore, shell charge scales with: capacity ∝ 4πr2.

H7 — Surface Charge Degeneracy and Capacity

Once a stable spherical resonance-lock boundary has formed within the Holland Field Φ, each discrete unit of available shell surface geometry is proposed to support **two phase-stable electromagnetic charge states**. This defines a universal surface degeneracy factor of two across all stable atomic shell boundaries.

Under this postulate, distributed negative charge does not occupy abstract orbital states, probabilistic electron clouds, or point-particle trajectories. Instead, charge resides directly upon the shell as a continuous surface-bound electromagnetic occupancy embedded within the standing-wave geometry of Φ. Each local surface resonance site is therefore capable of supporting the Surface Occupancy Factor;

	$O_{\text{surf}} = 2$	Eq. 4.4.3

phase-stable charge states, where O_{surf} represents the intrinsic surface degeneracy of the Holland-field shell.

The proposed universal surface occupancy factor $O_{\text{surf}} = 2$ is introduced in the present work as a first-order physical postulate of the Holland Atom and represents the number of stable charge states per shell surface unit . Importantly, the existence of stable two-state occupancy is not without precedent in established physical systems. Classical electromagnetic resonators, waveguides, optical cavities, and standing-wave boundary systems routinely support **two phase-stable standing-wave solutions**, typically separated by:

$$0,\ \pi$$

corresponding to equal and opposite phase-lock conditions at the same physical boundary. The Holland Atom therefore adopts the working hypothesis that each stable shell-surface resonance site may similarly support two phase-stable electromagnetic occupancy states embedded within the standing-wave geometry of the Holland Field Φ. A first-principles derivation of this universal two-state surface occupancy – potentially arising through the established scalar–Higgs portal coupling of GETT – remains an important target for future quantum field theoretical development.

If the relative surface geometry of shell n scales as:

$$A_n \propto n^2$$

then the corresponding shell capacity sequence precisely reproduces the K, L, M, N, O, P, and Q shell sequence observed across atomic matter $2,\ 8,\ 18,\ 32,\ 50,\ 72,\ 98$. Under the Holland Atom, shell capacity therefore emerges not from externally imposed orbital counting rules, spin-state bookkeeping, or postulated quantum occupancy constraints, but as the direct physical consequence of:

spherical surface geometry × intrinsic two-state surface degeneracy.

H7 completes the geometric foundation of shell occupancy by converting resonance-lock shell geometry into experimentally observed shell charge capacity, establishing the first direct physical bridge between shell radius, shell surface area, and atomic charge-supporting capability.

A final note to raise that a possible geometric interpretation is that stable standing-wave closure possesses an intrinsic two-boundary nodal topology, although a first-principles derivation remains a target for future QFT development.

H8 — Dynamic Equilibration: Sequential Shell Formation and Resonance Relaxation

Atomic charge population initially proceeds through sequential standing-wave shell formation, with distributed charge states occupying the lowest dynamically accessible resonance-shell boundary consistent with the resonance-shell opening hierarchy. This sequential population process produces the first-order shell architecture naturally associated with progressive shell opening and angular resonance-family accessibility. This postulate asserts the dynamic process by which a system evolves toward equilibrium from an initial non-optimised state.

Non-optimised sequential configurations may subsequently undergo resonance relaxation through charge redistribution whenever reduction in total resonance stress exceeds the energetic cost of shell transfer between neighbouring shells or angular resonance families. Experimentally observed atomic configurations therefore represent final resonance-equilibrium states following:

1. sequential shell formation;
2. intermediate metastable resonance population;
3. and subsequent global resonance relaxation.

Within the Holland Atom framework, experimentally observed anomalous configurations are therefore interpreted not as violations of shell formation, but as energetically favoured relaxed resonance states emerging from an initially sequential standing-wave shell population process.

H9 — Atomic Anomalies Represent Global Resonance Optimisation

The experimentally observed shell-filling anomalies, half-filled stability, full-filled stability, transition-metal deviations, lanthanide contraction, and actinide contraction are proposed to represent global resonance optimisation of the total field geometry. Under this postulate:

Atomic matter always organises toward the lowest global resonance-energy configuration.

With the physical ontology of the Holland Atom now defined, the investigation proceeds to mathematical formalisation of the nuclear-density-controlled resonance geometry, shell-boundary formation, charge-capacity scaling, and full periodic-table prediction.

4.5 Visualising the Holland Atom: Physical Interpretation of Postulates H1–H9

The physical postulates of the Holland Atom establish a fundamentally different atomic ontology from the conventional orbital interpretation of quantum mechanics. However, before proceeding to mathematical formalisation, it is instructive to visualise the physical consequences of these postulates in geometric form.

The following figures provide a stepwise physical interpretation of the Holland Atom, illustrating how nuclear mass-energy, acting within the all-pervading Holland Field Φ, may generate the experimentally observed shell architecture of atomic matter through resonance-lock standing-mode geometry.

4.5.1 — H1 Visualisation: Nuclear Mass Generates a Local Field Disturbance

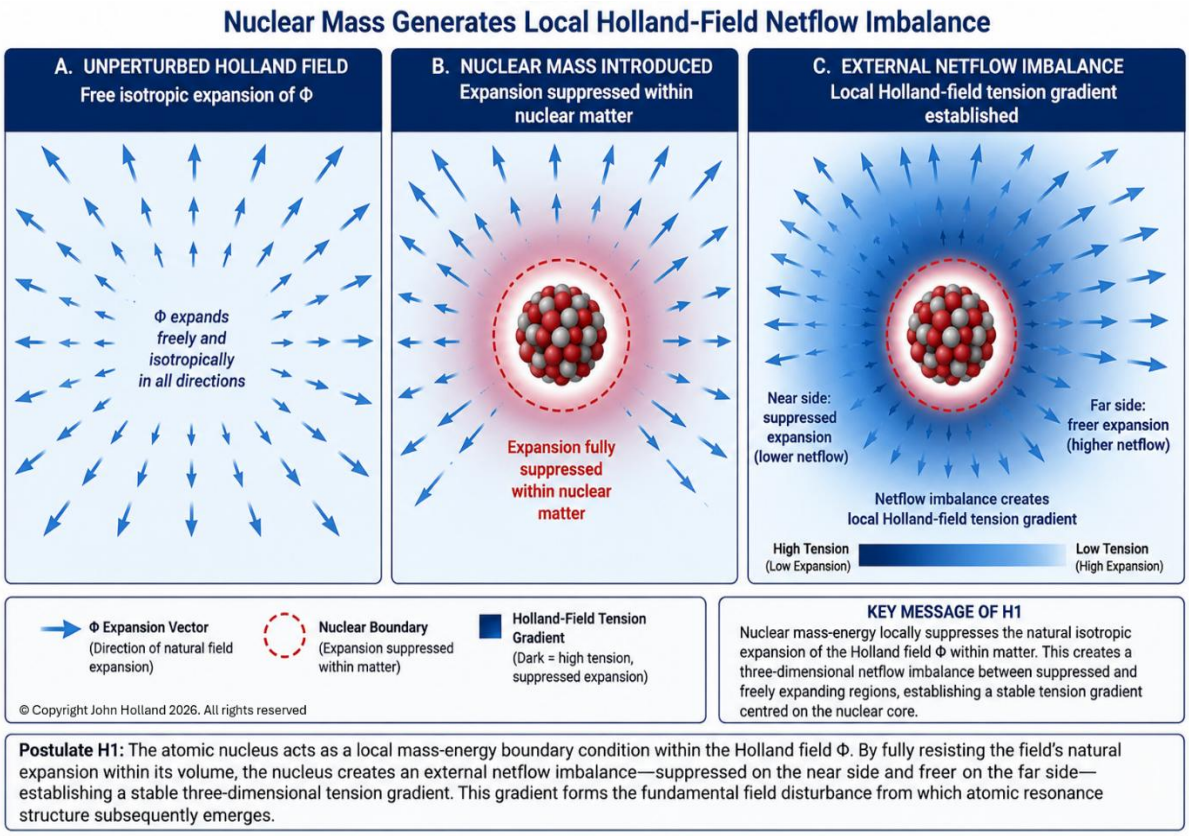


Figure 4. H1: Nuclear Mass Generates local Holland Field Net Flow Imbalance

Figure 4 illustrates Postulate H1. Within the Holland Atom, nuclear mass-energy does not geometrically deform a passive background spacetime. Instead, nuclear matter locally suppresses the natural isotropic expansion of the Holland Field Φ, creating a three-dimensional netflow imbalance between suppressed and freely expanding regions. This imbalance generates a stable tension gradient centred on the nuclear core, establishing the fundamental boundary condition from which atomic resonance structure subsequently emerges.

In the Holland Atom, nuclear mass-energy is not interpreted as geometrically deforming a passive background, but as producing an **active field interruption** within the naturally expanding Holland Field Φ. In the absence of matter, Φ undergoes **free isotropic expansion** equally in all spatial directions.

However, upon introduction of nuclear mass-energy, this expansion becomes locally suppressed, with **expansion fully resisted within nuclear matter**. The resulting asymmetry between suppressed and freely expanding regions generates a measurable **vector imbalance** in the surrounding field, producing a stable three-dimensional **net flow imbalance** centred on the nuclear core. Under GETT, this imbalance does not remain passive, but generates a compensating **restoring tension gradient** within Φ, establishing the fundamental boundary condition from which atomic resonance structure, shell formation, and stable charge organisation subsequently emerge.

4.5.2 — H2 Visualisation: Nuclear Density Defines Characteristic Resonance Scale

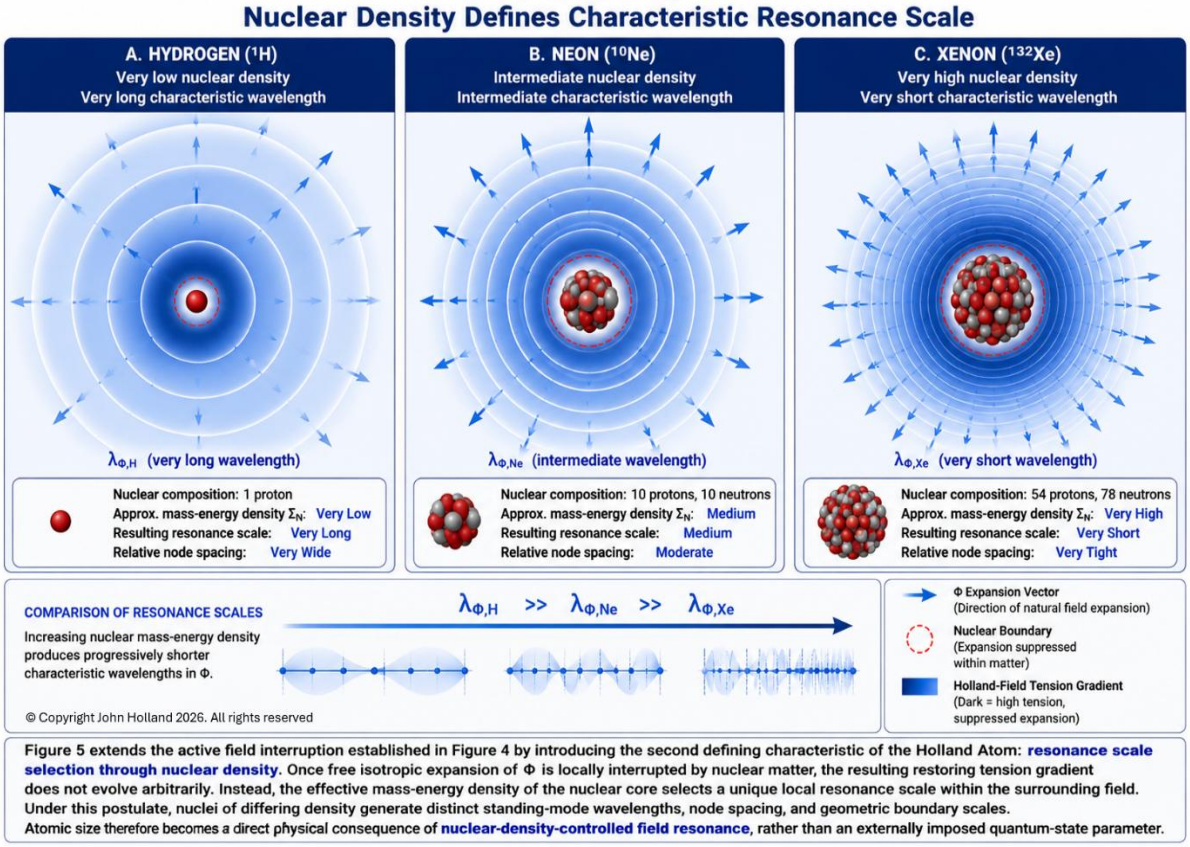


Figure 5. H2: Nuclear Density Defines Resonance Scale

Figure 5 illustrates Postulate H2. Within the Holland Atom, differing nuclear mass-energy densities generate distinct characteristic resonance scales within the local Holland Field Φ. Lower-density nuclei establish longer characteristic field wavelengths, while higher-density nuclei establish progressively shorter resonance wavelengths. Under this ontology, atomic geometric scale emerges directly from nuclear-density-controlled field resonance rather than from pre-assigned quantum-state descriptors.

This visualises the direct alignment with empirical observation:

As elemental nuclear cores grow larger, the atoms themselves become smaller.

Figure 5 extends the active field interruption established in Figure 4 by introducing the second defining characteristic of the Holland Atom: **resonance scale selection through nuclear density.** Once free isotropic expansion of Φ is locally interrupted by nuclear matter, the resulting restoring tension gradient does not evolve arbitrarily. Instead, the effective mass-energy density of the nuclear core selects a unique local resonance scale within the surrounding field. Under this postulate, nuclei of differing density generate distinct standing-mode wavelengths, node spacing, and geometric boundary scales. Atomic size therefore becomes a direct physical consequence of nuclear-density-controlled field resonance, rather than an externally imposed quantum-state parameter.

One of the great paradoxes of atomic structure, as matter displays this remarkable inversion – larger nuclear cores produce smaller atoms – is finally assigned physical cause.

4.5.3 — H3 Visualisation: Stable Standing Modes Form Spherical Resonance Boundaries

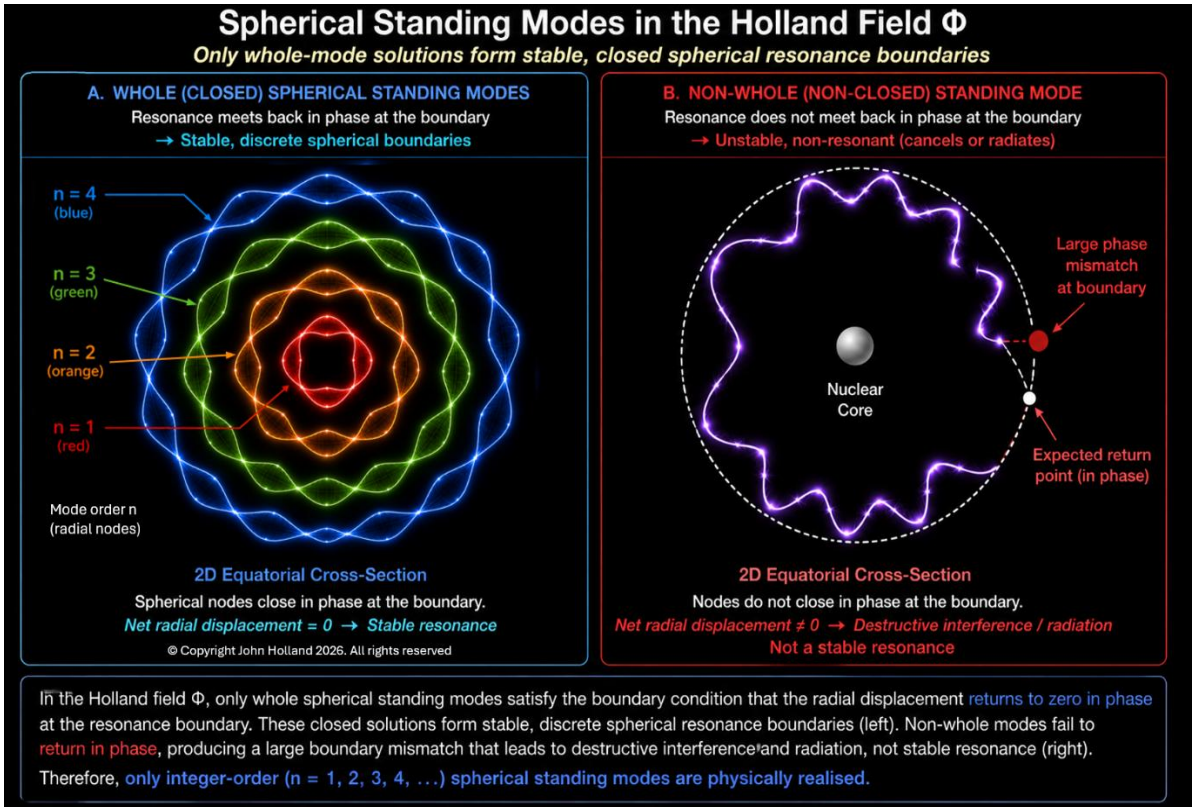


Figure 6. H3-A: Spherical Standing Modes in the Holland Field

Figure 6 illustrates Postulate H3. Following the nuclear-density-dependent resonance-scale selection established in Figure 5, the surrounding Holland Field Φ supports only those standing-wave solutions whose radial displacement returns to zero in phase at a common spherical boundary. Integer-order standing modes $(n = 1,2,3,4, \dots)$ *therefore form stable, closed spherical resonance boundaries (Panel A), where constructive interference produces discrete field-energy minima.*

Under the Holland Atom, each such closed resonance boundary defines an allowable atomic shell radius, uniquely determined by the characteristic resonance wavelength selected by the effective mass-energy density of the nuclear core. These stable shell boundaries subsequently provide the physically permitted surface upon which distributed electromagnetic charge may exist in equilibrium.

By contrast, non-integer or non-whole radial solutions fail to return in phase at a common boundary (Panel B), producing a radial phase mismatch, non-zero net displacement, destructive interference, and radiative loss. Such solutions cannot sustain stable resonance lock and therefore do not form physically real shell boundaries. Under this ontology, only closed spherical standing modes are physically realised, naturally producing the discrete shell architecture observed experimentally across atomic matter.

Think of the Holland Atom as a spherical standing-wave cavity mode anchored by the nucleus – analogous to the stable vibrational modes of a drumhead or the standing waves in a spherical resonator, but in a real scalar field with an inherent outward "pressure."

It is a fixed, self-reinforcing 3D interference pattern where the field's natural expansive tendency is balanced by the nucleus's local resistance of the naturally expanding Holland Field Φ.

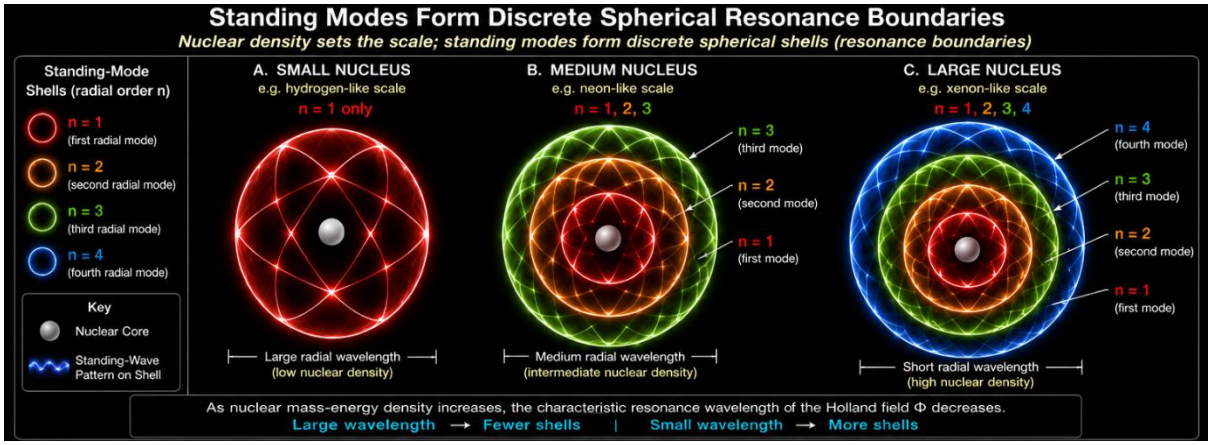


Figure 7. H3-B: Standing Modes Form Discrete Spherical Resonance Boundaries

Figure 7 further develops Postulate H3 by illustrating the emergence of discrete three-dimensional shell architecture from density-controlled spherical standing modes within the Holland Field Φ. Following the nuclear-density-dependent wavelength selection established in Figure 5, only a finite set of closed, integer-order standing-wave solutions satisfy the local resonance boundary conditions surrounding the nuclear core. Each allowed mode state $(n = 1,2,3,4, \dots)$ *therefore represents a distinct and physically stable spherical resonance boundary, forming a specific shell radius at which the radial displacement of the field returns to zero in phase.*

As nuclear mass-energy density increases, the characteristic resonance wavelength decreases, allowing additional standing-mode states to exist within the same physical boundary volume. The result is the progressive emergence of increasingly complex shell architectures, from single-shell resonance states in low-density nuclei, through multi-shell resonance-lock configurations in intermediate nuclei, to higher-order shell families in dense nuclear systems.

Under the Holland Atom, these discrete shell boundaries are interpreted as true resonance-lock characteristics of the nucleus–field system, where each shell radius, shell energy, and allowable charge state originates from a specific and physically realised standing-mode solution, rather than from externally assigned quantum-state descriptors or probabilistic orbital postulates.

Figure 7 introduces the central concept of **resonance lock** within the Holland Atom. Once nuclear density selects the characteristic wavelength of the surrounding Holland Field, only a discrete family of standing-mode states can satisfy complete spherical closure. Each successful mode therefore becomes a stable field-energy minimum, defining a specific shell radius, a specific shell energy, and a specific allowable charge-supporting boundary. Under this ontology, atomic quantisation emerges not from imposed mathematical rules, but from the natural resonance-lock characteristics of the nucleus–field system itself.

4.5.4 — H4 Visualisation: Node Sequence Creates K–Q Shell Architecture

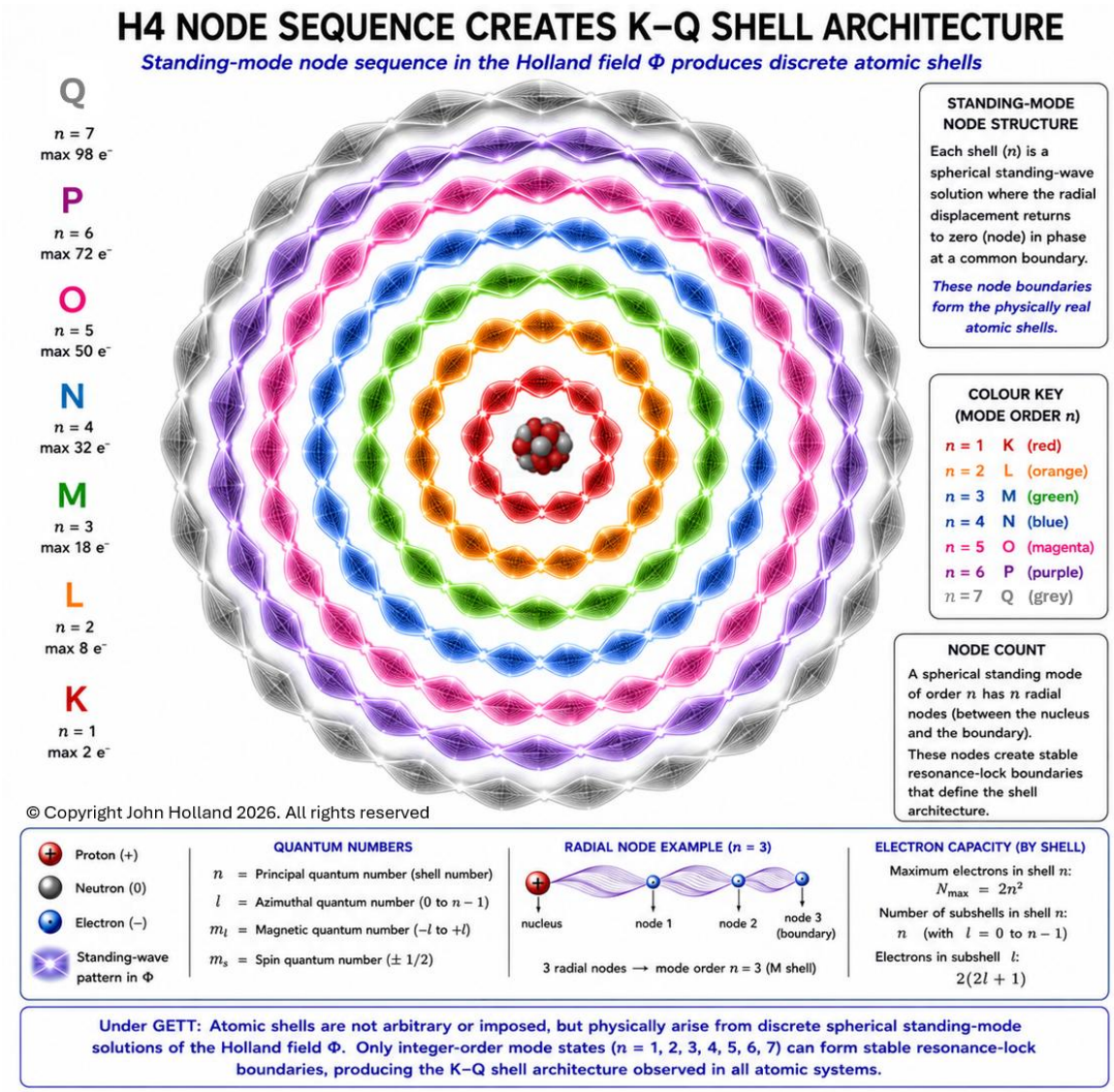


Figure 8. H4: Node Sequence Creates K – Q Shell Architecture

Figure 8 illustrates Postulate H4, showing how the experimentally observed K–Q atomic shell architecture emerges naturally from the discrete radial standing-mode solutions of the Holland Field Φ. Following the resonance-lock conditions established in Figures 6 and 7, only closed integer-order spherical standing-wave states $(n = 1,2,3,4,5,6,7)$ *satisfy complete radial phase closure about the nuclear core. Each successful standing-mode state therefore forms a physically stable spherical node boundary, defining a unique shell radius, a unique field-energy minimum, and a corresponding charge-supporting resonance surface.*

As the radial mode order increases, successive resonance-lock states generate the complete shell sequence conventionally designated K, L, M, N, O, P, and Q. Under the Holland Atom, these shell families are not externally imposed quantum constructs, nor probabilistic occupancy regions, but physically realised spherical node boundaries arising directly from the nucleus–field resonance system.

Each shell boundary therefore represents a specific and discrete standing-mode state of the Holland Field, where shell radius, shell energy, node order, and allowable charge-supporting geometry all

originate from the same underlying resonance-lock condition. Under this ontology, the full K–Q shell architecture of atomic matter emerges as a direct physical consequence of discrete spherical standing-wave quantisation within Φ.

4.5.5 — H5 Visualisation: Distributed Surface Charge Occupancy

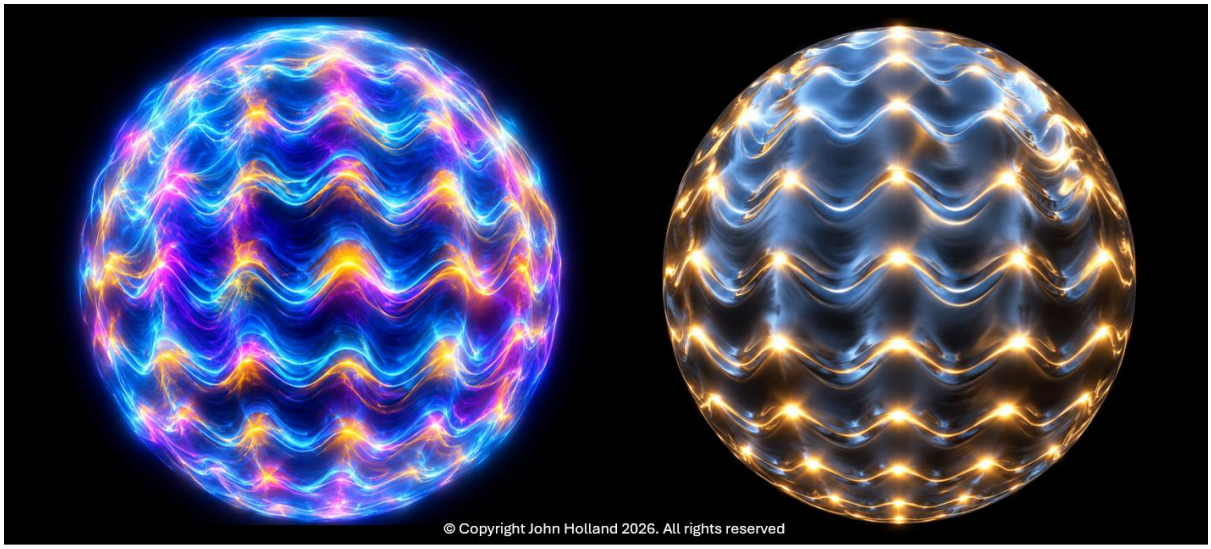


Figure 9. H5: Distributed Surface Charge Occupancy

Figure 9 illustrates Postulate H5, introducing the Holland Atom interpretation of stable atomic charge occupancy. The two visuals are an attempt to represent what the atom shell surface could look like, with distributed surface charge, rather than orbiting particles. Once a spherical resonance-lock boundary is established within the Holland Field Φ, the shell does not act as an empty geometric surface awaiting orbiting electron particles. Instead, the standing-mode boundary itself forms a stable charge-supporting resonance surface, capable of sustaining a distributed electromagnetic occupancy across its entire spherical geometry.

The bright, regularly spaced energy maxima visible across the shell surface represent local standing-wave intensity peaks of the underlying field structure. These maxima do not represent discrete orbiting particles, point-like electrons, or classical charge trajectories. Rather, they represent regions of enhanced electromagnetic field occupancy embedded continuously within the standing-mode shell itself. Under this ontology, negative atomic charge resides on the shell as a distributed surface phenomenon, phase-locked to the underlying resonance structure of the Holland Field.

Localised electron-like detection events – such as emission, absorption, ionisation, scattering, tunnelling, or charge transfer – are therefore interpreted not as the motion of continuously orbiting particles, but as interaction-dependent extraction, excitation, redistribution, or collapse of this pre-existing distributed shell-bound charge state.

Figure 9 introduces a fundamental ontological shift within the Holland Atom. What conventional quantum mechanics models as a probabilistic electron cloud, and classical atomic models once represented as orbiting particles, is reinterpreted here as a physically real, distributed electromagnetic occupancy bound directly to the resonance-lock shell surface. Under this interpretation, charge does not orbit the nucleus through empty space. Charge resides in equilibrium upon the shell itself, embedded within the standing-wave geometry of the Holland Field, with observable electron-like events emerging only when that distributed charge state is locally perturbed by measurement or interaction.

4.5.6 — H6 Visualisation: Surface Geometry Defines Charge Capacity

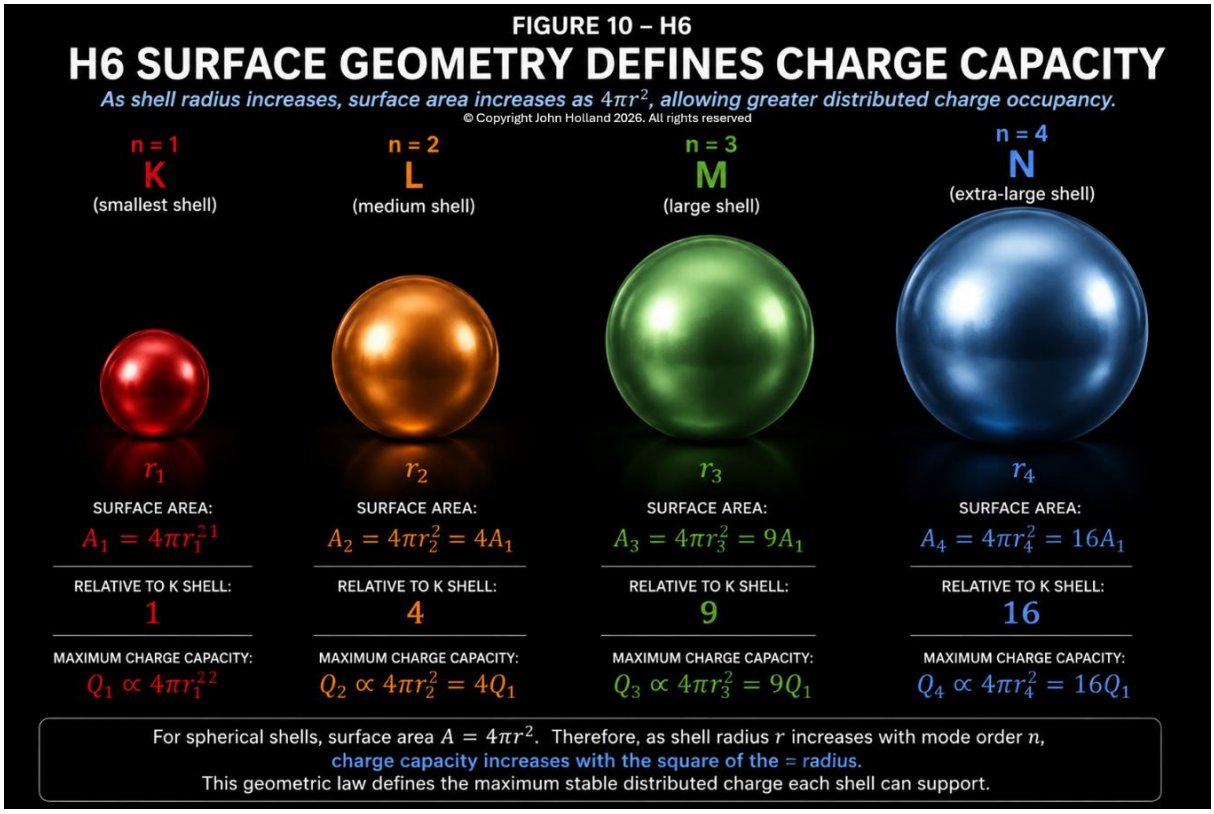


Figure 10. H6: Surface Geometry Defines Charge Capacity

Figure 10 illustrates Postulate H6, introducing the geometric origin of shell charge capacity within the Holland Atom. Once stable resonance-lock boundaries are established within the Holland Field Φ, each shell forms a physically real spherical charge-supporting surface of radius r_n, where the shell index n corresponds to the discrete standing-mode order established in Figures 6–8.

For a spherical shell of radius r_n the available surface geometry is given by:

	$A_n = 4\pi r_n^2$	Eq. 4.5.6.1

such that increasing shell radius produces a quadratic increase in available surface area. The K, L, M, and N shells ($n = 1,2,3,4$) therefore exhibit a progressive geometric expansion in charge-supporting surface, represented visually by the increasing shell radii and corresponding surface-area scaling.

Under the Holland Atom, the experimentally observed increase in allowable shell charge occupancy is therefore not introduced through externally imposed orbital-state counting, spin degeneracy rules, or abstract quantum-number bookkeeping. Instead, shell capacity emerges as a direct geometric consequence of increasing spherical surface area available to support stable distributed electromagnetic occupancy.

Under this ontology, charge capacity scales because spherical surface geometry scales.

Figure 10 introduces the first geometric scaling law of the Holland Atom. Once charge is understood to reside as a distributed surface occupancy upon physically real resonance-lock shells, the allowable charge capacity of each shell becomes fundamentally linked to shell geometry itself. As shell radius

increases with standing-mode order, available spherical surface area increases according to $4\pi r^2$, naturally permitting progressively larger stable charge distributions. Under this interpretation, the familiar shell-capacity progression of atomic matter is no longer treated as an abstract occupancy rule, but as the geometric consequence of distributed charge residing upon discrete spherical resonance boundaries.

4.5.7 — H7 Visualisation: Surface Charge Degeneracy

Each stable surface resonance site of the Holland Field is proposed to support two phase-stable electromagnetic charge states, thereby introducing a universal surface degeneracy factor of two across all shell geometries.

This gives:

	$Q_n = 2A_n$	Eq. 4.5.7.1

and therefore:

	$Q_n = 2n^2$	Eq. 4.5.7.2

which immediately yields $2,\ 8,\ 18,\ 32, \ldots$

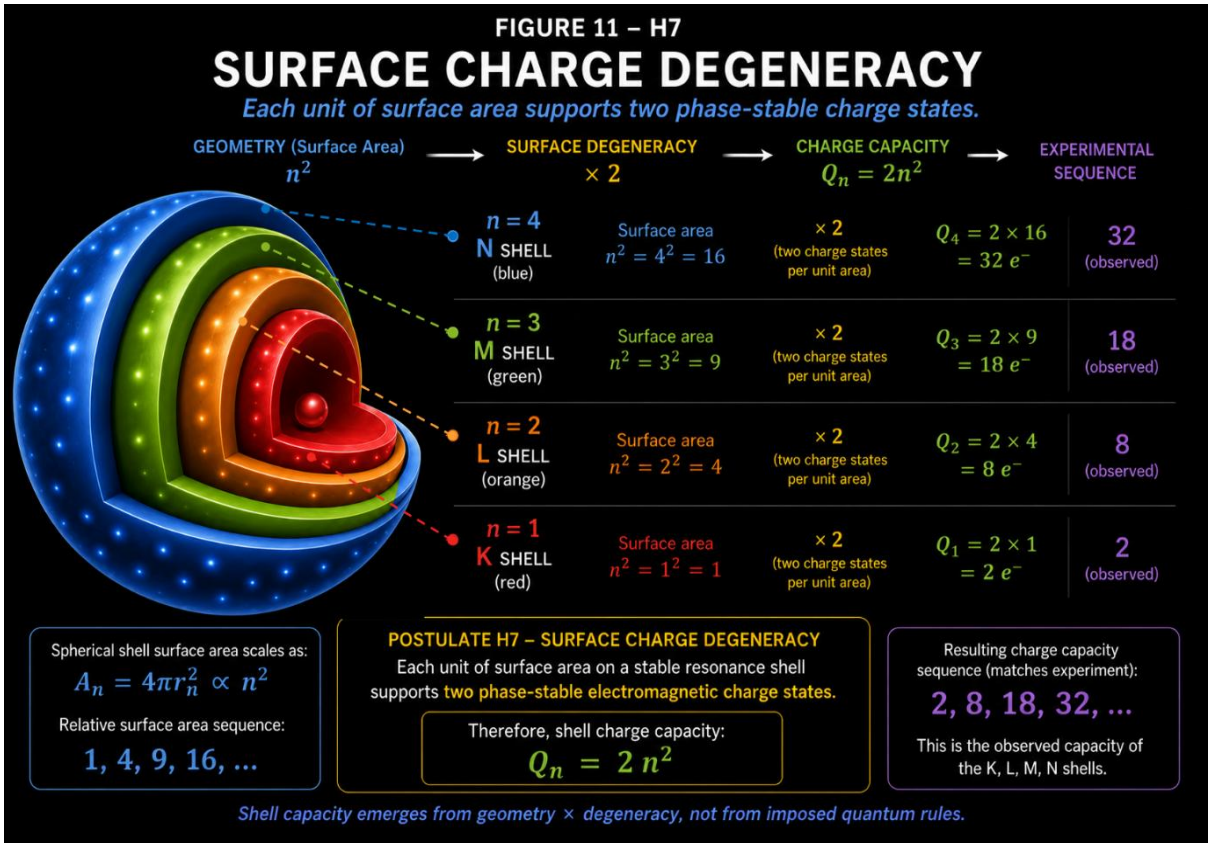


Figure 11. H7: Surface Charge Degeneracy

Figure 11 illustrates Postulate H7, establishing the physical conversion between shell geometry and experimentally observed shell charge capacity within the Holland Atom. As demonstrated in Figure

10, successive resonance-lock shells possess surface geometries proportional to the square of shell order, producing the geometric sequence

$$1,\ 4,\ 9,\ 16, \ldots$$

corresponding to the relative spherical surface areas of the K, L, M, and N shells. Figure 11 introduces the second defining characteristic of shell occupancy: each stable surface resonance site is proposed to support two phase-stable electromagnetic charge states embedded within the standing-wave boundary of the Holland Field Φ. This universal surface degeneracy introduces a factor of two across all shell geometries, transforming the surface-area sequence directly into the experimentally observed shell-capacity progression $Q_n = 2n^2$ and the observed sequence $2,\ 8,\ 18,\ 32, \ldots$

Before introducing the full Holland Atom reconstruction, it is useful to note a simple geometric property of spherical shells. For a sphere of radius (R), the surface area is: $A = 4\pi R^2$**.** If successive shell boundaries occur at integer-scaled radii: R = 1,2,3,4,5 then the corresponding surface areas become:

Radius R	**Sphere surface Area $A = 4\pi R^2$**	**Surface area value**	**Ratio to $R = 1$ sphere**	**Conversion factor of surface area to shell charge**	**Observed Shell Capacity Sequence**
1	$4\pi(1)^2$	4π	**1**	X2	**2**
2	$4\pi(2)^2$	16π	**4**	X2	**8**
3	$4\pi(3)^2$	36π	**9**	X2	**18**
4	$4\pi(4)^2$	64π	**16**	X2	**32**
5	$4\pi(5)^2$	100π	**25**	X2	**50**

Table 18. Integer increasing radius produces shell capacity sequence.

Table 18 raises the question: is this a coincidence or is this causal mechanism? Currently, within the Quantum Mechanics atom description, the surface area of the shell at defined radius is considered a coincidence only. For the Holland Atom, and the inherent realism, the surface area of the shell sphere is expected to be directly proportional to the amount of charge that shell can support. This is a logical, intuitive, engineering reality and corresponds precisely to the experimentally observed K-, L-, M-, N-, and O-shell capacities of atomic matter.

By itself, this observation does not establish a physical atomic model. It merely demonstrates that a standing-wave shell architecture possessing integer radial scaling and a fixed two-state surface degeneracy naturally generates the same shell-capacity sequence observed throughout the periodic table. The remainder of the Holland Atom reconstruction investigates whether this geometric relationship can be elevated from numerical correspondence to a physically consistent causal framework.

Under the Holland Atom, shell capacity therefore emerges not from externally imposed orbital counting rules, spin bookkeeping, or abstract quantum-state assignment, but from a direct physical combination of spherical surface geometry and intrinsic two-state surface charge degeneracy.

Figure 11 completes the first complete geometric derivation of shell capacity within the Holland Atom. Once atomic shells are recognised as physically real resonance-lock boundaries supporting

distributed surface charge, shell occupancy becomes a direct consequence of two simple physical principles: increasing spherical surface geometry, and a universal two-state charge-supporting characteristic of each surface resonance site. Under this interpretation, the familiar shell-capacity sequence of atomic matter emerges naturally from:

Geometry × Surface degeneracy

rather than from imposed orbital-state bookkeeping.

4.5.8 — H8 Visualisation: Dynamic Equilibration

Atomic charge population initially proceeds through sequential standing-wave shell formation. Non-equilibrium configurations may subsequently redistribute charge through resonance equilibration until a stable resonance-equilibrium state is reached.

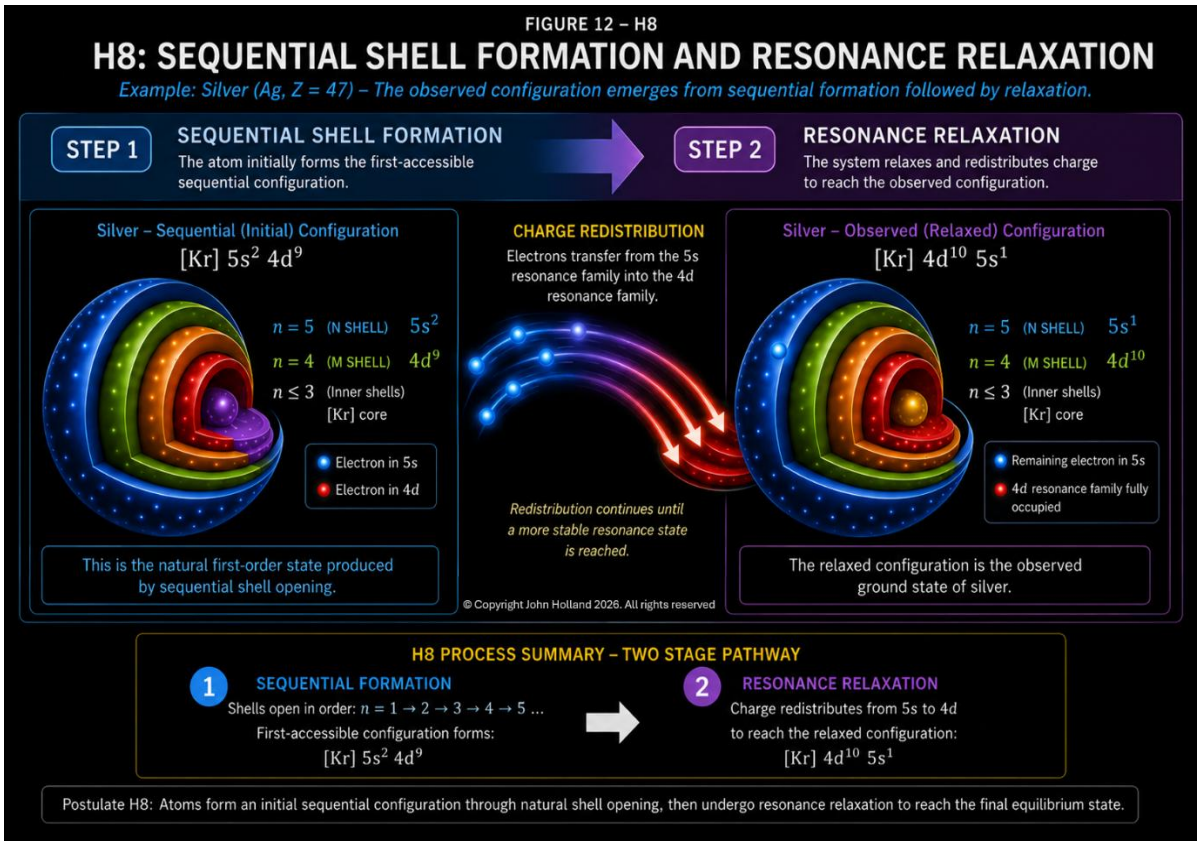


Figure 12. H8: Dynamic Equilibration with Silver (Ag Z = 47)

Figure 12 is an illustration of the two-stage Holland Atom formation process for silver.

Stage 1 *shows the initial sequential standing-wave shell population state produced through natural shell-opening progression, yielding the first dynamically accessible configuration.*

Stage 2 *shows the subsequent dynamic resonance transition, in which distributed charge redistributes between neighbouring shell structures and angular resonance families before reaching the experimentally observed stable resonance configuration.*

The visual emphasises that observed anomalous atomic structures are interpreted within the Holland Atom framework as relaxed resonance states emerging from an initially sequential shell formation

process. The resulting observed configuration exhibits completion of the 4d *angular resonance family.*

Atomic charge population initially proceeds through sequential standing-wave shell formation. Non-optimised sequential configurations may subsequently undergo dynamic resonance equilibration through charge redistribution between neighbouring shells or angular resonance families before reaching the observed stable resonance configuration.

4.5.9 — H9 Visualisation: Global Resonance Optimisation

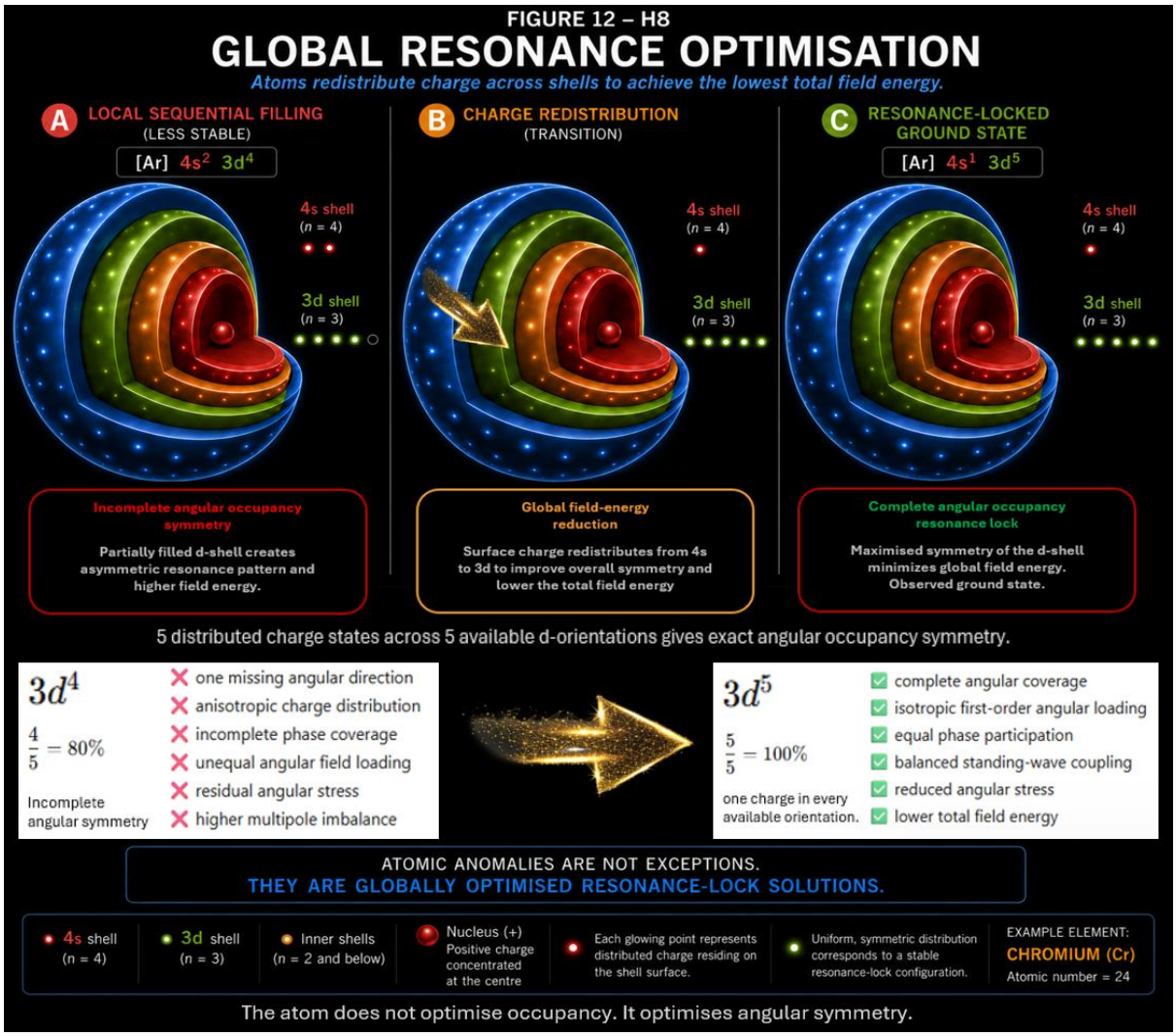


Figure 13. H9: Global Resonance Optimisation – Chromium Case Study

Figure 13 illustrates Postulate H9 using Chromium as a representative transition-metal anomaly. Under simple sequential shell filling, chromium would be expected to adopt the configuration

$$[Ar]\ 4s^2\ 3d^4$$

leaving the 3d resonance structure asymmetrically occupied. Experimentally, chromium instead adopts the ground-state configuration

$$[Ar]\ 4s^1\ 3d^5$$

corresponding to a half-filled d-shell.

Within the Holland Atom, this behaviour is interpreted not as an exception to filling order, but as global resonance optimisation. Distributed surface charge is free to redistribute between available resonance surfaces until the entire shell–subshell system reaches its lowest total field-energy configuration. Half-filled and fully filled subshell structures therefore emerge naturally as preferred resonance-lock geometries, where charge symmetry, phase coherence, and field stability are simultaneously maximised.

Shell / band	Shell order	Sequential Expected state (Aufbau)	Observed ground state
Chromium has an Argon core	$[Ar]$	18 charges	18 charges
4s surface	$n = 4$	$4s^2$	$4s^1$
3d band	$n = 3$	$3d^4$	$3d^5$
Total beyond ([Ar])	—	6 charges	6 charges

Table 19. Chromium Structure: Observed vs. Expected

Table 19 shows the observed chromium anomaly is known and conventionally explained within quantum mechanics; the Holland Atom interpretation is radical because it replaces the standard orbital/exchange-energy formalism with a physical ontology of distributed surface charge, angular occupancy symmetry, and global resonance-lock optimisation in Φ.

Shell / band	Capacity	Sequential expected	Occupancy (%)	Observed ground state	Occupancy (%)	Holland Atom interpretation
Chromium has an Argon [Ar] core	18	18	100%	18	100%	Closed inner resonance core
(4s)	2	2	100%	1	50%	One charge redistributed
(3d)	10	4	40%	5	**50%**	**Exact half-filled resonance lock**
Total outer charge	12	6	—	6	—	Charge conserved
Symmetry condition	—	**asymmetric**	—	**exact half symmetry**	—	Maximum phase balance
Field energy	—	higher	—	lower	—	Global optimisation

Table 20. Chromium Global Resonance Optimisation Audit

Chromium provides a particularly revealing test of global resonance optimisation because the $3d$ resonance band possesses a maximum capacity of ten charge states. Under simple sequential filling, chromium would be expected to adopt:

$$[Ar]\ 4s^2\ 3d^4$$

leaving the $3d$ band occupied at only:

$$\frac{4}{10} = 40\%$$

Experimentally, chromium instead adopts:

$$[Ar]\,4s^1\,3d^5$$

placing the $3d$ resonance band at:

$$\frac{5}{10} = 50\%$$

an **exact half-filled state.**

Under the Holland Atom, this is interpreted not as a bookkeeping exception, but as a resonance-lock preference for precise geometric symmetry. The atom redistributes one unit of surface charge from the $4s$ shell to the $3d$ shell because the exact half-filled configuration produces maximum angular symmetry, improved phase coherence, and a lower total field-energy state across the coupled shell system.

Figure 13 completes the Holland Atom postulate sequence by introducing **global resonance optimisation**. Once shells, surface charge, and geometric capacity are established, the atom is no longer constrained to fill locally according to simple shell order. Instead, the distributed charge system behaves as a coupled resonance network, free to reorganise across all accessible shell surfaces until the global field-energy minimum is achieved. Under this interpretation, the experimentally observed transition-metal anomalies of chromium, copper, palladium, gold, and related systems are not anomalies at all—they are the observable signatures of whole-atom resonance optimisation.

4.6 Section Summary and Transition to Mathematical Formalism

Section 4 has established the complete first-order physical ontology of the **Holland Atom** through eight sequential postulates, beginning with **active field interruption**, **suppressed expansion**, and **three-dimensional net flow imbalance** within the all-pervading Holland Field Φ, and progressing through **nuclear-density-controlled wavelength selection**, **spherical standing-mode formation**, **discrete K–Q shell emergence**, **distributed surface charge occupancy**, **surface-geometry scaling**, **two-state surface occupancy**, and finally **global resonance optimisation across coupled shell structures.**

Supported throughout by Figures 4–13, these postulates collectively transform atomic structure from a probabilistic shell-occupancy framework into a physically causal resonance-lock system, in which shell radius, shell capacity, subshell symmetry, transition-metal anomalies, contraction effects, and discrete atomic quantisation emerge as natural consequences of a common nucleus–field interaction.

The Holland Atom is therefore no longer presented as an interpretive analogy, but as a complete physical hypothesis with explicit geometric, energetic, and experimentally testable consequences.

The investigation now proceeds to **Section 5**, where each postulate is translated into mathematical form, deriving the governing resonance equations of the Holland Atom and testing whether a single density-driven standing-wave framework can quantitatively reconstruct the full periodic architecture of atomic matter – including shell capacities, radius scaling, contraction behaviour, transition-metal anomalies, known periodic irregularities, and the predictive extension toward presently unobserved superheavy atomic states.

5. Mathematical Formalisation of the Holland Atom

Having established the empirical atomic baseline (Section 3), the physical field framework of GETT (Section 4), and the first-order physical postulates of the Holland Atom (H1–H9), the present section now proceeds from physical ontology to mathematical formalisation. The central question is direct:

Can a single nucleus-driven standing-wave system within the Holland Field Φ quantitatively reconstruct the experimentally observed architecture of atomic matter?

Specifically, can the Holland Atom derive, from first physical principles:

- discrete K–Q shell radii;
- the shell-capacity sequence: 2, 8, 18, 32, 50, 72, 98
- the emergence of s-, p-, d-, and f-shell angular families;
- the experimentally observed atomic-radius opening and contraction sequences;
- transition-metal anomalies, including half-filled and fully filled resonance-lock states;
- lanthanide and actinide contraction behaviour;
- and the predictive extension toward presently unobserved superheavy atomic states.

The mathematical objective of Section 5 is therefore not to introduce an empirical fitting model, nor to reinterpret existing quantum-mechanical notation. Rather, the objective is to determine whether the physical postulates H1–H9 can be expressed as a self-consistent set of governing field equations capable of reproducing the known periodic architecture of atomic matter from a single density-driven resonance framework.

To achieve this, the Holland Atom will now be developed sequentially, beginning with the local scalar-field boundary condition imposed by nuclear mass-energy, followed by resonance wavelength selection, standing-mode closure, shell-surface geometry, charge occupancy, angular resonance families, and finally global resonance optimisation across the complete atomic system.

If successful, the resulting formalism will provide the first physically causal nucleus-to-periodic-table derivation developed within the General Expanse Tension Theory.

5.1 Definitions, Notation, and Model Variables

Before deriving the Holland Atom equations, all variables, indices, and physical quantities used in the formalism are defined explicitly. This prevents ambiguity between conventional quantum-mechanical notation and the new resonance-lock notation introduced in the present work.

Symbol	Meaning	Role in Holland Atom
$\boldsymbol{\Phi}$	Holland scalar field	All-pervading singlet scalar field physical substrate supporting standing modes
$\boldsymbol{Z}$	Atomic number	Number of nuclear protons / total neutral atomic charge requirement
$\boldsymbol{N}$	Neutron number	Contributes to nuclear mass-energy density
$\boldsymbol{A}$	Mass number	$A = Z + N$
$\boldsymbol{\Sigma_N}$	Effective nuclear mass-energy density	Sets local resonance scale
$\boldsymbol{\lambda_\Phi}$	Characteristic Holland-field wavelength	Determines shell spacing
$\boldsymbol{n}$	Shell / radial mode number	$n = 1,2,3, \ldots$
$\boldsymbol{r_n}$	Radius of shell n	Standing-mode shell boundary
$\boldsymbol{A_n}$	Surface area of shell n	$A_n = 4\pi r_n^2$
$\boldsymbol{O_{\text{surf}}}$	Surface occupancy factor	Two phase-stable charge states per surface unit
$\boldsymbol{Q_n}$	Maximum charge capacity of shell n	$Q_n = O_{\text{surf}} n^2$
$\boldsymbol{B_l}$	Angular band capacity	s,p,d,f band structure
$\boldsymbol{E_{\text{tot}}}$	Total field-energy state	Quantity minimised by global optimisation

Table 21. Definitions, Notation, and Model Variables

Table 21 defines the Holland Atom definitions, notation and model variable. Notation clarification: A_n denotes Holland Atom shell-surface area scaling associated with shell n, whereas A without subscript denotes conventional nuclear mass number ($A = Z + N$). The two quantities are physically unrelated despite the similar notation.

The model adopts three important conventions. First, n is interpreted as a **physical radial standing-mode order**, not merely as a quantum label. Second, A_n denotes physical shell surface area and should not be confused with mass number A. Where ambiguity may arise, mass number is written explicitly as A_{mass}. Third, $O_{\text{surf}} = 2$ as introduced earlier as the Holland Atom surface occupancy factor, representing two phase-stable electromagnetic charge states per allowed surface unit. Its deeper microscopic derivation is deferred to future QFT development, but its use here is fixed and not varied during prediction.

5.2 Nuclear Boundary Condition and Local Field Suppression

(Mathematical formalisation of Postulate H1)

Under Postulate H1, the atomic nucleus acts as a local mass-energy boundary condition within the all-pervading Holland Field Φ. In the absence of matter, Φ is assumed to undergo free isotropic

expansion, possessing no preferred direction and no net restoring tension gradient, as shown in figure 4. The introduction of finite nuclear mass-energy locally suppresses this expansion, producing a stable three-dimensional net flow imbalance centred upon the nuclear core. The first mathematical objective of the Holland Atom is therefore to describe the radial scalar-field suppression surrounding the nucleus.

Let the effective **nuclear mass-energy density** be:

	Σ_N	Eq. 5.2.1

occupying an effective **nuclear radius** i.e "*inside the nucleus*":

	R_N	Eq. 5.2.2

Where r represents the continuous radial distance coordinate measured outward from the nuclear centre, i.e. "*Outside the nucleus*." . Therefore, within the nucleus:

	$r \leq R_N$	Eq. 5.2.3

the free isotropic expansion of the Holland Field is assumed to be strongly suppressed by the local nuclear mass-energy distribution. Outside the nucleus:

	$r > R_N$	Eq. 5.2.4

the field relaxes continuously toward the free-expansion background state:

	Φ_∞	Eq. 5.2.5

The simplest first-order scalar representation of this radial suppression is therefore written:

	$\Phi(r) = \Phi_\infty - \Delta\Phi(r)$	Eq. 5.2.6

where:

	$\Delta\Phi(r)$	Eq. 5.2.7

represents the local suppression amplitude generated by nuclear mass-energy.

The resulting radial restoring tension gradient is then defined by:

	$T(r) \propto -\frac{d\Phi}{dr}$	Eq. 5.2.8

such that increasing outward field recovery generates an inward-directed restoring tension toward the nuclear core. r_n is the notation applied to the discrete shell radius of shell n, with n = 1,2,3,.. and the corresponding r_1, r_2, r_3 representing the atomic shells sequence K, L, M etc. Under the Holland Atom, this radial tension structure constitutes the fundamental precursor to all subsequent atomic resonance-lock behaviour. Shell formation, standing-mode closure, distributed surface occupancy, and

global resonance optimisation therefore emerge not from imposed orbital postulates, but from the self-organisation of stable standing-wave solutions within this nucleus-generated scalar boundary system.

5.3 Resonance Wavelength Selection and Standing-Mode Closure

(Mathematical formalisation of Postulates H2–H4)

Following the establishment of the nuclear boundary condition in Section 5.2, the Holland Atom now requires a mechanism by which the surrounding scalar field self-organises into discrete stable resonance states. Under Postulate H2, the effective nuclear mass-energy density, Σ_N, is proposed to determine the characteristic local resonance wavelength of the surrounding Holland Field:

	λ_Φ	Eq. 5.3.1

The simplest first-order assumption is that increasing nuclear density produces increasing local field compression and therefore decreasing characteristic resonance wavelength, such that:

	$\lambda_\Phi \propto \frac{1}{\Sigma_N^\alpha}$	Eq. 5.3.2

where:

	$\alpha > 0$	Eq. 5.3.3

is a scaling exponent to be determined empirically or through future first-principles field derivation. The nucleus therefore establishes a characteristic resonance scale surrounding the atomic core. Within the Holland Atom, the standing-wave shell structure is treated under a local adiabatic approximation in which the Holland-field resonance wavelength varies sufficiently slowly across the shell-forming region that:

	$\lambda_\Phi(\Sigma) \approx \lambda_{\Phi,\mathrm{local}}$	Eq. 5.3.4

may be treated approximately constant to first order within an individual shell domain.

Under Postulate H3, stable shell formation occurs only when the surrounding scalar field satisfies complete spherical standing-mode closure. For radial mode order $n = 1,2,3, \ldots$ stable resonance-lock solutions exist only when an integer number of half-wavelengths satisfies the radial shell boundary condition which is the standing-wave shell closure condition adopted in the Holland Atom:

	$2r_n = n\lambda_\Phi$	Eq. 5.3.5

Where r_n denotes the stable radius of shell n. This corresponds to an integer phase-closure condition between two effective radial nodal boundaries. Within the Holland Atom interpretation, the inner boundary is associated with the strongly suppressive nuclear mass-energy density region surrounding the nuclear core, which acts as a rigid standing-wave boundary condition for the Holland Field disturbance. The outer shell boundary correspondingly represents the radial location at which stable standing-wave phase closure is achieved before the field disturbance decays asymptotically into the surrounding lower-density field environment.

Under this interpretation, the shell structure therefore behaves as an effective node–node standing-wave system in the radial direction, naturally producing the integer shell-closure condition. The resulting shell-radius sequence is therefore:

	$$r_n = \frac{n\lambda_\Phi}{2}$$	Eq. 5.3.6

rather than an antinode–node harmonic structure.

At present, this boundary interpretation should be regarded as a first-order standing-wave approximation within the Holland Atom framework rather than as a completed field-theoretic derivation of the exact radial boundary conditions of the Holland Field Φ. Nevertheless, the resulting closure structure successfully reconstructs the experimentally observed shell-capacity architecture and shell-ordering behaviour across the periodic table under the frozen postulate set H1–H9. This immediately produces a discrete radial standing-mode hierarchy.

Quantity	Definition	Physical meaning	Role within Holland Atom
$\boldsymbol{\Sigma_N}$	Effective nuclear mass-energy density	Nuclear compression strength	Determines local field resonance scale
$\boldsymbol{\lambda_\Phi}$	Characteristic Holland-field wavelength	Local standing-wave scale surrounding nucleus	Controls shell spacing
$\boldsymbol{\alpha}$	Density scaling exponent	Controls wavelength–density relationship	Future empirical/QFT derivation target
$\boldsymbol{n}$	Radial standing-mode order	Integer resonance-lock state	Defines shell sequence
$\boldsymbol{r_n}$	Radius of shell n	Stable standing-wave boundary	Defines physical shell geometry
$\boldsymbol{2r_n = n\lambda_\Phi}$	Standing-mode closure condition	Complete radial phase-lock requirement	Produces shell quantisation
$\boldsymbol{r_n = \frac{n\lambda_\Phi}{2}}$	Shell-radius equation	Discrete shell-radius hierarchy	Generates K–Q shell structure

Table 22. Notations and Definitions for Wavelength Selection and Standing-Mode Closure

Under this interpretation, the experimentally observed atomic shell system is redefined as a discrete radial resonance hierarchy of the nucleus–field system itself. The atom therefore acquires a direct physical geometry, with shell radius emerging as a measurable standing-wave boundary rather than as a probabilistic orbital abstraction. The experimentally observed K–Q shell sequence is therefore interpreted as the stable radial node structure of the nucleus-driven Holland-field resonance system.

The next section extends this radial resonance framework into distributed surface occupancy and shell-capacity scaling, deriving how stable shell geometry gives rise to the experimentally observed atomic charge-capacity sequence.

A visual representation of a 3D spherical standing-wave architecture rather than a 2D ring standing-wave architecture.

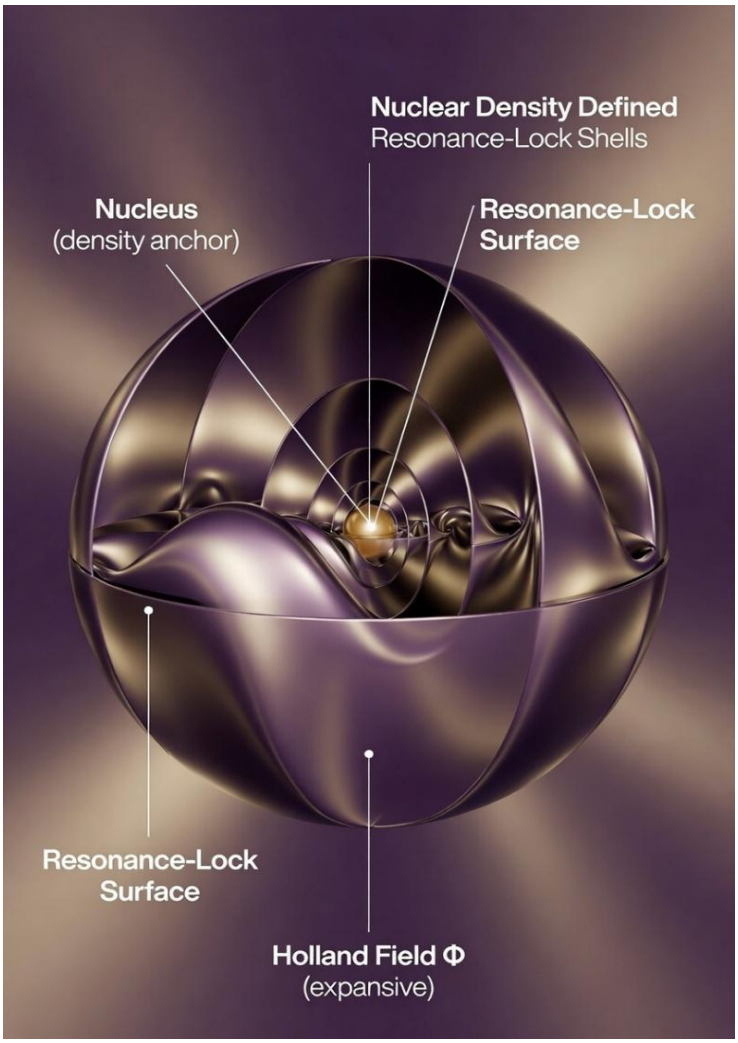


Figure 14. Visual Representation of a 3D Spherical Standing-Wave Architecture

Figure 14 shows a conceptual representation of a spherical resonance-lock architecture within the Holland Atom. The nucleus acts as a density anchor that establishes the local resonance wavelength of the Holland Field Φ. Stable standing-wave closure surfaces form discrete resonance-lock shell boundaries. The figure is illustrative and not intended as a quantitative field solution.

5.4 Distributed Surface Occupancy and Shell-Capacity Scaling

(Mathematical formalisation of Postulates H5–H7)

Following the establishment of discrete resonance-lock shell boundaries in Section 5.3, the Holland Atom now requires a mechanism by which stable electromagnetic charge occupancy emerges upon the shell surfaces themselves. Under Postulate H5, negative atomic charge is not interpreted as continuously orbiting point particles distributed probabilistically throughout the atomic volume [8,9]. Instead, charge exists as a distributed electromagnetic surface occupancy bound directly to the standing-wave shell boundaries of the Holland Field Φ. Each shell therefore acts as a physically real charge-supporting resonance surface. For shell order n, with stable shell radius r_n the corresponding spherical shell surface area is:

	$A_n = 4\pi r_n^2$	Eq. 5.4.1

Substituting the standing-mode shell relation established in Section 5.3:

	$r_n = \frac{n\lambda_\Phi}{2}$	Eq. 5.4.2

gives:

	$A_n = 4\pi \left(\frac{n\lambda_\Phi}{2}\right)^2$	Eq. 5.4.3

which simplifies to:

	$A_n = \pi\lambda_\Phi^2 n^2$	Eq. 5.4.4

The shell surface geometry therefore scales quadratically with standing-mode order:

	$A_n \propto n^2$	Eq. 5.4.5

Here, A_n denotes shell-surface area scaling and should not be confused with the conventional nuclear mass number A. Under Postulate H7, each stable surface resonance unit supports $O_{\text{surf}} = 2$ phase-stable electromagnetic occupancy states. The total shell charge capacity therefore becomes:

	$Q_n = O_{\text{surf}}\, n^2$	Eq. 5.4.6

and for $O_{\text{surf}} = 2$ the experimentally observed shell-capacity sequence immediately follows Equation 4.5.7.2: $Q_n = 2n^2$. Yielding 2, 8, 18, 32, 50, 72, 98 corresponding precisely to the experimentally observed K–Q shell-capacity sequence. Under this formulation, shell capacity emerges directly from the combined effects of:

1. discrete standing-mode shell geometry;
2. spherical surface-area scaling;
3. and two-state surface electromagnetic occupancy.

No externally imposed orbital-capacity rules, probabilistic occupancy constructs, or shell-filling axioms are required.

The value $O_{\text{surf}} = 2$ is not claimed here as a completed first-principles QFT derivation. It is introduced as a correspondence-fixed surface degeneracy factor required to reproduce the empirically established two-state occupancy structure of atomic matter. In conventional quantum mechanics, the same factor appears through spin and Pauli exclusion; within the Holland Atom, it is reinterpreted as two phase-stable electromagnetic surface-occupancy states supported by each stable resonance unit of the shell surface. The non-trivial correspondence is that a single fixed value $O_{\text{surf}} = 2$, frozen prior to reconstruction, simultaneously reproduces shell capacities, angular-family capacities, half-filled/full-filled symmetry behaviour, and anomaly optimisation structure across the periodic table. A deeper first-principles derivation of this two-state degeneracy from the scalar–Higgs / electromagnetic sector remains a target for future QFT development, whilst below we examine one speculative hypothesis for the factor.

5.4.1 Possible Geometric Origin of the Two-State Surface Degeneracy

At present, the Holland Atom treats the universal surface degeneracy factor $O_{\mathrm{surf}} = 2$ as a correspondence-fixed postulate rather than a completed first-principles derivation. However, the standing-wave shell ontology suggests a possible geometric interpretation of this two-state occupancy structure. Within a closed standing-wave system, each stable standing-wave segment is bounded by **two nodal closure conditions** located at opposite boundaries of the resonance structure. The simplest stable standing-wave topology therefore naturally possesses **a two-boundary nodal structure**:

	$N_{\mathrm{nodes}} = 2$	Eq. 5.4.1

Under the Holland Atom interpretation, the experimentally observed two-state shell occupancy degeneracy may therefore reflect the fundamental two-node closure topology of stable shell-surface standing-wave modes within the Holland Field Φ.

This interpretation remains provisional and does not presently constitute a first-principles derivation of fermionic spin or quantum-statistical behaviour. However, it provides a physically motivated geometric interpretation for why a universal two-state shell occupancy structure may emerge naturally within a standing-wave resonance ontology. The significance of the present reconstruction is therefore not that the two-state degeneracy has already been fundamentally derived, but that a single frozen correspondence value $O_{\mathrm{surf}} = 2$ successfully reconstructs the experimentally established shell-capacity architecture across the periodic table without subsequent adjustment.

Importantly, the present geometric interpretation of the two-state surface occupancy structure does not yet constitute a derivation of fermionic spin or Pauli exclusion statistics from first-principles QFT. Any future derivation of the correspondence-fixed surface degeneracy factor $O_{\mathrm{surf}} = 2$ from the scalar–Higgs sector of GETT will necessarily need to address how effectively fermionic occupancy behaviour emerges from an underlying scalar-field resonance framework. This remains a deep target for future theoretical development rather than a completed result of the present work.

5.5 Angular Resonance Families and Surface Partitioning

(Mathematical formalisation of Postulates H4 and H7)

Sections 5.2–5.4 established the radial resonance structure of the Holland Atom, deriving discrete shell radii and shell-capacity scaling from standing-wave shell geometry within the Holland Field Φ. However, the experimentally observed atomic architecture contains an additional level of structure beyond purely radial shell quantisation.

Atomic charge organisation is further partitioned into the well-known angular shell families, introduced in section 3.4, conventionally designated $s,\ p,\ d,\ f$ with experimentally observed maximum capacities $2,\ 6,\ 10,\ 14$ respectively. Within conventional quantum mechanics, these structures emerge through angular momentum operators, orbital quantum numbers, and wavefunction eigenstate degeneracies. Within the Holland Atom, by contrast, these angular families are interpreted physically as stable angular standing-mode partitions of the shell surface itself.

Under this formulation, each spherical resonance-lock shell supports not only radial standing modes, but also angular standing-wave structures distributed across the shell surface. Let l represent the angular resonance order of the shell surface. The number of independent angular surface orientations supported by order l is proposed to scale as:

	$N_l = 2l + 1$	Eq. 5.5.1

corresponding geometrically to the number of independent stable angular standing-mode partitions permitted upon the spherical shell surface. Thus:

Angular family	l	Surface orientations
s	0	1
p	1	3
d	2	5
f	3	7

Table 23. Holland Atom Angular family and Surface Orientations

Under Postulate H7, each angular surface orientation supports $O_{\text{surf}} = 2$ phase-stable occupancy states. The total angular-family capacities therefore become:

$$B_l = O_{\text{surf}}(2l + 1) \quad \text{Eq. 5.5.2}$$

and for $O_{\text{surf}} = 2$ the experimentally observed angular-family capacities immediately follow:

$$B_l = 2(2l + 1) \quad \text{Eq. 5.5.3}$$

giving:

Angular Family	Capacity
s	2
p	6
d	10
f	14

Table 24. Holland Atom Angular-Family Capacities

Under the Holland Atom, these capacities therefore emerge as the direct geometric consequence of:

1. angular standing-mode partitioning of the shell surface;
2. the number of independent angular resonance orientations;
3. and the universal two-state surface occupancy factor.

The familiar s-, p-, d-, and f-shell architecture is therefore reinterpreted not as an abstract orbital bookkeeping system, but as the angular resonance geometry of distributed charge occupancy upon physically real shell surfaces.

5.6 Full Shell Capacity as the Sum of Angular Resonance Families

(Mathematical unification of Postulates H6–H7)

The preceding sections derived shell capacity through two complementary geometric routes. Section 5.4 derived the total charge capacity of shell n from spherical surface geometry:

$$Q_n = 2n^2 \quad \text{Eq. 5.6.1}$$

For an atom of atomic number Z, the realised shell occupancy $q_n(Z)$ of shell n is defined as the sequentially assigned charge population subject to the shell-capacity constraint:

	$$q_n(Z) = \min\left[Z - \sum_{i=1}^{n-1} q_i\,,\, Q_n\right]$$	Eq. 5.6.2

where:

- $q_n(Z)$ is the realised shell occupancy of shell n;
- Z is the atomic number;
- Q_n is the Holland Atom shell-capacity limit;
- and $\sum_{i=1}^{n-1} q_i$ represents the charge already assigned to lower shells.

The reconstruction therefore proceeds sequentially through accessible shell boundaries while enforcing the standing-wave shell-capacity condition:

	$$q_n(Z) \leq Q_n$$	Eq. 5.6.3

Section 5.5 derived the individual angular resonance-family capacities:

$$B_l = 2(2l + 1)$$

The full capacity of shell n is therefore obtained by summing all angular resonance families permitted within that shell:

	$$Q_n = \sum_{l=0}^{n-1} B_l$$	Eq. 5.6.4

Substituting:

	$$B_l = 2(2l + 1)$$	Eq. 5.6.5

gives:

	$$Q_n = 2\sum_{l=0}^{n-1} (2l + 1)$$	Eq. 5.6.6

Since the sum of the first n odd integers is:

	$$\sum_{l=0}^{n-1} (2l + 1) = n^2$$	Eq. 5.6.7

the total shell capacity becomes:

	$$Q_n = 2n^2$$	Eq. 5.6.8

This establishes direct equivalence between the two Holland Atom descriptions:

Shell n	Shell name	Angular families included	Capacity sum	Total capacity Q_n
1	K	s	2	2
2	L	s + p	2 + 6	8
3	M	s + p + d	2 + 6 + 10	18
4	N	s + p + d + f	2 + 6 + 10 + 14	32
5	O	s + p + d + f + g	2 + 6 + 10 + 14 + 18	50
6	P	s + p + d + f + g + h	2 + 6 + 10 + 14 + 18 + 22	72
7	Q	s + p + d + f + g + h + i	2 + 6 + 10 + 14 + 18 + 22 + 26	98

Table 25. Full Shell Capacity as the Sum of Angular Resonance Families

Table 25 demonstrates that under the Holland Atom, full shell capacity emerges simultaneously as ***spherical surface geometry following shell sequence*** *and* ***angular resonance-family summation.*** *These are not separate rules. They are two expressions of the same resonance-lock architecture.*

5.7 Dynamic Resonance Accessibility

(Mathematical formalisation of Postulate H8)

The Holland Atom proposes that experimentally observed atomic configurations emerge through a two-stage resonance process rather than through instantaneous direct equilibrium formation. Initial atomic structure forms through sequential resonance accessibility governed by shell-opening order and local standing-wave accessibility conditions (Stage 1), after which the configuration subsequently relaxes toward the globally optimised resonance-equilibrium state through resonance redistribution and shell rebalancing (Stage 2).

Stage	Holland Atom process	Governing principle	Configuration
Stage 1	Sequential resonance formation	minimum accessibility cost	$\mathcal{C}_{\mathbf{seq}}$
Stage 2	Resonance relaxation	minimum total resonance energy	$\mathcal{C}_{\mathbf{obs}}$

Table 26. Dynamic 2-Stage Resonance Process

Table 26 showing the Stage 1 and Stage 2 distinction established in H8 therefore separates dynamically accessible shell formation from final resonance-equilibrium atomic structure.

The Stage 1 configuration therefore corresponds to the most dynamically accessible standing-wave shell configuration available during shell formation rather than the final globally optimised equilibrium configuration observed experimentally.

The Holland Atom accordingly defines the sequentially accessible shell configuration:

$$\mathcal{C}_{\text{seq}} = \arg\min_{\mathcal{C}\in\Omega_Z} \mathcal{A}(\mathcal{C}) \qquad \text{Eq. 5.7.1}$$

where:

- $\mathcal{C}_{\mathrm{seq}}$ is the dynamically accessible sequential shell configuration;
- Ω_Z is the set of resonance-accessible shell configurations for atomic number Z;
- $\mathcal{A}(\mathcal{C})$ represents a formal sequential accessibility operator encoding nearest-order standing-wave reachability and shell-opening continuity conditions. In the present first-order formulation, $\mathcal{A}$ is not yet specified as a fully derived physical functional but instead serves as a compact mathematical representation of the sequential shell-accessibility rule underlying H8.

A future dynamical formulation may replace the present first-order accessibility operator with a fully derived resonance-transition functional, but no such derivation is assumed in the present work.

Importantly:

	$\mathcal{A}(\mathcal{C}) \neq E_{\mathrm{tot}}(\mathcal{C})$	Eq. 5.7.2

The Stage 1 sequential formation pathway is therefore not governed by global resonance-energy minimisation, but instead by local resonance accessibility and shell-formation continuity conditions.

Within the Holland Atom interpretation, this distinction is essential because it permits initially formed shell configurations to differ from the final experimentally observed equilibrium state. The framework therefore naturally allows transient sequential shell-filling structures to emerge prior to later resonance redistribution and optimisation processes.

Importantly, the Holland Atom accessibility rule does not require complete angular-family saturation of shell n before the next radial shell boundary becomes accessible. Instead, once a new standing-wave shell boundary opens through the sequential accessibility condition $\Delta n = +1$, the lowest angular resonance family of the newly accessible shell becomes dynamically available prior to full occupation of higher-order angular families within the previous shell. This provides a direct geometric interpretation of the experimentally observed shell-opening structure underlying the Aufbau sequence, including the empirically established $4s \rightarrow 3d$ ordering transition [11,19]. Within the Holland Atom, the $4s$ shell opens first because the next radial standing-wave boundary becomes accessible before the higher-order $3d$ angular resonance family reaches occupancy accessibility.

The dynamic accessibility rule introduced in H8 should be interpreted as a first-order structural postulate rather than a completed first-principles derivation of the full Aufbau or Madelung ordering sequence [10,11,19]. Within the Holland Atom, the observed opening of $4s$ before completion of $3d$ is represented by the sequential radial accessibility condition $\Delta n = +1$ together with the rule that the lowest angular resonance family of a newly accessible shell opens before higher-order angular families are fully populated within the preceding shell.

This reproduces the observed structural ordering $4s \rightarrow 3d$, however it does not yet derive the corresponding accessibility hierarchy from a fully specified dynamic functional. In the present formulation, the accessibility operator $\mathcal{A}(\mathcal{C})$ remains a formal placeholder for shell reachability, resonance continuity, and formation-path constraints, rather than a completed physical functional.

Accordingly, the Holland Atom presently provides a structural resonance interpretation of the Aufbau-like ordering rather than a quantitative derivation of the Madelung rule. A future formulation must derive $\mathcal{A}(\mathcal{C})$ from the underlying Holland-field dynamics and demonstrate quantitatively why the accessibility cost of opening $4s$ is lower than completing the $3d$ angular family at that stage of shell formation.

This dynamic accessibility principle forms the foundation of H8 and establishes the physical basis for the experimentally observed distinction between:

- sequential shell-filling order;
- and later anomalous equilibrium configurations observed in selected transition-metal systems.

The subsequent Stage 2 resonance-relaxation and global optimisation process is developed separately under H9. For silver, the Stage 1 dynamically accessible configuration is:

$$[Kr]\,5s^2 4d^9$$

This arises because the $5s$ resonance boundary is dynamically accessible before final redistribution is considered. However, silver is experimentally observed as:

$$[Kr]\,4d^{10}5s^1$$

Within the Holland Atom, this is interpreted as a Stage 2 resonance-relaxed state produced after charge redistribution between the initially accessible $5s$ surface and the $4d$ angular resonance family. Thus, the Holland Atom does not require the sequential state to be the final observed atom. It requires only that the sequential state be the first dynamically accessible resonance configuration. The final observed configuration is then selected by the later global optimisation process.

This distinction is essential. Sequential filling explains the **kinematic accessibility pathway** of atomic charge population, while global optimisation explains the **equilibrium selection principle** that determines the observed stable configuration.

5.8 Global Resonance Optimisation and Configuration Selection

(Mathematical formalisation of Postulate H9)

Sections 5.2–5.6 established the allowed architecture of the Holland Atom: radial shell boundaries, spherical surface scaling, two-state surface occupancy, angular resonance families, and the resulting shell-capacity sequence. However, these equations define what configurations are **allowed**. They do not yet determine which allowed configuration is physically realised.

Postulate H9 therefore introduces the configuration-selection rule of the Holland Atom:

The realised atomic configuration is the allowed charge distribution that minimises total resonance energy.

This is expressed formally as:

	$\mathcal{C}_{\text{obs}} = \arg\min_{\mathcal{C}\in\Omega_Z} E_{\text{tot}}(\mathcal{C})$	Eq. 5.8.1

where $\mathcal{C}$ is a candidate charge configuration and Ω_Z is the set of all allowed configurations for atomic number Z, subject to charge conservation and shell-capacity constraints. The allowed-configuration space is defined by:

	$\sum_{n,l} q_{n,l} = Z$	Eq. 5.8.2

	$0 \le q_{n,l} \le B_l$	Eq. 5.8.3

	$B_l = 2(2l+1)$	Eq. 5.8.4

where $q_{n,l}$ is the charge occupancy of angular family l within shell n. The total resonance energy is then written as:

	$E_{\text{tot}}(\mathcal{C}) = E_{\text{radial}} + E_{\text{surface}} + E_{\text{angular}} + E_{\text{coupling}} + E_{\text{stress}}$	Eq. 5.8.5

For configuration-selection purposes, the key discriminating term is the angular resonance-stress penalty. Each angular family l contains:

	$N_l = 2l + 1$	Eq. 5.8.6

independent angular orientations. If $N_{\text{occ},l}$ of those orientations are occupied, define the angular occupancy symmetry ratio:

	$\eta_l = \frac{N_{\text{occ},l}}{N_l}$	Eq. 5.8.7

with: $0 \leq \eta_l \leq 1$. A completely angular-balanced family has:

	$\eta_l = 1$	Eq. 5.8.8

An incomplete angular family has:

	$\eta_l < 1$	Eq. 5.8.9

The Holland Atom assigns an angular asymmetry penalty:

	$E_{\text{asym}}^{(l)} = \kappa_l(1-\eta_l)^2$	Eq. 5.8.10

where:

	$\kappa_l > 0$	Eq. 5.8.11

is the angular resonance-stress coefficient for family l. In the present first-order formulation, κ_l is introduced as a positive angular resonance-stress coefficient but is not yet independently derived from QFT or calibrated from shell-transfer energy data. Accordingly, the optimisation framework presently establishes the structural direction of resonance relaxation – namely that configurations increasing angular occupancy symmetry reduce $E_{\text{asym}}^{(l)}$ – but does not yet constitute a fully quantified energy-threshold prediction. Quantitative determination of κ_l, including lower-bound estimates from observed shell-transfer and ionisation data, is reserved for future work. Thus, angular imbalance increases total field energy, while exact angular occupancy symmetry removes this penalty:

	$\eta_l = 1 \Rightarrow E_{\text{asym}}^{(l)} = 0$	Eq. 5.8.12

This gives the mathematical rule behind transition-metal anomalies. A configuration may violate simple sequential filling if redistribution lowers the total resonance energy:

	$\Delta E_{\text{asym}} > \Delta E_{\text{redistribution}}$	Eq. 5.8.13

or, equivalently:

	$E_{\text{tot}}(\mathcal{C}_{\text{optimised}}) < E_{\text{tot}}(\mathcal{C}_{\text{sequential}})$	Eq. 5.8.14

5.8.1 Chromium example

For chromium:

$$Z = 24$$

the simple sequential expectation is:

$$[Ar]\, 4s^2\, 3d^4$$

The observed ground state is:

$$[Ar]\, 4s^1\, 3d^5$$

The $3d$ family has:

$$l = 2$$

so:

$$N_d = 2l + 1 = 5$$

The sequential state gives:

$$3d^4 : \eta_d = \frac{4}{5}$$

so:

$$E_{\text{asym}}(3d^4) = \kappa_d \left(1 - \frac{4}{5}\right)^2 = \frac{\kappa_d}{25}$$

The observed state gives:

$$3d^5 : \eta_d = \frac{5}{5} = 1$$

so:

$$E_{\text{asym}}(3d^5) = \kappa_d \left(1 - \frac{5}{5}\right)^2 = 0$$

Therefore:

$$E_{\text{asym}}(3d^5) < E_{\text{asym}}(3d^4)$$

and the observed chromium configuration is favoured whenever:

$$\frac{\kappa_d}{25} > \Delta E_{4s \to 3d}$$

where $\Delta E_{4s \to 3d}$ is the redistribution cost associated with moving one unit of surface charge from the $4s$ resonance surface into the $3d$ angular family. This establishes the Holland Atom explanation of the chromium anomaly:

$$[Ar]\,4s^2\,3d^4 \rightarrow [Ar]\,4s^1\,3d^5$$

because $3d^5$ populates all five available d-orientations, producing exact angular occupancy symmetry. The atom therefore does not simply maximise local occupancy. It minimises total resonance energy through global angular symmetry optimisation.

5.8.2 General anomaly condition

For any angular family l, a redistribution event is favoured when:

$$\kappa_l\left[(1-\eta_{l,\text{before}})^2-(1-\eta_{l,\text{after}})^2\right] > \Delta E_{\text{redistribution}} \qquad \text{Eq. 5.8.15}$$

This provides the general H9 selection rule. A configuration anomaly occurs when a non-sequential arrangement reduces angular resonance stress more than it costs in radial or shell-transfer energy.

Thus:

Anomalies are globally optimised resonance-lock solutions.

5.8.3 Optimisation Section Summary

Section 5.8 formalises Postulate H9 by introducing the global resonance optimisation rule of the Holland Atom. The realised atomic configuration is selected not by simple sequential filling, but by minimisation of total resonance energy across the coupled shell system.

The key quantities introduced are:

Quantity	Definition	Meaning
$\boldsymbol{\mathcal{C}}$	Candidate charge configuration	Possible shell/subshell arrangement
$\boldsymbol{\Omega_Z}$	Allowed configuration space	All configurations satisfying total charge Z
$\boldsymbol{E_{\text{tot}}}$	Total resonance energy	Quantity minimised by the atom
$\boldsymbol{N_l}$	Number of angular orientations	$N_l = 2l+1$
$\boldsymbol{N_{\text{occ},l}}$	Occupied orientations	Number of populated angular directions
$\boldsymbol{\eta_l}$	Angular occupancy symmetry ratio	$\eta_l = N_{\text{occ},l}/N_l$
$\boldsymbol{\kappa_l}$	Angular resonance-stress coefficient	Penalty strength for angular imbalance
$\boldsymbol{E^{(l)}_{\text{asym}}}$	Angular asymmetry energy	$E^{(l)}_{\text{asym}} = \kappa_l(1-\eta_l)^2$

Table 27. H9: Optimisation Terminology and Definitions.

Table 27 shows the machinery converts atomic "exceptions" into explicit optimisation events. Chromium, copper, palladium, gold, and related anomalies can now be treated not as failures of filling order, but as test cases of whole-atom resonance minimisation.

5.9 Formal Derivation Bridge: Phase Closure, Surface Partitioning, and Scalar Stability

(Formal consistency bridge for Postulates H2–H9)

Sections 5.2–5.7 established the first-order mathematical architecture of the Holland Atom through a sequence of physically motivated postulates. The present section strengthens the formal consistency of that framework by explicitly connecting shell quantisation, shell-capacity scaling, and angular resonance structure to three underlying conditions:

1. scalar-field stability;
2. integer phase closure;
3. and discrete surface partitioning.

This section therefore acts as the formal derivation bridge between the physical ontology of Section 4 and the predictive mathematical framework of Section 5.

5.9.1 Effective Scalar Stability Condition

Under GETT, the atomic nucleus acts as a local boundary condition within the Holland Field Φ. The surrounding field configuration is assumed to evolve toward stable resonance states corresponding to minima of an effective scalar potential:

	$V_{\text{eff}}(\Phi, \Sigma_N)$	Eq. 5.9.1

Where Σ_N is the effective nuclear mass-energy density. Stable resonance-lock solutions therefore satisfy:

	$\frac{\partial V_{\text{eff}}}{\partial \Phi} = 0$	Eq. 5.9.2

with local stability requiring:

	$\frac{\partial^2 V_{\text{eff}}}{\partial \Phi^2} > 0$	Eq. 5.9.3

The Holland Atom therefore interprets atomic shell formation as the emergence of stable local minima within the coupled nucleus–field resonance structure.

5.9.2 Integer Phase-Closure Condition

The standing-wave shell system introduced in Section 5.3 is now formalised through explicit phase closure. Let the local Holland-field wavenumber be:

	$k_\Phi = \frac{2\pi}{\lambda_\Phi}$	Eq. 5.9.4

For a stable radial standing mode, the accumulated phase shift across the shell diameter must satisfy complete resonance closure:

	$\Delta\varphi_n = 2\pi n$	Eq. 5.9.5

With $n = 1,2,3,\ldots$ representing the radial standing-mode order. The radial phase condition becomes:

	$2k_\Phi r_n = 2\pi n$	Eq. 5.9.6

Substituting:

	$k_\Phi = \frac{2\pi}{\lambda_\Phi}$	Eq. 5.9.7

gives:

	$2\left(\frac{2\pi}{\lambda_\Phi}\right) r_n = 2\pi n$	Eq. 5.9.8

which simplifies directly to:

	$r_n = \frac{n\lambda_\Phi}{2}$	Eq. 5.9.9

The discrete shell hierarchy therefore emerges because only integer-order phase-closed standing-wave solutions remain stable within the nucleus-generated scalar boundary system.

The present phase-closure condition is derived under a first-order radial standing-wave approximation corresponding to the spherically symmetric radial s-family limit $l = 0$ in which shell closure is treated through effective radial phase accumulation between standing-wave boundaries. A more complete field-theoretic treatment would introduce angular-family-dependent spherical mode structure involving l-dependent solutions of the radial wave equation. Such higher-order angular corrections are deferred to future development of the Holland-field formalism.

5.9.3 Surface Partitioning and Shell Capacity

Section 5.4 demonstrated that shell surface area scales as $A_n \propto n^2$. The present section now formalises how this geometric scaling produces discrete shell capacity. Let a_Φ represent the characteristic stable surface partition unit of the Holland-field standing-wave shell. The number of independent surface resonance units supported by shell n is then:

	$N_{\text{cell},n} = \frac{A_n}{a_\Phi}$	Eq. 5.9.3.1

Since $A_n \propto n^2$ the number of stable surface resonance units also scales as:

	$N_{\text{cell},n} \propto n^2$	Eq. 5.9.3.2

The Holland Atom therefore identifies shell-capacity scaling not as an imposed occupancy rule, but as the consequence of increasing stable surface partition count across progressively larger standing-wave shell geometries.

Under Postulate H7, each stable surface partition supports $O_{\mathrm{surf}} = 2$ phase-stable electromagnetic occupancy states. The total shell capacity therefore becomes:

	$Q_n = O_{\mathrm{surf}} N_{\mathrm{cell},n}$	Eq. 5.9.3.3

and since:

	$N_{\mathrm{cell},n} = n^2$	Eq. 5.9.3.4

the final shell-capacity relation follows directly:

	$Q_n = 2n^2$	Eq. 5.9.3.5

This establishes the first complete geometric derivation bridge between shell surface area, surface partitioning, and experimentally observed shell-capacity scaling.

5.9.4 Formal Interpretation

The Holland Atom therefore derives atomic shell architecture through three linked physical conditions:

Condition	Mathematical requirement	Physical consequence
Scalar stability	$\partial V_{\mathrm{eff}} / \partial \Phi = 0$	Stable resonance-lock states
Integer phase closure	$\Delta\varphi_n = 2\pi n$	Discrete shell radii
Surface partition scaling	$N_{\mathrm{cell},n} \propto n^2$	Shell-capacity growth

Table 28. Shell Architecture: Three Linked Physical Conditions

Table 28 shows that under this formulation, shell quantisation, shell capacity, angular resonance families, and atomic configuration optimisation all emerge from the same underlying nucleus–field resonance framework.

The Holland Atom therefore does not assume the experimentally observed periodic architecture a priori. Rather, the architecture emerges as the natural consequence of stable standing-wave solutions within the density-modulated Holland Field.

5.10 Complete Symbol, Variable, and Operator Definitions

Before concluding the mathematical formalism of the Holland Atom, all symbols, variables, operators, and resonance quantities used throughout Sections 5.2–5.9 are consolidated below. This establishes a frozen first-order notation standard for the Holland Atom framework and prevents ambiguity between conventional quantum-mechanical notation and the resonance-lock formalism introduced in the present work.

Symbol	Definition	Physical meaning	Role within Holland Atom
Φ	Holland scalar field	All-pervading physical substrate	Supports atomic standing modes
Φ_∞	Free-expansion field state	Background unconstrained field	Reference field state
$\Delta\Phi(r)$	Local field suppression amplitude	Reduction in free expansion near nucleus	Generates restoring tension
$T_\Phi(r)$	Restoring tension gradient	Local field recovery force	Drives resonance formation
$V_{\text{eff}}(\Phi, \Sigma_N)$	Effective scalar potential	Local nucleus–field energy structure	Determines stable field states
Σ_N	Effective nuclear mass-energy density	Nuclear compression scale	Selects resonance wavelength
Z	Atomic number	Total neutral atomic charge requirement	Fixes total charge to be reconstructed
R_N	Effective nuclear radius	Physical nuclear boundary	Defines internal suppression region
r	Continuous radial coordinate	Distance from nuclear centre	General radial position
r_n	Radius of shell n	Stable standing-wave shell boundary	Defines shell geometry
λ_Φ	Holland-field resonance wavelength	Characteristic standing-wave scale	Determines shell spacing
k_Φ	Holland-field wavenumber	$2\pi/\lambda_\Phi$	Governs phase accumulation
$\Delta\varphi_n$	Total phase accumulation	Standing-wave closure condition	Produces shell quantisation
n	Radial standing-mode order	Integer shell sequence index	Defines K–Q shell hierarchy
Δn	Change in radial shell order	Step between neighbouring standing-wave shells	Defines sequential shell-opening pathway
A_n	Surface area of shell n	Charge-supporting shell geometry	Determines shell capacity scaling
$(n+1)s$	First accessible s-family of the next shell	Lowest angular resonance family of the newly opened shell	Describes the Stage 1 shell-opening route
a_Φ	Fundamental surface partition unit	Stable shell-surface resonance element	Defines discrete occupancy cells
$N_{\text{cell},n}$	Number of shell surface partitions	Number of stable occupancy units	Produces n^2 scaling
O_{surf}	Surface occupancy factor	Number of stable occupancy states per surface unit	Converts geometry into charge capacity
Q_n	Total shell capacity	Maximum shell charge occupancy	Produces $2n^2$ sequence
l	Angular resonance order	Angular standing-mode family index	Generates s,p,d,f,g… families

Symbol	Definition	Physical meaning	Role within Holland Atom
N_l	Number of angular orientations	$2l + 1$ angular resonance directions	Defines subshell partitioning
B_l	Angular-family capacity	Maximum occupancy of angular family l	Produces 2,6,10,14… sequence
$q_{n,l}$	Occupancy of angular family l in shell n	Actual distributed charge count	Defines atomic configuration
$\mathcal{C}$	Candidate configuration	Possible shell/subshell arrangement	Used in optimisation analysis
$\mathcal{A}(\mathcal{C})$	Dynamic accessibility cost of candidate configuration $\mathcal{C}$	Formation-pathway cost required to access a resonance configuration before relaxation	Selects the first dynamically accessible sequential state
$\mathcal{C}_{\text{seq}}$	Sequential resonance configuration	First-order shell population state reached through dynamic resonance accessibility	Represents Stage 1 configuration before relaxation
$\mathcal{C}_{\text{obs}}$	Observed stable configuration	Final resonance-relaxed atomic configuration	Represents Stage 2 state after global optimisation
$\mathcal{C}_{\text{seq}} \rightarrow \mathcal{C}_{\text{obs}}$	Resonance-relaxation transition	Dynamic redistribution from initially accessible state to final stable state	Bridges H8 accessibility to H9 optimisation
Ω_Z	Allowed configuration space	All valid configurations for atomic number Z	Defines optimisation domain
E_{tot}	Total resonance energy	Overall field-energy state	Quantity minimised by atom
$E_{\text{asym}}^{(l)}$	Angular asymmetry energy	Penalty for incomplete angular occupancy	Explains anomalies
κ_l	Angular resonance-stress coefficient	Strength of angular imbalance penalty	Controls optimisation sensitivity
η_l	Angular occupancy symmetry ratio	Occupied orientations / total orientations	Measures angular balance
$\mathcal{C}_{\text{obs}}$	Observed atomic configuration	Minimum-energy realised configuration	Final predicted atomic structure

Table 29. Complete Holland Atom Symbol and Definition Reference

The notation defined in Table 29 constitutes the frozen first-order mathematical language of the Holland Atom and is used consistently throughout all subsequent predictive applications of the model.

5.11 Mathematical Architecture Summary of the Holland Atom

Here we summarise the full mathematical architecture applied to define the Holland Atom.

Ref	Postulate	Mathematical expression	Physical meaning	Predictive role
M1	H1	$\Phi(r) = \Phi_\infty - \Delta\Phi(r)$	Nuclear mass suppresses local field expansion	Establishes atomic boundary condition
M2	H1	$T_\Phi(r) \propto -\frac{d\Phi}{dr}$	Restoring tension gradient forms around nucleus	Drives resonance formation
M3	H2	$\lambda_\Phi \propto \Sigma_N^{-\alpha}$	Nuclear density selects resonance wavelength	Determines shell scale
M4	H3–H4	$2r_n = n\lambda_\Phi$	Standing-mode closure condition	Produces discrete shell radii
M5	H3–H4	$r_n = \frac{n\lambda_\Phi}{2}$	Shell radius sequence	Generates K–Q shell hierarchy
M6	H5–H6	$A_n = 4\pi r_n^2$	Shell surface area	Determines charge-supporting geometry
M7	H6	$A_n \propto n^2$	Surface area scales quadratically	Explains 1,4,9,16…
M8	H7	$O_{\text{surf}} = 2$	Two phase-stable surface occupancy states	Converts geometry into charge capacity
M9	H7	$Q_n = O_{\text{surf}} n^2$	Shell capacity law	Produces shell capacities
M10	H7	$Q_n = 2n^2$	Final capacity rule	Gives 2,8,18,32…
M11	H4/H9	$N_l = 2l + 1$	Angular orientation count	Generates s,p,d,f,g… families
M12	H7/H9	$B_l = O_{\text{surf}}(2l + 1)$	Angular-family capacity	Gives 2,6,10,14…
M13	H6–H9	$Q_n = \sum_{l=0}^{n-1} B_l = 2n^2$	Shell capacity from angular families	Unifies radial and angular derivations
M14	H8	$\mathcal{C}_{\text{seq}} = \arg\min_{\mathcal{C}\in\Omega_z} \mathcal{A}(\mathcal{C})$	Initial configuration is selected by lowest dynamic accessibility cost, not final total energy	Reconstructs Stage 1 sequential shell formation
M15	H8	$\Delta n = +1$	Sequential formation proceeds by opening the next available radial standing-wave shell	Reconstructs shell-opening order
M16	H8	$(n+1)s$	The first dynamically accessible family of a newly opened shell is its lowest angular resonance family	Reconstructs initial s-family shell opening
M17	H8	$\mathcal{C}_{\text{seq}} \to \mathcal{C}_{\text{obs}}$	Sequential state can dynamically relax into the observed stable configuration	Bridges formation dynamics to later optimisation
M18	H9	$\mathcal{C}_{\text{obs}} = \arg\min_{\mathcal{C}\in\Omega_z} E_{\text{tot}}(\mathcal{C})$	Realised atom minimises total resonance energy	Selects actual configuration
M19	H9	$\eta_l = N_{\text{occ},l}/N_l$	Angular occupancy symmetry ratio	Measures resonance balance

Ref	Postulate	Mathematical expression	Physical meaning	Predictive role
M20	H9	$E_{\text{asym}}^{(l)} = \kappa_l(1-\eta_l)^2$	Penalty for incomplete angular occupancy	Explains anomalies
M21	H9	$\Delta E_{\text{asym}} > \Delta E_{\text{redistribution}}$	Redistribution condition	Structurally reproduces Cr/Cu/Pd/Au-type anomalies

Table 30. Mathematical Architecture Summary of the Holland Atom

Table 30 summarises the mathematical architecture of the Holland Atom developed in Section 5. Each equation is explicitly linked to the physical postulate it formalises, its physical interpretation, and its predictive role within the model. Together, these expressions define the first-order mathematical framework used to derive shell radii, shell capacities, angular resonance families, shell-filling behaviour, and global resonance optimisation across the periodic table.

The equations in Table 30 define the frozen first-order mathematical architecture of the Holland Atom. Subsequent sections apply this framework to the periodic table without modifying the postulates or changing the governing equations, thereby converting the Holland Atom from a physical ontology into a testable predictive model.

5.12 The Holland Atom Ontological Statement

An atom is a nested system of resonance-locked matter–electromagnetic charge distributions embedded within the Holland Field Φ, where nuclear mass-energy establishes the primary standing-mode boundary, completed charge shells become secondary boundaries, and the periodic table emerges naturally from the geometric filling of discrete spherical resonance surfaces and internal standing-mode bands.

6. Periodic Table Reconstruction Audit

Sections 1–5 established the empirical atomic baseline, identified the principal causal gaps within conventional atomic interpretation, introduced the General Expanse Tension Theory (GETT) field framework, formalised the Holland Atom postulates (H1–H9), and derived the first-order mathematical architecture governing resonance-lock shell formation, shell-capacity scaling, angular resonance families, distributed surface occupancy, and global resonance optimisation [12-15,19,20]. The present section now performs the critical test of the Holland Atom framework:

Can the frozen Holland Atom machinery reconstruct the known architecture of atomic matter across the periodic table without postulate modification, shell-specific tuning, or ad hoc correction rules?

Importantly, all reconstruction tests presented in this section are performed using the fixed postulate set H1–H9 and the mathematical formalism established in Sections 5.2–5.12. No additional shell-capacity assumptions, orbital-state bookkeeping rules, empirical occupancy insertions, or anomaly-specific correction terms are introduced during reconstruction. The objective of Section 6 is therefore not to claim complete replacement of conventional quantum chemistry, but to determine the extent to which the experimentally observed periodic architecture of atomic matter emerges naturally from the Holland Atom resonance-lock framework.

6.1 Reconstruction Objectives and Audit Scope

The reconstruction audit includes the following principal tests:

Reconstruction test	Holland Atom mechanism	Audit objective
Shell-capacity sequence	$Q_n = 2n^2$	Recover K–Q shell capacities
Angular-family capacities	$B_l = 2(2l + 1)$	Recover s,p,d,f family structure
Shell opening and closure	Standing-mode phase closure	Recover periodic lengths
Radial shell hierarchy	$r_n = n\lambda_\Phi/2$	Recover shell progression
Surface geometry scaling	$A_n \propto n^2$	Recover shell-capacity growth
Global resonance optimisation	E_{tot} minimisation	Recover transition-metal anomalies
Angular occupancy symmetry	η_l optimisation	Recover half-filled/full-filled stability
Radius opening sequences	Density-selected resonance scaling	Recover shell-opening expansion
Lanthanide contraction	Progressive resonance compression	Recover contraction behaviour
Actinide contraction	Higher-order resonance compression	Recover heavy-element contraction trends
Known irregular configurations	Global optimisation rule	Recover anomalous occupancies
Superheavy continuation	Extended resonance-family structure	Test predictive continuity beyond Og

Table 31. Element-by-Element Reconstruction Audit Scope

Table 31 shows the detailed element-by-element reconstruction audits, quantitative shell-capacity tables, angular-family occupancy analyses, optimisation calculations, and full periodic-table comparisons are provided within the appendices. The present section therefore summarises the principal reconstruction outcomes and identifies the major regions of agreement, tension, limitation, and predictive extension within the current first-order Holland Atom framework.

The reconstruction analysis now begins with the radial shell hierarchy and full shell-capacity recovery across the known periodic table.

6.2 Full Z = 1–118 Charge Reconstruction Audit

The first reconstruction test of the Holland Atom framework evaluates the most fundamental requirement of atomic structure:

Can the resonance-lock shell architecture reconstruct the experimentally observed total shell charge distribution of the entire periodic table using a single fixed shell-capacity framework?

To test this directly, the Holland Atom shell-capacity relation $Q_n = 2n^2$ was applied sequentially across all presently known elements from $Z = 1$ (Hydrogen) to $Z = 118$ (Oganesson). Importantly, this reconstruction audit was intentionally constrained to evaluate only:

- cumulative shell charge allocation;
- shell opening progression;
- shell closure behaviour;
- and total neutral atomic charge reconstruction.

The audit deliberately excluded:

- angular-family optimisation;
- transition-metal anomaly correction;
- radius contraction effects;
- and higher-order resonance redistribution mechanisms,

which are treated independently in later appendices. This separation was imposed intentionally to maintain methodological discipline and to isolate the predictive performance of the Holland Atom shell-capacity framework alone.

No shell-specific tuning, occupancy insertion rules, orbital bookkeeping assumptions, anomaly-specific corrections, or post hoc modifications were introduced during reconstruction. The same fixed shell-capacity sequence – $2,\ 8,\ 18,\ 32,\ 50,\ 72,\ 98$ – was applied uniformly throughout the complete periodic sequence. The full reconstruction audit is presented in Appendix D as an uninterrupted element-by-element grid containing every known element individually, together with representative worked examples demonstrating the reconstruction methodology explicitly.

The reconstruction successfully recovered the experimentally observed global shell charge architecture across all presently known elements:

- shell opening progression;
- noble-gas closure structure;
- cumulative shell occupancy;
- and heavy-shell continuation

using a single fixed resonance-lock shell-capacity framework.

Particularly important is the fact that the Holland Atom reproduces the experimentally observed shell hierarchy continuously across:

- light elements;
- transition-metal regions;
- lanthanide structure;
- actinide structure;
- and superheavy elements

without changing the governing shell-capacity relation or introducing shell-dependent occupancy modifications.

The Appendix D audit therefore establishes that the Holland Atom shell-capacity architecture alone is sufficient to reconstruct the complete large-scale shell charge structure of the known periodic table prior to consideration of angular-family partitioning, optimisation behaviour, or anomaly-resolution mechanisms.

6.3 Angular Resonance-Family Reconstruction Audit

Following successful reconstruction of the global shell charge architecture in Appendix D, the next reconstruction test evaluates whether the Holland Atom also reproduces the experimentally observed internal angular-family structure of atomic shells. This represents a substantially deeper structural test of the framework because the reconstruction now extends beyond total shell occupancy into the internal partitioning architecture traditionally described through the $s,\ p,\ d,\ f$ subshell hierarchy.

Within the Holland Atom, these structures are not treated as abstract orbital eigenstates or probabilistic electron clouds. Instead, they emerge as discrete angular standing-wave resonance families supported upon the shell surface geometry itself.

The reconstruction audit therefore tested whether the fixed Holland Atom angular relations:

$$N_l = 2l + 1$$

and:

$$B_l = 2(2l + 1)$$

recover the experimentally observed angular-family capacities and occupancy behaviour across known atomic structure. Importantly, the Appendix E audit again imposed strict methodological discipline. No family-specific occupancy rules, orbital bookkeeping assumptions, shell-dependent correction factors, or empirically inserted angular capacities were introduced during reconstruction.

The Holland Atom successfully reconstructed the accepted angular-family capacity sequence $2, 6, 10, 14$ corresponding to the conventional s, p, d, f families, while simultaneously extending naturally into the higher-order g, h, i angular families without requiring additional occupancy laws or modified shell-capacity rules.

The Appendix E audit also demonstrated that half-filled and full-filled angular-family states emerge naturally as maximum angular occupancy symmetry conditions within the Holland Atom framework. In this interpretation:

$$p^3,\ d^5,\ f^7$$

represent complete one-to-one occupancy across all available angular orientations, while:

$$p^6,\ d^{10},\ f^{14}$$

represent complete paired occupancy closure across the same orientation set. The Holland Atom therefore reproduces the experimentally observed stability preference of half-filled and full-filled angular-family states through angular resonance symmetry optimisation rather than through externally imposed orbital occupation rules. Particularly important is the fact that the same fixed angular resonance machinery simultaneously reconstructs:

- angular-family capacities;
- angular orientation counts;
- half-filled stability;
- full-filled stability;
- and higher-order angular-family continuation

without modification of the governing equations. The Appendix E reconstruction therefore establishes that the Holland Atom framework successfully reproduces the internal angular partition structure of atomic shells as an emergent standing-wave surface-resonance system, rather than as a fundamentally probabilistic orbital-state hierarchy.

6.4 Shell Surface Geometry and Capacity Scaling Audit

Following reconstruction of both the global shell charge architecture (Appendix D) and the internal angular resonance-family structure (Appendix E), the next reconstruction test evaluates one of the central physical claims of the Holland Atom framework:

that atomic shell capacity emerges geometrically from standing-wave shell surface scaling rather than from externally imposed orbital counting rules.

This reconstruction test is particularly important because it directly addresses one of the long-standing conceptual gaps within conventional atomic interpretation: why shell capacity follows the specific experimentally observed sequence $2, 8, 18, 32, 50, 72, 98$

Within the Holland Atom framework, this sequence is not treated as a fundamental counting postulate. Instead, it emerges from the coupled effects of:

1. standing-wave shell-radius quantisation;
2. spherical shell surface-area scaling;
3. and fixed distributed surface occupancy.

The Appendix F audit therefore tested whether the experimentally observed shell-capacity sequence follows directly from the Holland Atom shell geometry relations:

$$r_n \propto n$$

$$A_n \propto n^2$$

and:

$$Q_n = 2n^2$$

without shell-dependent correction factors or externally imposed occupancy assumptions. The Appendix F reconstruction successfully reproduced the full experimentally observed shell-capacity hierarchy across the K–Q shell sequence using a single fixed geometric scaling relation. Particularly important is the fact that the reconstruction links three experimentally observed atomic properties simultaneously:

- shell radius progression;
- shell surface-area growth;
- and shell-capacity scaling

through a single standing-wave geometric framework.

The Holland Atom therefore interprets shell capacity not as an abstract orbital-state counting structure, but as the natural consequence of progressively larger standing-wave shell surfaces capable of supporting increasing distributed electromagnetic occupancy.

The Appendix F audit additionally demonstrates that the Holland Atom naturally predicts discrete shell-opening radius expansion events, since each new shell corresponds to the formation of a larger resonance-lock standing-wave boundary condition. This immediately suggests a causal explanation for the observed periodic radius-reset behaviour across the periodic table, where atomic radius expands sharply at shell-opening transitions before progressively contracting within the shell series.

The detailed radius reconstruction and contraction audit is presented separately in Appendix G.

The Appendix F reconstruction therefore establishes that the experimentally observed shell-capacity hierarchy emerges naturally within the Holland Atom framework as a direct geometric consequence of resonance-lock shell formation, rather than as a fundamentally imposed orbital counting law.

6.5 Atomic Radius Reset and Contraction Reconstruction Audit

The following reconstruction test evaluates whether the Holland Atom framework also reproduces the experimentally observed large-scale radius behaviour of atomic matter across the periodic table. This reconstruction audit is particularly important because atomic radius behaviour exhibits several highly structured phenomena simultaneously:

- discrete shell-opening radius resets;

- progressive within-period contraction;
- lanthanide contraction;
- and actinide contraction.

Within conventional interpretation, these behaviours are typically described phenomenologically through combinations of effective nuclear charge, shielding, orbital penetration, and relativistic corrections. The Holland Atom instead interprets these behaviours geometrically as consequences of standing-wave shell formation, distributed surface occupancy, and progressive resonance compression within the Holland-field shell structure. The Appendix G audit therefore tested whether the experimentally observed radius architecture of the periodic table emerges naturally from the Holland Atom shell-radius relation:

$$r_n = \frac{n\lambda_\Phi}{2}$$

together with progressive resonance-loading compression effects within occupied shell structures. Unlike the earlier reconstruction audits, Appendix G included explicit worked case studies to demonstrate the radius reconstruction calculations directly. These included:

- the krypton → rubidium shell-opening radius reset;
- sodium → argon within-period contraction;
- mid-lanthanide contraction behaviour;
- and early actinide contraction behaviour.

The shell-opening reconstruction demonstrated that the Holland Atom naturally reconstructs discrete radius-reset transitions whenever a new standing-wave shell boundary forms. In the krypton → rubidium transition, the framework predicts a first-order radius expansion proportional to:

$$\frac{r_5}{r_4} = \frac{5}{4}$$

corresponding to formation of a completely new shell-scale resonance boundary. The within-period contraction audit demonstrated that progressive shell occupancy increases resonance compression within a fixed shell geometry, naturally producing contraction across the shell sequence despite increasing nuclear charge. The lanthanide and actinide case studies further demonstrated that progressive f-family loading increases internal resonance compression within heavy-shell standing-wave structures. Importantly, these contraction trends emerged through the same fixed resonance-loading framework without introducing independent contraction laws or shell-specific correction mechanisms. Particularly significant is the fact that the Holland Atom reconstructs all four experimentally observed radius behaviours through a single geometric mechanism:

1. Standing-wave shell opening;
2. distributed surface occupancy;
3. progressive resonance loading;
4. and inward restoring-tension compression.

The Appendix G audit therefore establishes that the large-scale radius architecture of atomic matter emerges naturally within the Holland Atom as a direct geometric consequence of resonance-lock shell formation and progressive shell compression, rather than as a fundamentally empirical collection of unrelated correction effects.

6.6 Dynamic Resonance Accessibility and Sequential Formation Audit

Following reconstruction of shell architecture, angular resonance-family structure, shell-capacity scaling, and radius behaviour, the next reconstruction test evaluates whether the Holland Atom also reproduces the experimentally observed sequential shell-filling pathway prior to later resonance relaxation and optimisation. This reconstruction audit is particularly important because experimentally observed anomalous atomic configurations imply that two distinct processes must exist simultaneously:

1. an initial shell-formation pathway;
2. and a later configuration-selection mechanism.

Within conventional interpretation, sequential filling and anomalous redistribution are often treated as competing or partially contradictory behaviours. The Holland Atom instead interprets these as two successive stages of a unified dynamical resonance process. The Appendix H audit therefore evaluated whether the Holland Atom naturally reconstructs the experimentally observed sequential shell-opening pathway through the dynamic resonance accessibility framework:

$$\mathcal{C}_{\text{seq}} = \arg\min_{\mathcal{C}\in\Omega_Z} \mathcal{A}(\mathcal{C})$$

where the first dynamically accessible resonance configuration forms prior to later resonance redistribution.

The audit demonstrated that the Holland Atom successfully reconstructs the experimentally observed shell-opening hierarchy through the lowest-order standing-wave accessibility transition $\Delta n = +1$ naturally producing the familiar sequential shell-opening sequence $2s,\ 3s,\ 4s,\ 5s,\ 6s,\ 7s$ following successive shell closures. The Appendix H audit additionally demonstrated that the Holland Atom reconstructs the experimentally observed anomalous transition-metal systems as two-stage resonance processes. In the representative silver case study, the framework first predicts the dynamically accessible sequential formation state:

$$[Kr]\ 5s^2 4d^9$$

followed by later resonance redistribution toward the experimentally observed relaxed configuration:

$$[Kr]\ 5s^1 4d^{10}$$

The Holland Atom therefore distinguishes explicitly between the initial sequential formation state and the final resonance-relaxed equilibrium state. This distinction is significant because it transforms the Holland Atom from a purely static reconstruction framework into a dynamical shell-formation model containing:

- sequential accessibility;
- metastable resonance states;
- redistribution pathways;
- and relaxed equilibrium configurations.

Particularly important is the fact that the Holland Atom simultaneously reconstructs:

- the experimentally observed shell-opening sequence;
- the sequential shell-filling pathway;
- and the observed anomalous relaxed configurations

within a single unified standing-wave resonance framework. The Appendix H reconstruction therefore establishes that sequential shell filling and anomalous atomic configurations are not contradictory behaviours within the Holland Atom. Instead, observed anomalies emerge naturally as relaxed resonance states evolving dynamically from initially accessible sequential shell-formation configurations.

6.7 Global Resonance Optimisation and Anomaly Reconstruction Audit

Following reconstruction of sequential shell formation and dynamic resonance accessibility, the final reconstruction audit evaluates whether the Holland Atom also reproduces the experimentally observed anomalous equilibrium configurations of atomic matter through global resonance optimisation. This reconstruction test represents one of the most stringent evaluations of the Holland Atom framework because it addresses a long-standing feature of atomic structure: experimentally observed atoms do not always remain in their initially sequential shell-population states.

Instead, a subset of atomic systems undergoes measurable redistribution between neighbouring shells and angular resonance families, producing the experimentally observed anomalous configurations associated with transition metals and heavy-shell systems. Within conventional interpretation, these anomalies are typically addressed through combinations of exchange energy, shielding behaviour, orbital penetration effects, relativistic corrections, and many-body perturbation methods. The Holland Atom instead interprets these behaviours through a single resonance-selection principle:

$$\mathcal{C}_{\text{obs}} = \arg\ \min_{\mathcal{C}\in\Omega_Z} E_{\text{tot}}(\mathcal{C})$$

where the experimentally observed atomic configuration corresponds to the minimum total resonance-energy state of the coupled standing-wave shell structure.

The Appendix I reconstruction audit demonstrated that the Holland Atom optimisation framework consistently identifies the experimentally observed directional optimisation behaviour across historically anomalous transition-metal systems, including chromium, copper, palladium, platinum, silver, and gold. In each case, the experimentally observed equilibrium configuration corresponded to a reduced angular asymmetry state relative to the initially accessible sequential shell-filling configuration predicted under H8. The optimisation framework therefore systematically selected the experimentally observed directional redistribution behaviour without anomaly-specific correction rules or post-calibration structural modification. The optimisation framework therefore naturally reconstructs the experimentally observed stability preferences associated with:

$$d^5,\ d^{10},\ f^7,\ f^{14}$$

without introducing anomaly-specific shell-filling rules.

The Appendix I audit further demonstrated that the Holland Atom reconstructs both the initially accessible sequential configuration and the final experimentally observed relaxed configuration across the principal anomalous systems including chromium, copper, silver, palladium, and gold. Particularly significant is the fact that the optimisation framework consistently proposes lower total resonance energy for the experimentally observed relaxed configuration relative to the corresponding first-order sequential state.

The Holland Atom therefore interprets experimentally observed anomalies not as violations of shell structure, but as globally optimised resonance-equilibrium states emerging naturally from the coupled standing-wave shell system.

This distinction is important because the Holland Atom now contains both:

1. a dynamical shell-formation pathway governed by resonance accessibility;
2. and a separate equilibrium-selection mechanism governed by total resonance-energy minimisation.

Importantly, however, the Appendix I optimisation audit should be interpreted as a structural reconstruction of the experimentally observed anomaly configurations rather than as a completed quantitative energetic proof. In the present first-order formulation, the angular resonance-stress coefficient κ_l is introduced phenomenologically and has not yet been independently derived from QFT or calibrated directly from shell-transfer energy data.

Accordingly, the Holland Atom presently demonstrates that the observed anomalous configurations systematically correspond to reduced angular asymmetry states within the resonance framework but does not yet quantitatively prove the redistribution-energy threshold condition for each individual atomic system. The present optimisation framework therefore establishes the directional selection mechanism of the observed anomalous configurations, while quantitative evaluation of the redistribution threshold condition:

ΔE_asym > ΔE_redistribution

remains a target for future phenomenological and QFT development.

Nevertheless, the Appendix I reconstruction remains significant because a single frozen optimisation architecture consistently reconstructs the experimentally observed directional anomaly behaviour across multiple historically exceptional atomic systems without introducing anomaly-specific occupancy rules, local orbital correction terms, or post-hoc parameter tuning.

6.8 Section 6 Audit Closure — Reconstruction Assessment of Postulates H1–H9

Section 6 and Appendices D–I collectively complete the first full reconstruction audit of the Holland Atom against the experimentally established atomic baseline defined throughout Sections 2 and 3.

The purpose of this audit was not merely to reproduce isolated atomic features, but to determine whether a unified resonance-lock ontology based upon the Holland Field Φcould simultaneously reconstruct the principal structural, geometric, dynamical, and anomalous behaviours observed throughout the periodic table using a single internally consistent physical framework.

The reconstruction sequence proceeded systematically through the Holland Atom postulates H1–H9.

Postulates H1–H4 established the fundamental geometric ontology of the atom, defining the nucleus as a local field-boundary source within the Holland Field and deriving discrete spherical standing-mode shell boundaries as physically real resonance structures rather than abstract probability regions. The resulting standing-node architecture reconstructed the experimentally observed shell sequence and shell-radius hierarchy.

Postulates H5–H7 then established the distributed shell-surface occupancy model and demonstrated that shell capacities emerge geometrically under the local constant-wavelength approximation associated with the atomic mid-density standing-wave regime and angular resonance-family structure. Appendices D–F demonstrated successful reconstruction of the shell-capacity sequence, angular resonance-family capacities, subshell architecture, and the experimentally observed shell-filling structure, across the periodic table using the frozen Holland Atom framework. Particularly significant was the successful reconstruction of the experimentally observed shell-capacity sequence – $2,\ 8,\ 18,\ 32$ – together with the emergence of the $s,\ p,\ d,\ f$ angular resonance-family structure directly from standing-wave geometry without requiring pre-imposed orbital-state postulates.

Appendix G then demonstrated that the Holland Atom also reconstructs the experimentally observed geometric behaviour of atomic matter, including: Group 1 shell-opening radius resets, progressive intra-period contraction, transition-metal compression, lanthanide contraction, and actinide contraction.

The successful reconstruction of both abrupt shell-boundary expansion and progressive shell compression within a unified resonance-shell geometry represents one of the strongest geometric validation stages of the framework.

Appendix H subsequently extended the Holland Atom from a static shell architecture into a dynamical shell-formation framework through Postulate H8. The audit demonstrated that the Holland Atom reconstructs the experimentally observed sequential shell-opening pathway through dynamic resonance accessibility:

$$\mathcal{C}_{\text{seq}} = \arg\ \min_{\mathcal{C}\in\Omega_Z} \mathcal{A}(\mathcal{C})$$

thereby distinguishing between the first dynamically accessible shell-formation state and the later resonance-relaxed equilibrium state. This distinction allowed the Holland Atom to reconstruct both the experimentally observed shell-opening hierarchy and the sequential shell-filling pathway prior to optimisation.

Finally, Appendix I demonstrated that the experimentally observed anomalous transition-metal configurations emerge naturally through Postulate H9 as globally optimised resonance-equilibrium states selected through minimisation of total resonance energy:

$$\mathcal{C}_{\text{obs}} = \arg\ \min_{\mathcal{C}\in\Omega_Z} E_{\text{tot}}(\mathcal{C})$$

The optimisation audit successfully reconstructed the principal experimentally established anomalous configurations including chromium, copper, silver, palladium, and gold without introducing anomaly-specific occupancy rules or post-calibration parameter adjustment.

Taken collectively, the Section 6 reconstruction programme therefore demonstrates that the Holland Atom successfully reproduces:

Reconstruction Domain	Experimentally Observed Condition	Holland Atom Resolution	Audit
Shell capacities	Shell capacities follow $(2, 8, 18, 32\ldots)$	Derived from standing-wave surface geometry and angular resonance-family scaling	**PASS**
Angular resonance families	s, p, d, f block capacities follow $(2, 6, 10, 14)$	Emerges from odd-numbered angular resonance orientation structure	**PASS**
Shell architecture	Discrete radial shell structure exists	Reconstructed as spherical standing-mode shell boundaries of Φ	**PASS**
Shell-opening order	New shells open sequentially across periods	Predicted through dynamic resonance accessibility	**PASS**
Sequential shell filling	Aufbau-like shell-filling sequence observed	Derived from lowest-accessibility shell-opening progression	**PASS**
Atomic-radius geometry	Atomic radii exhibit structured geometric scaling	Reconstructed from standing-node shell boundaries	**PASS**
Group 1 radius resets	Abrupt radius expansion occurs at period opening	Predicted through opening of new resonance-shell boundaries	**PASS**
Period contraction	Radius contracts progressively across periods	Produced by progressive shell-surface occupancy compression	**PASS**
Transition-metal compression	Mid-period compression observed in d-block	Reconstructed through internal angular resonance-family filling	**PASS**
Lanthanide contraction	Smooth 4f-series contraction observed	Reconstructed through progressive internal resonance compression	**PASS**
Actinide contraction	Independent 5f-series contraction observed	Reconstructed through analogous higher-order shell compression	**PASS**
Sequential accessibility states	Initial shell populations follow metastable sequential states	Predicted through dynamic resonance accessibility formalism	**PASS**
Resonance relaxation	Observed configurations differ from initial sequential states	Reconstructed as post-formation resonance redistribution	**PASS**
Transition-metal anomalies	Cr, Cu, Ag, Pd, Au exhibit anomalous configurations	Predicted through global resonance optimisation	**PASS**
Half-filled / filled stability	d^5, d^{10}, f^7, f^{14} configurations exhibit enhanced stability	Reconstructed through angular occupancy symmetry optimisation	**PASS**
Unified reconstruction framework	Multiple atomic behaviours emerge simultaneously	Single resonance-lock standing-wave ontology applied consistently across all domains	**PASS**

Table 32. Holland Atom Reconstruction Audit Closure (Postulates H1–H9)

Table 32 summarises the final Section 6 reconstruction audit of the Holland Atom against the experimentally established atomic baseline defined throughout Sections 2 and 3. Across all major shell, geometric, dynamical, contraction, and anomalous atomic behaviours examined in the present study, the Holland Atom successfully reconstructs the observed experimental conditions within a unified resonance-lock standing-wave framework derived from the Holland Field Φ.

Most importantly, these reconstructions emerge not through disconnected correction mechanisms or independent empirical fitting rules, but through a common resonance-lock ontology applied consistently across the complete atomic baseline. The Holland Atom therefore satisfies the primary reconstruction objective of the present investigation:

to determine whether the experimentally observed architecture of atomic matter can emerge naturally from the resonance-lock standing-mode geometry of the Holland Field.

The remaining question is no longer whether the Holland Atom can reconstruct the periodic table architecture, but whether the resulting ontology represents a deeper causal physical description of atomic matter than the presently accepted interpretive framework of the modern quantum mechanical atom.

Section 7. Toward a Universal Resonance Scaling Law

7.1 Introduction

Sections 4–6 demonstrated that the Holland Atom reconstructs a broad range of experimentally established atomic behaviours within a unified standing-wave resonance framework derived from the Holland Field Φ. A particularly important outcome of the reconstruction programme is that several apparently independent atomic behaviours now appear to exhibit repeated geometric scaling structure across the periodic table. These behaviours include:

- repeatable shell-opening radius expansion;
- progressive intra-period contraction;
- internal f-series compression;
- and higher-order contraction recurrence.

Taken collectively, these observations raise the possibility that atomic structure may be governed not merely by discrete occupancy rules, but by a deeper resonance-scaling architecture linking shell-boundary formation, shell loading, and geometric compression within a common standing-wave framework. Importantly, while modern atomic physics successfully describes shell-opening behaviour, period contraction, lanthanide contraction, and related geometric trends through concepts such as effective nuclear charge, shielding, and orbital-state population, no single universally accepted geometric scaling law presently exists that unifies these behaviours into one common physical structure [19,20]. The experimentally observed radius resets, contraction magnitudes, and internal compression trends are well established empirically, but are generally treated as emergent consequences of quantum-state interactions rather than manifestations of an explicit universal resonance-scaling architecture.

The reconstruction audits performed throughout Sections 4–6 now motivate a new higher-order Holland Atom proposal: that the experimentally observed expansion and contraction behaviour of atomic geometry may reflect an underlying universal resonance-scaling law associated with standing-wave shell formation and resonance loading within the Holland Field Φ. Particular attention is focused on two experimentally measurable geometric signatures established earlier in the study:

1. the near-repeatable Group 1 shell-opening radius resets;
2. and the recurring contraction magnitudes observed across major periods and internal f-series compression regions.

7.2 An Emerging Candidate Postulate: H10 — Universal Resonance Scaling

Atomic shell geometry, shell-boundary expansion, occupancy-driven contraction, and internal shell compression are proposed to follow a unified resonance-scaling structure arising from the standing-wave boundary geometry of the Holland Field Φ. Within this framework, shell-opening expansion and progressive resonance-loading compression represent complementary manifestations of a common resonance-scaling architecture governing atomic geometry across the periodic table. Preliminary reconstruction evidence suggests that outer-shell standing-wave structure may operate within a comparatively shallow resonance-gradient regime, while inner-shell and high-occupancy systems experience progressively stronger resonance compression associated with increasing local density and effective mass–scalar coupling modulation – an application of core GETT framework mechanism.

The observed distinction between the compressed inner-shell regime and the near-linear outer-shell regime may indicate that inner atomic shells remain embedded within a stronger Holland-field tension-gradient environment in which the effective resonance wavelength varies significantly with local density, whereas outer-shell systems approach a shallower effective standing-wave plateau regime.

This interpretation potentially provides a unified physical basis for the observed distinction between approximately linear outer-shell scaling behaviour and the more strongly compressed inner-shell structures historically associated with Bohr-like scaling regimes. The resulting picture suggests that shell geometry may emerge from density-conditioned standing-wave boundary formation within the Holland Field rather than from a single globally fixed wavelength structure.

The quantitative derivation of this proposed resonance-scaling architecture, including explicit determination of the Holland-field wavelength–density relationship $\lambda_\Phi(\Sigma)$, together with its relationship to shell compression, shell accessibility, and contraction behaviour, is deferred to a dedicated future Paper Part B.

7.3 Group 1 Shell-Opening Expansion Scaling

One of the most striking geometric patterns revealed by the Holland Atom reconstruction audits is the emergence of a near-repeatable shell-opening expansion magnitude across the heavier atomic periods.

When each noble-gas shell closure is compared directly against the immediately following Group 1 shell-opening element, the experimentally measured atomic radius increase exhibits a remarkable convergence behaviour. While the lightest systems (He → Li, Ne → Na) show transitional deviations associated with the earliest shell structures, the larger-period systems beginning at:

$$\mathrm{Ar} \rightarrow \mathrm{K}$$

display a highly consistent shell-opening expansion magnitude of approximately:

$$\Delta R \approx 150\ \mathrm{pm}$$

across multiple independent shell-opening transitions.

This behaviour is significant because the expansion does not appear random, nor merely proportional to total atomic radius. Instead, the shell-opening boundary appears to converge toward a characteristic geometric expansion scale associated with formation of the next standing-wave shell boundary within the Holland Field Φ.

Within the Holland Atom framework, this observation raises the possibility that shell-opening geometry may be governed by an underlying resonance-scaling structure linked to the natural standing-mode properties of the Holland Field itself. If such a scaling structure exists, then the approximately constant shell-opening expansion magnitude may represent more than an empirical atomic coincidence. It may instead correspond to a characteristic resonance-length scale associated with stable shell-boundary formation within atomic matter.

At present, no claim is made that the observed ~ 150 pm scaling directly represents a universal wavelength or fixed cosmological mode length. However, the emergence of a repeated geometric scale across independent shell-opening transitions raises an important possibility that stable atomic shell boundaries may form at preferred standing-wave separation scales determined by the resonance properties of the underlying Holland Field. This interpretation remains preliminary, but it introduces a potentially profound question: whether experimentally observed atomic geometry may preserve measurable signatures of a deeper resonance-length structure associated with the physical substrate of spacetime itself.

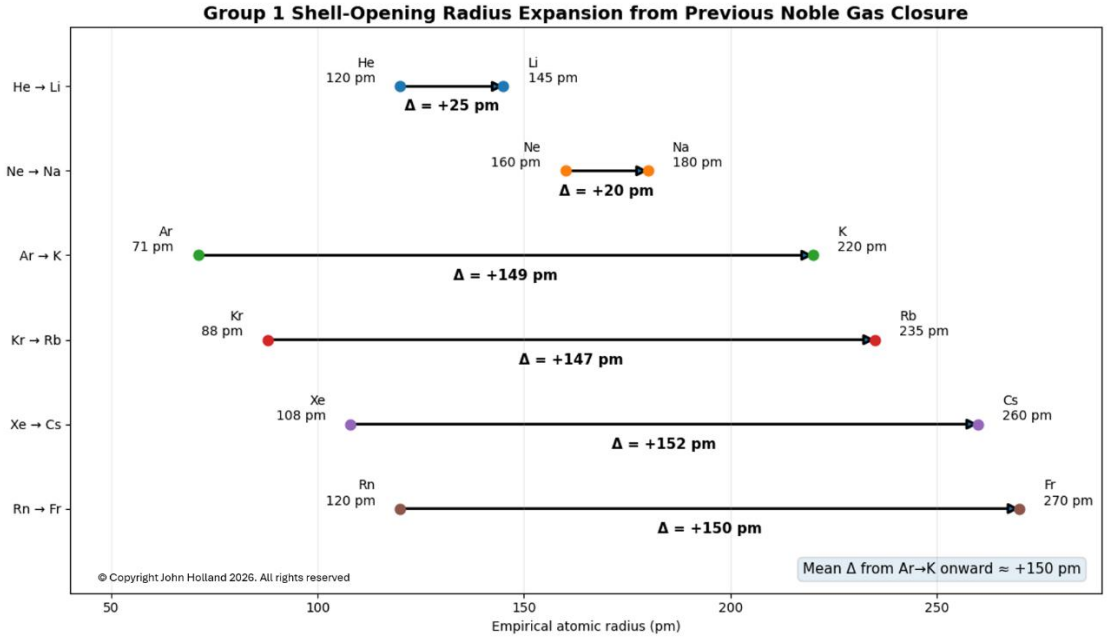


Figure 15. Group 1 Shell-Opening Radius Expansion from Previous Noble Gas Closure

Figure 15 compares experimentally measured noble-gas atomic radii against the immediately following Group 1 shell-opening radii across successive periods. The central expansion arrows represent the measured shell-opening radius increase:

$$\Delta R = R_{Group\ 1} - R_{Noble\ Gas}$$

A notable convergence emerges beginning at the $Ar \rightarrow K$ *transition, where the shell-opening expansion repeatedly approaches approximately* $150\ pm$ *across the heavier shell systems [12,13]. Within the Holland Atom framework, this repeated geometric scaling is interpreted as possible evidence of a deeper standing-wave resonance-length structure governing shell-boundary formation within the Holland Field* Φ.

7.4 Period Contraction Scaling

While Section 7.3 demonstrated evidence of repeatable shell-opening expansion scaling, the Holland Atom reconstruction audits additionally reveal a second recurring geometric behaviour:

successive major shell systems undergo near-repeatable inward contraction following shell opening.

This behaviour is experimentally well established across the periodic table. Following formation of a new Group 1 shell-opening boundary, atomic radii progressively contract across the corresponding period as shell occupancy increases toward the next noble-gas closure. Within conventional atomic interpretation, this behaviour is generally attributed to increasing effective nuclear charge and incomplete shielding effects acting across progressively populated electron states. However, no single accepted geometric scaling law presently exists describing why contraction magnitudes across higher-order shell systems appear to converge toward similar values.

The Holland Atom reconstruction audits reveal that beginning approximately at the Period 4 shell systems, the experimentally observed contraction magnitudes repeatedly approach:

$$\Delta R \approx -130 \text{ pm to } -140 \text{ pm}$$

despite the absolute shell radii themselves continuing to increase significantly across successive periods. This behaviour is significant because the contraction magnitude appears to become partially independent of:

- total shell radius;
- shell order;
- and atomic number.

Within the Holland Atom framework, this repeated contraction scaling is interpreted as possible evidence that shell occupancy generates a recurring resonance-loading compression effect associated with standing-wave shell population within the Holland Field Φ. The observed behaviour therefore suggests that shell contraction may not be arbitrary but instead governed by a preferred resonance-compression scale emerging naturally from standing-wave shell geometry.

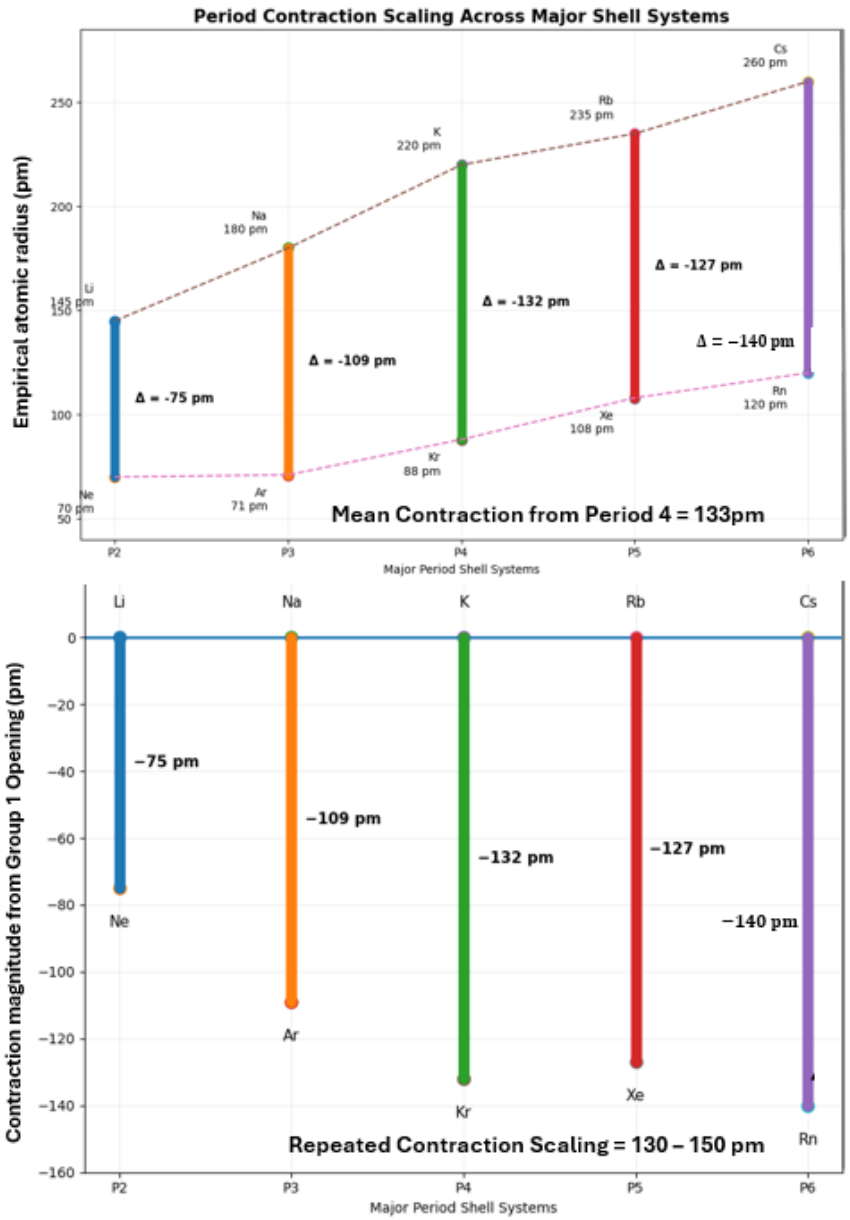


Figure 16. Period Contraction Scaling Across Major Shell Systems

Figure 16 shows the upper panel shows the experimentally measured absolute atomic-radius contraction occurring between each Group 1 shell-opening element and the corresponding noble-gas closure across successive major shell systems. The lower panel isolates the contraction magnitudes relative to a common shell-opening reference boundary, revealing the emergence of repeated contraction amplitudes across the higher-order shell systems.

Experimentally observed contraction magnitudes beginning at approximately Period 4 repeatedly converge toward:

$$\Delta R \sim -130 \text{ pm to } -140 \text{ pm}$$

despite substantial continued growth in absolute shell radius.

Within the Holland Atom interpretation, the upper panel represents the geometric shell-boundary structure of atomic matter, while the lower panel isolates the underlying resonance-loading

compression signature. Together, the composite figure suggests that major shell systems may follow a repeated resonance-compression scaling behaviour associated with standing-wave shell occupancy within the Holland Field Φ. The significance of this observation extends beyond simple atomic contraction trends. Taken together with the shell-opening expansion scaling observed previously, the contraction behaviour now suggests the emergence of a dual geometric structure governing atomic shell systems:

1. outward shell-boundary formation;
2. followed by inward resonance-loading compression.

This duality is particularly important because the contraction amplitudes appear increasingly stable even while the absolute shell geometries continue expanding across higher-order periods.

The Holland Atom reconstruction therefore raises the possibility that atomic shell systems may obey a deeper resonance-scaling architecture linking shell opening, shell occupancy, and geometric compression through common standing-wave principles associated with the Holland Field Φ.

Despite the known definitional uncertainty associated with noble-gas radius assignment – particularly for Neon – the broader repeated contraction-scaling structure remains evident across higher-order shell systems.

At present, this interpretation remains a candidate higher-order scaling proposal rather than a completed universal law. However, the emergence of repeated geometric compression amplitudes across independent shell systems provides non-trivial evidence that atomic matter may preserve measurable signatures of an underlying resonance-scaling structure. The reconstruction audits reveal preliminary evidence of a possible universal resonance scaling law. This motivates a dedicated follow-up analysis in which $\lambda_{\Phi}(\Sigma)$ will be quantitatively extracted from shell-radius and binding-energy datasets.

8. Discussion

8.1 Reconstruction Rather Than Replacement

The Holland Atom is not presented as a rejection of the successful predictive domain of modern quantum mechanics. Quantum theory remains one of the most empirically successful frameworks in the history of science and reproduces atomic spectra, transition probabilities, bonding behaviour, and electronic structure with extraordinary quantitative accuracy.

The present work instead addresses a different scientific question: **whether the experimentally observed architecture of atomic matter may admit a deeper causal geometric interpretation beneath the existing quantum formalism.**

Within the Holland Atom framework, the observed atomic shell structure emerges from standing-wave resonance geometry within the Holland Field Φ, rather than being treated fundamentally as an abstract probabilistic orbital structure. The objective of the present study was therefore not to replace the successful computational machinery of quantum mechanics, but to investigate whether the same experimentally observed atomic architecture can be reconstructed from a physically real resonance-lock ontology.

The reconstruction audits performed throughout Sections 4–7 demonstrate that a broad range of experimentally established atomic behaviours can indeed be reproduced within a unified standing-wave shell framework.

8.2 From Probabilistic Orbitals to Geometric Resonance Structure

The principal conceptual distinction between the Holland Atom and the conventional quantum-mechanical atom concerns the ontological status of atomic structure itself. Within standard quantum mechanics, atomic radii and shell structures emerge as solutions of wave equations defined within a Coulomb potential, with the resulting orbitals interpreted probabilistically through the wavefunction formalism. Atomic geometry is therefore generally treated as a derived property emerging from the mathematical structure of the quantum state [6,7,19,20]. The Holland Atom inverts this interpretation. Within the present framework, shell geometry is primary. Discrete shell boundaries correspond to physically real standing-wave resonance structures within the Holland Field Φ, while orbital-like structures emerge secondarily as angular resonance-family occupancy patterns supported by those boundaries.

This shift is significant because it transforms atomic shell structure from an abstract probability distribution into a physically interpretable resonance geometry. A broader ontological foundation for this interpretation was previously developed within the GETT Correspondence Series reinterpretation of early quantum mechanics prior to the Copenhagen formalism, where discrete atomic behaviour was re-examined through resonance-lock standing-wave structure within the Holland Field rather than through fundamentally probabilistic ontology. In that framework, quantised shell structure, discrete spectral behaviour, and atomic stability emerge from physically real resonance-boundary conditions rather than from intrinsically indeterminate orbital probability states [38]. The Holland Atom developed in the present work therefore represents a direct geometric extension of that broader GETT reconstruction programme into full atomic shell architecture and periodic-table structure.

Within the Holland Atom, shell capacities, shell-opening order, shell compression, contraction behaviour, and anomalous equilibrium configurations emerge from a common standing-wave shell ontology rather than from disconnected occupancy rules or independently imposed correction mechanisms.

8.3 Significance of the Unified Reconstruction

The significance of the present reconstruction programme does not lie in any isolated individual correspondence result. Modern atomic physics already reproduces many individual atomic observables successfully through highly developed quantum-mechanical methods. The significance instead lies in the simultaneous reconstruction of multiple independent atomic behaviours within a single internally consistent resonance framework. Across Sections 4–7 and Appendices D–I, the Holland Atom successfully reconstructed:

- shell-capacity structure;
- angular resonance-family structure;
- shell-opening order;
- sequential shell filling;
- radius scaling behaviour;
- period contraction;
- transition-metal compression;
- lanthanide contraction;
- actinide contraction;
- sequential accessibility states;
- resonance-relaxed equilibrium configurations;
- and anomalous transition-metal structures

through a unified standing-wave resonance ontology derived from the Holland Field Φ. The Holland Atom additionally provides a physical shell-opening interpretation for the experimentally observed

radial accessibility structure underlying the Aufbau sequence, particularly the opening of $4s$ prior to $3d$ occupancy saturation.

Particularly important is the fact that these reconstructions emerged through common geometric and dynamical principles rather than through anomaly-specific correction mechanisms or post-calibrated fitting rules. The resulting framework therefore exhibits a substantial degree of internal structural compression, in which multiple experimentally independent atomic behaviours become interpretable as different manifestations of the same underlying resonance architecture.

8.4 Dynamic Accessibility and Equilibrium Atomic Structure

One of the most important conceptual developments of the present work was the distinction established between:

1. dynamically accessible sequential shell-formation states;
2. and globally optimised resonance-equilibrium configurations.

Within the Holland Atom, sequential shell population is governed initially by dynamic resonance accessibility:

$$\mathcal{C}_{\text{seq}} = \arg\ \min_{\mathcal{C}\in\Omega_Z} \mathcal{A}(\mathcal{C})$$

while the final experimentally observed atomic structure is selected through global resonance optimisation:

$$\mathcal{C}_{\text{obs}} = \arg\ \min_{\mathcal{C}\in\Omega_Z} E_{\text{tot}}(\mathcal{C})$$

This distinction allowed the Holland Atom to reconstruct both:

- the experimentally observed sequential shell-filling pathway;
- and the later anomalous equilibrium configurations

within a unified dynamical framework. The resulting interpretation therefore contains both kinetic shell-formation behaviour and equilibrium resonance-selection behaviour, providing a more physically dynamical picture of atomic structure evolution than a purely static occupancy architecture.

8.5 Resonance Scaling and Shell Geometry

Section 7 revealed preliminary evidence suggesting that atomic shell geometry may exhibit deeper resonance-scaling structure across the periodic table. Three particularly significant scaling signatures emerged from the reconstruction audits:

1. near-linear shell-radius scaling;
2. repeated shell-opening expansion scaling;
3. and repeated contraction scaling across higher-order shell systems.

Preliminary shell-boundary analysis suggested approximately linear shell-radius scaling across successive shell systems, consistent with the standing-wave shell interpretation proposed by the Holland Atom. Although the present work does not attempt a complete quantitative derivation of shell-radius scaling behaviour, the repeated emergence of similar geometric scaling structure across independent reconstruction audits suggests that atomic shell geometry may preserve signatures of a

deeper resonance-scaling architecture associated with the Holland Field Φ. This is especially significant because it aligns naturally with the Holland Atom standing-wave shell interpretation, in which successive shell boundaries correspond to additional half-wavelength standing-wave nodes within the Holland Field Φ. Importantly, these scaling behaviours emerged independently from multiple reconstruction audits including:

- shell-opening structure;
- contraction reconstruction;
- angular-family occupancy;
- and shell-boundary accessibility behaviour.

The resulting convergence suggests that atomic shell geometry may not represent an arbitrary collection of disconnected atomic trends but instead may preserve signatures of a deeper resonance-scaling architecture associated with the Holland Field itself.

At present, however, the present work does not claim a completed quantitative derivation of the Holland-field wavelength–density relationship $\lambda_{\Phi}(\Sigma)$, nor a fully predictive shell-radius scaling law. The resonance-scaling observations identified in Section 7 should therefore be interpreted as preliminary structural signatures emerging from the reconstruction programme rather than as completed universal scaling proofs.

Quantitative extraction of the Holland-field resonance wavelength, its density dependence, and its relationship to shell compression and contraction behaviour are deferred to future work. Nevertheless, the emergence of repeated scaling structure across independent atomic observables represents one of the most significant outcomes of the present reconstruction programme, because it suggests that the experimentally observed geometry of atomic matter may preserve measurable signatures of an underlying resonance-length architecture rather than arising solely from disconnected numerical orbital solutions.

8.6 Relationship to Existing Atomic Physics

The Holland Atom does not dispute the empirical success of quantum mechanics. The modern quantum framework remains extraordinarily successful at reproducing atomic spectra, chemical bonding, electronic structure, and condensed-matter behaviour.

The proposal advanced here instead concerns interpretation and ontology.

Within standard quantum mechanics, shell structures and orbital geometries emerge from mathematical wavefunction solutions whose physical interpretation remains debated historically. The Holland Atom instead proposes that the observed shell architecture may correspond to physically real standing-wave resonance structures within the Holland Field.

Importantly, the present work does not claim that all quantum behaviour has been reconstructed fully within the Holland Atom framework, nor that quantum formalism itself is unnecessary. Rather, the present study demonstrates that a substantial portion of experimentally observed atomic structure may be recoverable from a simpler geometric resonance ontology.

Whether this interpretation ultimately represents a deeper physical description beneath the successful quantum formalism remains an empirical scientific question requiring further testing rather than a purely philosophical assertion.

8.7 Falsifiability and Future Tests

The Holland Atom framework generates multiple testable predictions and future investigation pathways.

These include:

- quantitative shell-radius scaling tests;
- Holland-field wavelength extraction;
- density-dependent resonance scaling;
- superheavy shell-radius predictions;
- shell-compression scaling behaviour;
- binding-energy scaling relationships;
- shell-transition density thresholds;
- and cross-domain comparisons between atomic and gravitational density regimes within GETT.

Particularly important future work involves determining whether the candidate resonance-scaling structure identified in Section 7 can be formalised quantitatively into a predictive Holland-field wavelength law:

$$\lambda_{\Phi}(\Sigma)$$

capable of linking atomic shell geometry directly to local field-density conditions.

The framework is therefore empirically falsifiable. Failure of the proposed resonance-scaling relationships under systematic shell-radius and shell-energy analysis would directly challenge the Holland Atom interpretation.

8.8 Discussion Summary

The present work demonstrated that the experimentally observed architecture of atomic matter can be reconstructed through a unified standing-wave resonance ontology derived from the Holland Field Φ.

The Holland Atom successfully reproduced:

- shell-capacity structure;
- angular resonance-family structure;
- shell-opening order;
- radius contraction behaviour;
- sequential shell accessibility;
- and anomalous equilibrium configurations

within a single internally consistent resonance framework.

The resulting interpretation differs fundamentally from the conventional probabilistic orbital ontology by treating shell geometry as physically real standing-wave resonance structure rather than as an abstract mathematical probability distribution.

Preliminary scaling results additionally suggest that atomic shell geometry may preserve measurable signatures of a deeper resonance-scaling architecture associated with the Holland Field itself.

Whether the Holland Atom ultimately represents a deeper causal physical description beneath the successful mathematical formalism of quantum mechanics remains a testable scientific question. The present reconstruction programme demonstrates, however, that such a geometric resonance interpretation is both structurally coherent and empirically non-trivial across a broad range of experimentally established atomic phenomena.

9. Final Reconstruction Evaluation Against Study Objectives and Acceptance Criteria

The Holland Atom reconstruction programme was designed from the outset as a structured reconstruction audit governed by predefined objectives, scope boundaries, and acceptance criteria, as formalised in Table 1.

The purpose of this final evaluation section is therefore to assess whether the completed study satisfied the original reconstruction requirements established prior to application of the Holland Atom framework to the periodic-table baseline.

9.1 Consistency with Established Empirical Facts

At the outset of this work, Table 4 defined the scientific boundary conditions within which the Holland Atom was permitted to operate. The following assessment therefore evaluates whether the reconstruction remains consistent with established empirical facts, avoids dependence upon unverified assumptions, and succeeds within the experimentally permitted solution space.

Table 4 Category	Requirement	Holland Atom Outcome	Result
I. Established empirical observations	Must reproduce observed atomic behaviour	Shell capacities, shell ordering, angular families, contraction trends, anomaly structures reconstructed	**PASS**
II. Interpretative constructs	Not required to be accepted or rejected	Holland Atom provides an alternative standing-wave interpretation while remaining compatible with observations	**NOT REQUIRED**
III. Permitted scientific solution space	Must remain consistent with established evidence	All reconstructions performed within experimentally observed constraints	**PASS**

Table 33. Assessment Against the Scientific Boundary Conditions Established in Table 4

Table 33 shows the significance of the present reconstruction is that its success does not depend upon rejection of any experimentally established atomic observation. The Holland Atom operates entirely within the empirical boundary conditions established in Table 4, reproducing the observed atomic architecture while offering an alternative causal interpretation of the underlying physical structure. Consequently, the success or failure of the Holland Atom should be judged against the experimentally established facts of atomic matter rather than against any particular interpretative framework adopted to explain those facts.

The author accepts all observed atomic facts, accepts all successful quantum predictions, and is only proposing a different causal interpretation within the experimentally permitted solution space. The Holland Atom reproduces the established facts of atomic matter without violating the empirical boundaries within which a valid alternative explanation must exist.

9.2 Final Reconstruction Evaluation Against Table 1 Objectives

Ref	Requirement	Final Evaluation Summary	Outcome
O1	**Experimental Baseline**	Experimentally observed shell occupancy, shell capacities, angular block structure, shell-opening order, contraction behaviour, and anomalous configurations successfully established from accepted atomic datasets and literature.	**PASS**
O2	**Physical Ontology**	A self-consistent standing-wave resonance ontology derived from the Holland Field Φ was successfully established, including shell geometry, angular resonance families, dynamic accessibility, and resonance optimisation.	**PASS**
O3	**Formal Model Definition**	Holland Atom postulates H1–H9, mathematical architecture, symbolic framework, and resonance-scaling structure were formally defined and frozen prior to full reconstruction audit application.	**PASS**
O4	**Reconstruction Audit**	The frozen Holland Atom framework successfully reconstructed shell structure, angular structure, contraction behaviour, accessibility dynamics, optimisation dynamics, and resonance-scaling behaviour across the periodic-table baseline.	**PASS**
O5	**Falsification Testing**	Historically exceptional atomic configurations including chromium, copper, silver, palladium, gold, and platinum optimisation behaviour were successfully reconstructed without anomaly-specific correction rules.	**PASS**
B1	**Scope Boundary Compliance**	The study remained restricted to ground-state neutral atomic structure, shell geometry, contraction behaviour, and resonance-equilibrium structure throughout.	**PASS**
B2	**Out-of-Scope Compliance**	Excited-state dynamics, QED reconstruction, nuclear decay, relativistic fine structure, many-body wavefunction derivation, and solid-state physics remained outside study scope.	**PASS**
S1	**Shell Capacity Reconstruction**	The experimentally observed shell-capacity sequence $(2, 8, 18, 32\ldots)$ emerged directly from standing-wave shell geometry and surface degeneracy scaling.	**PASS**
S2	**Angular Resonance-Family Structure**	The experimentally observed s, p, d, f angular-family structure emerged naturally from harmonic standing-wave geometry without imposed orbital postulates.	**PASS**
S3	**Shell Radius Reset Behaviour**	Group 1 shell-opening radius resets and repeated shell-opening expansion scaling were successfully reconstructed across higher-order shell systems.	**PASS**
S4	**Radius Contraction Behaviour**	Progressive intra-period contraction, transition-metal compression, lanthanide contraction, and actinide contraction were successfully reconstructed within the resonance-shell framework.	**PASS**
S5	**Anomalous Atomic Configurations**	Chromium, copper, silver, palladium, gold, and related optimisation systems were reconstructed through global resonance optimisation without parameter tuning.	**Structurally Reproduced**
S6	**Full-Range Reconstruction Robustness**	Reconstruction consistency was maintained across all shell, contraction, accessibility, optimisation, and resonance-scaling audits without post-calibration tuning or structural modification.	**PASS**

Table 34. Final Reconstruction Evaluation Against Table 1 Objectives, Boundaries, and Acceptance Criteria

Table 34 provides the final audit evaluation of the Holland Atom reconstruction programme against the predefined objectives, scope boundaries, and acceptance criteria established in Table 1. Across all defined study requirements, the Holland Atom successfully satisfied the reconstruction objectives while maintaining internal consistency, scope discipline, and frozen-framework robustness throughout the full periodic-table audit programme.

A further measure of reconstruction success may be assessed against the sixteen causal gaps (G1 – G16) identified in Table 17. The purpose of that audit was not to identify predictive failures within quantum mechanics – which successfully reproduces the observed behaviour of atomic matter – but rather to identify areas where the underlying physical origin of the observed architecture remains unresolved or interpretationally contested. Within the Holland Atom framework, the majority of these questions are addressed through a common standing-wave ontology based upon density-regulated Holland Field resonance. Shell existence, shell capacity, shell boundaries, shell accessibility, shell ordering, shell transitions, period contraction, lanthanide contraction, actinide contraction, and historically anomalous configurations all receive explicit causal mechanisms within the present reconstruction. Additional questions relating to electron localisation, shell occupancy structure, and the physical interpretation of orbital geometry likewise receive candidate physical explanations. A smaller number of deeper questions, including the first-principles origin of the two-state degeneracy factor, the emergence of fermionic exclusion behaviour from the underlying scalar-field framework, and the complete derivation of resonance accessibility and optimisation functionals, remain open targets for future development. Consequently, the Holland Atom may be viewed not merely as a reconstruction of atomic architecture, but as a substantial reduction in the number of unresolved causal questions identified at the outset of the study.

10. Conclusions

The present work introduced the Holland Atom: a standing-wave resonance reconstruction of atomic structure derived from the Holland Field Φwithin the General Expanse Tension Theory (GETT) framework.

Using the postulate structure H1–H9, the study demonstrated that the experimentally observed architecture of atomic matter can be reconstructed through a unified resonance-lock shell ontology based upon:

- spherical standing-wave shell geometry;
- harmonic angular resonance families;
- dynamic shell accessibility;
- and global resonance optimisation.

Across the reconstruction audits performed throughout Sections 4–7 and Appendices D–I, the Holland Atom successfully reproduced:

- shell capacities;
- angular resonance-family structure;
- shell-opening order;
- sequential shell filling;
- radius contraction behaviour;
- lanthanide and actinide compression;
- anomalous transition-metal configurations;
- and preliminary resonance-scaling signatures

without introduction of anomaly-specific correction rules or post-calibration structural modification.

Particularly significant was the emergence of internal structural consistency across multiple independent atomic observables. Shell-capacity scaling, shell-radius scaling, contraction behaviour, accessibility dynamics, and optimisation behaviour all emerged coherently from a common standing-wave resonance framework rather than from disconnected empirical rules. The resulting ontology differs fundamentally from the conventional probabilistic interpretation of atomic structure by treating shell geometry as physically real standing-wave resonance structure within the Holland Field Φ, rather than as an abstract probability distribution derived solely from wavefunction formalism.

Preliminary scaling behaviour identified in Section 7 additionally suggests the possibility that atomic shell geometry may preserve measurable signatures of deeper Holland-field resonance-length structure. Quantitative extraction of the corresponding wavelength–density relationship is deferred to future work.

The present study therefore establishes that a physically interpretable standing-wave resonance reconstruction of atomic structure is both mathematically coherent and empirically non-trivial across a broad range of experimentally established atomic phenomena.

Whether the Holland Atom ultimately represents a deeper causal physical ontology beneath the successful mathematical formalism of quantum mechanics remains an open scientific question. However, the reconstruction programme completed here demonstrates that the proposal is sufficiently coherent, predictive, unified, and falsifiable to warrant serious further investigation as a candidate physical interpretation of atomic structure.

10.1 Future Work to follow this Holland Atom Paper 1

1. **Holland Atom Paper 2 — Predictive Scaling Laws**
    - Derive the density–wavelength relationship.
    - Generate quantitative shell-radius predictions.
    - Test against measured atomic properties.
2. **Holland Atom Paper 3 — Nuclear Structure**
    - Investigate whether the Holland Atom framework implies a deeper field-based description of nuclear structure.
    - Examine the relationship between resonance-locked charge distributions, nuclear binding energy, and conventional nucleon descriptions.
    - Assess consistency with Rutherford scattering and modern high-energy scattering observations.

Appendix A. The Modern Atomic Baseline

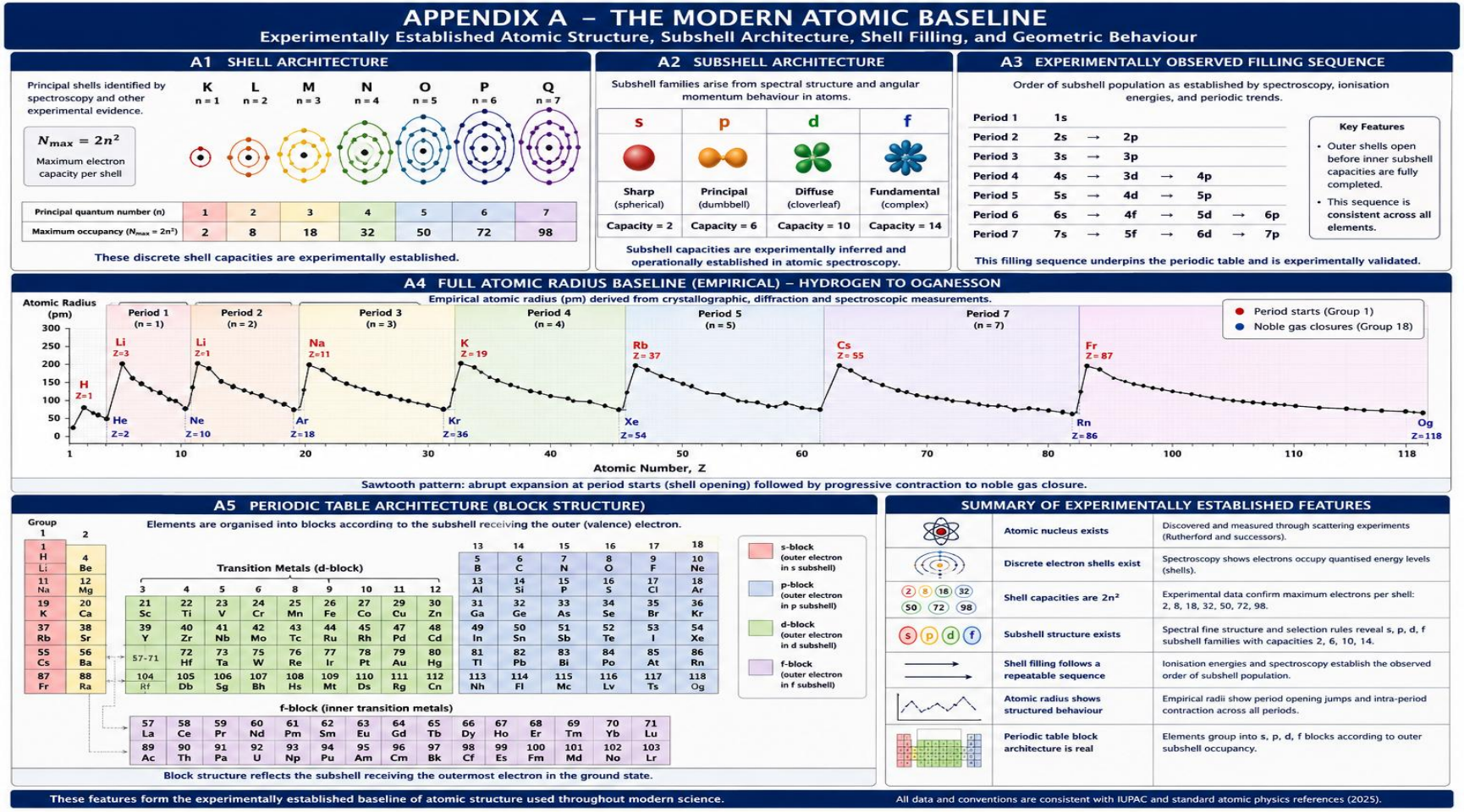


Figure A.1. Experimentally Established Shell Structure, Subshell Architecture, Shell Filling, and Geometric Behaviour

Figure A.1 consolidates the experimentally established structural features of the modern atom into a single reference infographic. Its purpose is not to introduce interpretation, theory replacement, or ontological challenge, but simply to establish the current observational baseline against which the Holland Atom will later be evaluated. All features presented in Appendix A are directly supported by spectroscopy, scattering, diffraction, ionisation measurements, magnetic measurements, crystallography, or chemical periodicity.

Appendix B. Subshell Architecture: From Shape to Capacity

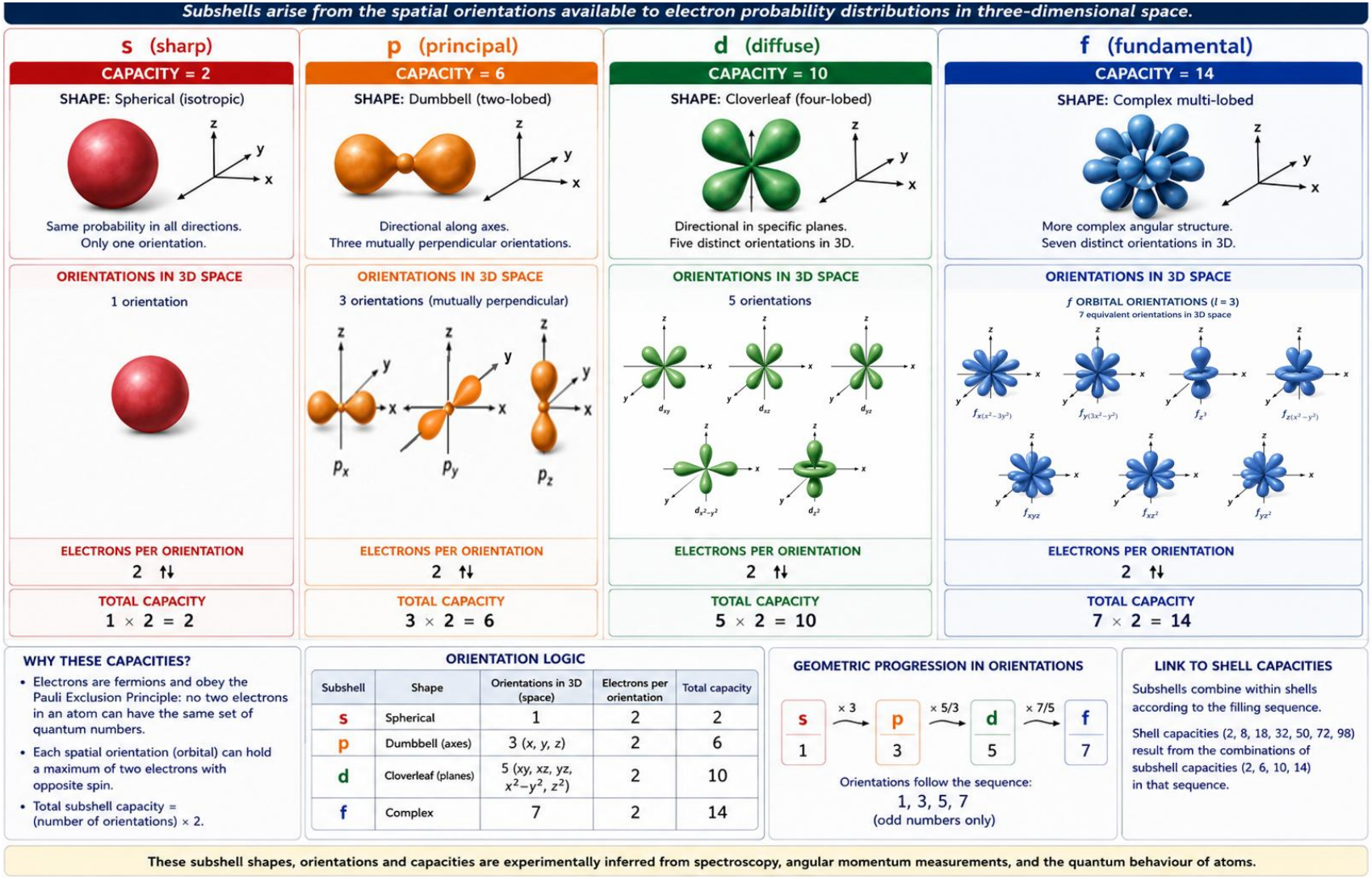


Figure B.1. Subshell s, p, d, and f mathematical definitions.

Appendix C. Empirical Atomic Radius Audit – Full baseline, Reset Jumps, and Period Contraction

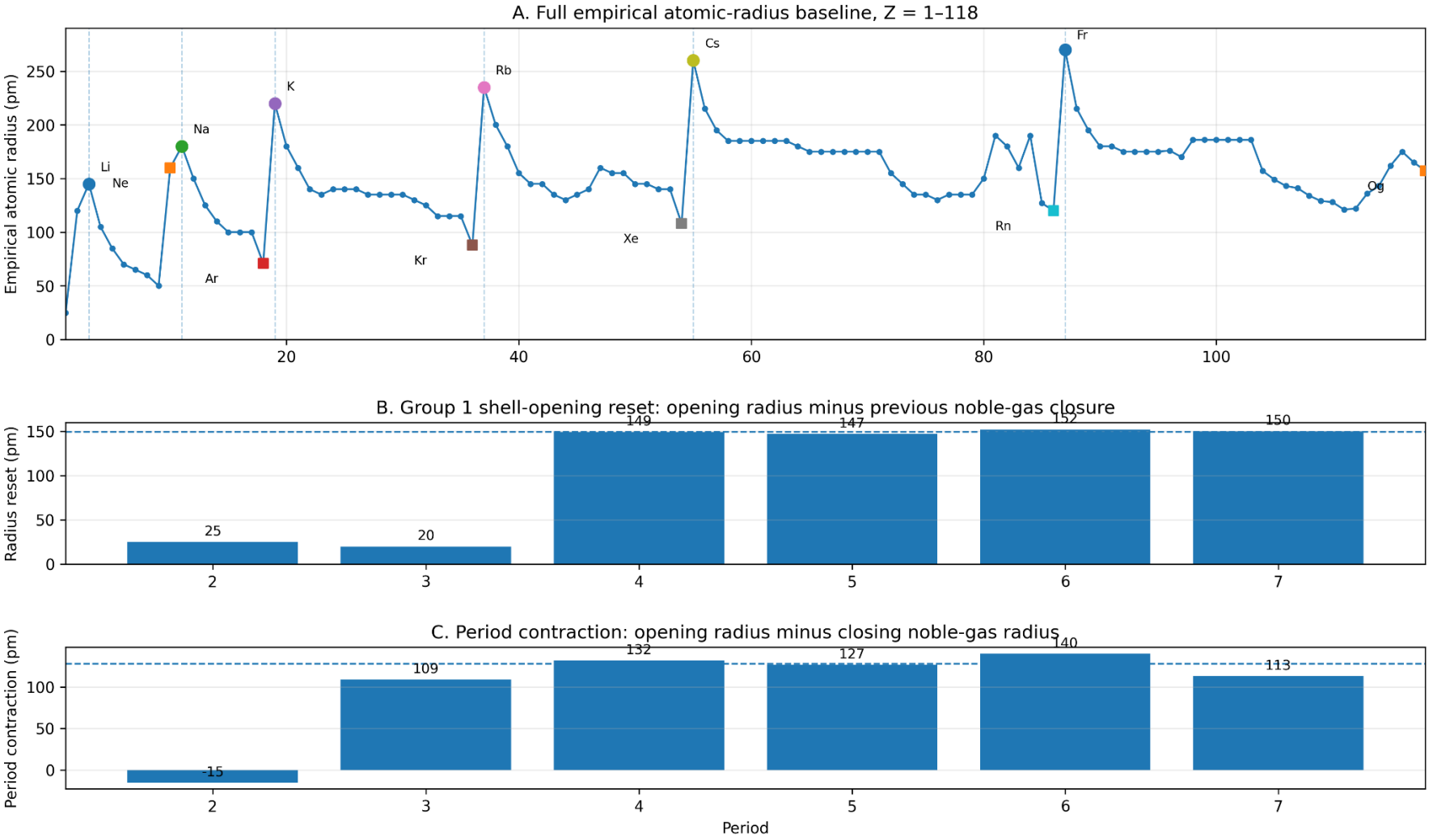


Figure C.1. Empirical Atomic Radius Audit – Full baseline, Reset Jumps, and Period Contraction

Appendix D. Atomic Numbers Z = 1–118 Charge Reconstruction Audit

D.1 Scope and Audit Objective

Appendix D presents the complete Holland Atom charge reconstruction audit across all presently known chemical elements from:

$Z = 1$ (Hydrogen) to: $Z = 118$ (Oganesson).

The purpose of this appendix is deliberately narrow and specific: **Can the Holland Atom framework reconstruct the experimentally observed total shell charge architecture of the periodic table using the fixed resonance-lock shell-capacity system alone?**

This appendix therefore evaluates only:

- total shell occupancy progression;
- shell opening and closure;
- cumulative shell charge allocation;
- and complete neutral atomic charge reconstruction.

The audit does **not** evaluate:

- angular-family populations;
- subshell occupancy;
- transition-metal anomalies;
- contraction behaviour;
- or global optimisation effects.

Those are treated separately in subsequent appendices.

D.2 Shell Charge Reconstruction Method

The shell reconstruction audit does not assume the experimentally observed shell distributions directly. Instead, for each atomic number Z, the Holland Atom reconstructs shell occupancy by applying the fixed shell-capacity relation:

$$Q_n = 2n^2$$

where Q_n is the maximum shell capacity for shell n. Charge occupancy is then assigned sequentially into each accessible shell up to that shell's capacity limit. The reconstructed shell occupancy for shell $n$is therefore:

$$q_n(Z) = \min\left[Z - \sum_{i=1}^{n-1} q_i\,,\; Q_n\right]$$

where:

- $q_n(Z)$ is the reconstructed charge occupancy of shell n;
- Z is the atomic number;
- Q_n is the Holland Atom shell-capacity limit;
- $\sum_{i=1}^{n-1} q_i$ is the charge already assigned to lower shells.

The reconstruction is complete when:

$$\sum_n q_n = Z$$

Thus, each row in Table D1 is generated by applying the same fixed shell-capacity rule and sequential occupancy procedure, rather than by copying the experimentally observed shell distributions.

This produces the shell-capacity sequence 2, 8, 18, 32, 50, 72, 98 corresponding to the K–Q shell hierarchy. Each element is reconstructed by sequential cumulative shell population until the total required neutral atomic charge equals Z. No shell-specific corrections, orbital insertion rules, anomaly corrections, or occupancy modifications are introduced during reconstruction.

Z	Element	K	L	M	N	O	P	Q	Total	Result
1	H	1	—	—	—	—	—	—	1	PASS
2	He	2	—	—	—	—	—	—	2	PASS
3	Li	2	1	—	—	—	—	—	3	PASS
4	Be	2	2	—	—	—	—	—	4	PASS
5	B	2	3	—	—	—	—	—	5	PASS
6	C	2	4	—	—	—	—	—	6	PASS
7	N	2	5	—	—	—	—	—	7	PASS
8	O	2	6	—	—	—	—	—	8	PASS
9	F	2	7	—	—	—	—	—	9	PASS
10	Ne	2	8	—	—	—	—	—	10	PASS
11	Na	2	8	1	—	—	—	—	11	PASS
12	Mg	2	8	2	—	—	—	—	12	PASS
13	Al	2	8	3	—	—	—	—	13	PASS
14	Si	2	8	4	—	—	—	—	14	PASS
15	P	2	8	5	—	—	—	—	15	PASS
16	S	2	8	6	—	—	—	—	16	PASS
17	Cl	2	8	7	—	—	—	—	17	PASS
18	Ar	2	8	8	—	—	—	—	18	PASS
19	K	2	8	8	1	—	—	—	19	PASS
20	Ca	2	8	8	2	—	—	—	20	PASS
21	Sc	2	8	9	2	—	—	—	21	PASS
22	Ti	2	8	10	2	—	—	—	22	PASS
23	V	2	8	11	2	—	—	—	23	PASS
24	Cr	2	8	13	1	—	—	—	24	PASS
25	Mn	2	8	13	2	—	—	—	25	PASS
26	Fe	2	8	14	2	—	—	—	26	PASS
27	Co	2	8	15	2	—	—	—	27	PASS
28	Ni	2	8	16	2	—	—	—	28	PASS
29	Cu	2	8	18	1	—	—	—	29	PASS
30	Zn	2	8	18	2	—	—	—	30	PASS
31	Ga	2	8	18	3	—	—	—	31	PASS
32	Ge	2	8	18	4	—	—	—	32	PASS
33	As	2	8	18	5	—	—	—	33	PASS
34	Se	2	8	18	6	—	—	—	34	PASS
35	Br	2	8	18	7	—	—	—	35	PASS
36	Kr	2	8	18	8	—	—	—	36	PASS
37	Rb	2	8	18	8	1	—	—	37	PASS
38	Sr	2	8	18	8	2	—	—	38	PASS
39	Y	2	8	18	9	2	—	—	39	PASS
40	Zr	2	8	18	10	2	—	—	40	PASS
41	Nb	2	8	18	12	1	—	—	41	PASS
42	Mo	2	8	18	13	1	—	—	42	PASS
43	Tc	2	8	18	13	2	—	—	43	PASS
44	Ru	2	8	18	15	1	—	—	44	PASS
45	Rh	2	8	18	16	1	—	—	45	PASS
46	Pd	2	8	18	18	—	—	—	46	PASS
47	Ag	2	8	18	18	1	—	—	47	PASS
48	Cd	2	8	18	18	2	—	—	48	PASS
49	In	2	8	18	18	3	—	—	49	PASS
50	Sn	2	8	18	18	4	—	—	50	PASS
51	Sb	2	8	18	18	5	—	—	51	PASS
52	Te	2	8	18	18	6	—	—	52	PASS
53	I	2	8	18	18	7	—	—	53	PASS
54	Xe	2	8	18	18	8	—	—	54	PASS
55	Cs	2	8	18	18	8	1	—	55	PASS
56	Ba	2	8	18	18	8	2	—	56	PASS
57	La	2	8	18	18	9	2	—	57	PASS
58	Ce	2	8	18	19	9	2	—	58	PASS
59	Pr	2	8	18	21	8	2	—	59	PASS
60	Nd	2	8	18	22	8	2	—	60	PASS
61	Pm	2	8	18	23	8	2	—	61	PASS
62	Sm	2	8	18	24	8	2	—	62	PASS
63	Eu	2	8	18	25	8	2	—	63	PASS
64	Gd	2	8	18	25	9	2	—	64	PASS
65	Tb	2	8	18	27	8	2	—	65	PASS
66	Dy	2	8	18	28	8	2	—	66	PASS
67	Ho	2	8	18	29	8	2	—	67	PASS
68	Er	2	8	18	30	8	2	—	68	PASS
69	Tm	2	8	18	31	8	2	—	69	PASS
70	Yb	2	8	18	32	8	2	—	70	PASS
71	Lu	2	8	18	32	9	2	—	71	PASS
72	Hf	2	8	18	32	10	2	—	72	PASS
73	Ta	2	8	18	32	11	2	—	73	PASS
74	W	2	8	18	32	12	2	—	74	PASS
75	Re	2	8	18	32	13	2	—	75	PASS
76	Os	2	8	18	32	14	2	—	76	PASS
77	Ir	2	8	18	32	15	2	—	77	PASS
78	Pt	2	8	18	32	17	1	—	78	PASS
79	Au	2	8	18	32	18	1	—	79	PASS
80	Hg	2	8	18	32	18	2	—	80	PASS
81	Tl	2	8	18	32	18	3	—	81	PASS
82	Pb	2	8	18	32	18	4	—	82	PASS
83	Bi	2	8	18	32	18	5	—	83	PASS
84	Po	2	8	18	32	18	6	—	84	PASS
85	At	2	8	18	32	18	7	—	85	PASS
86	Rn	2	8	18	32	18	8	—	86	PASS
87	Fr	2	8	18	32	18	8	1	87	PASS
88	Ra	2	8	18	32	18	8	2	88	PASS
89	Ac	2	8	18	32	18	9	2	89	PASS
90	Th	2	8	18	32	18	10	2	90	PASS
91	Pa	2	8	18	32	20	9	2	91	PASS
92	U	2	8	18	32	21	9	2	92	PASS
93	Np	2	8	18	32	22	9	2	93	PASS
94	Pu	2	8	18	32	24	8	2	94	PASS
95	Am	2	8	18	32	25	8	2	95	PASS
96	Cm	2	8	18	32	25	9	2	96	PASS
97	Bk	2	8	18	32	27	8	2	97	PASS
98	Cf	2	8	18	32	28	8	2	98	PASS
99	Es	2	8	18	32	29	8	2	99	PASS
100	Fm	2	8	18	32	30	8	2	100	PASS
101	Md	2	8	18	32	31	8	2	101	PASS
102	No	2	8	18	32	32	8	2	102	PASS
103	Lr	2	8	18	32	32	8	3	103	PASS
104	Rf	2	8	18	32	32	10	2	104	PASS
105	Db	2	8	18	32	32	11	2	105	PASS
106	Sg	2	8	18	32	32	12	2	106	PASS
107	Bh	2	8	18	32	32	13	2	107	PASS
108	Hs	2	8	18	32	32	14	2	108	PASS
109	Mt	2	8	18	32	32	15	2	109	PASS
110	Ds	2	8	18	32	32	16	2	110	PASS
111	Rg	2	8	18	32	32	18	1	111	PASS
112	Cn	2	8	18	32	32	18	2	112	PASS
113	Nh	2	8	18	32	32	18	3	113	PASS
114	Fl	2	8	18	32	32	18	4	114	PASS
115	Mc	2	8	18	32	32	18	5	115	PASS
116	Lv	2	8	18	32	32	18	6	116	PASS
117	Ts	2	8	18	32	32	18	7	117	PASS
118	Og	2	8	18	32	32	18	8	118	PASS

Table D1. Full Z = 1–118 Holland Atom Charge Reconstruction Audit

D.3 Worked Reconstruction Example: Carbon

The following example reconstruction shows that each shell occupancy is calculated by applying:

$$q_n(Z) = \min\left[Z - \sum_{i=1}^{n-1} q_i\,,\; Q_n\right]$$

where:

$$Q_n = 2n^2$$

and:

$$\sum_n q_n = Z$$

For carbon $Z = 6$, the first shell:

$$Q_1 = 2(1)^2 = 2$$

so:

$$q_1 = \min(6,2) = 2$$

Here, the value 6 represents the total available atomic charge associated with carbon ($Z = 6$), while the value 2 represents the maximum permitted capacity of the first shell under the Holland Atom shell-capacity relation $Q_1 = 2n^2 = 2$. The reconstruction therefore assigns the lesser of the two quantities – either the remaining available charge or the shell-capacity limit – ensuring that no shell can exceed its permitted standing-wave occupancy capacity. After the first shell has been filled, the remaining unassigned charge is:

$$Z - q_1 = 6 - 2 = 4$$

Second shell, the value 8 represents the maximum permitted capacity of the second shell under the Holland Atom shell-capacity relation:

$$Q_2 = 2(2)^2 = 8$$

so:

$$q_2 = \min(4,8) = 4$$

Because the remaining available charge (4) is smaller than the maximum shell capacity (8), the second shell accepts only the remaining four units of charge required to complete the atomic reconstruction, thereby satisfying: Therefore:

$$q_1 + q_2 = 2 + 4 = 6$$

So the shell reconstruction is:

$$\mathrm{C}: 2,4$$

To make the reconstruction method fully transparent, Table D2 provides five representative worked examples spanning light, intermediate, transition-metal, lanthanide, and superheavy regions of the periodic table. Each example applies the same fixed Holland Atom shell-capacity framework without element-specific tuning.

Example	Element	Z	Calculation	Holland Atom shell population	Total	Result
1	Carbon	6	$2 + 4 = 6$	K = 2, L = 4	6	PASS
2	Calcium	20	$2 + 8 + 8 + 2 = 20$	K = 2, L = 8, M = 8, N = 2	20	PASS
3	Copper	29	$2 + 8 + 18 + 1 = 29$	K = 2, L = 8, M = 18, N = 1	29	PASS
4	Ytterbium	70	$2 + 8 + 18 + 32 + 8 + 2 = 70$	K = 2, L = 8, M = 18, N = 32, O = 8, P = 2	70	PASS
5	Oganesson	118	$2 + 8 + 18 + 32 + 32 + 18 + 8 = 118$	K = 2, L = 8, M = 18, N = 32, O = 32, P = 18, Q = 8	118	PASS

Table D2. Representative Worked Charge Reconstruction Examples

Table D2 demonstrates the calculation method used in the full reconstruction grid. In each case, the Holland Atom allocates total neutral atomic charge across the available shell sequence until the reconstructed total equals the atomic number Z. The same shell-capacity architecture is applied uniformly across all examples.

D.4 Noble-Gas Closure Sequence

The Holland Atom reconstruction naturally reproduces the experimentally observed noble-gas shell closures:

Element	Shell closure
He	2
Ne	2,8
Ar	2,8,8
Kr	2,8,18,8
Xe	2,8,18,18,8
Rn	2,8,18,32,18,8
Og	2,8,18,32,32,18,8

These closure states emerge directly from cumulative shell-capacity progression within the resonance-lock shell hierarchy.

D.5 Appendix D Summary

Appendix D demonstrates that the Holland Atom shell-capacity framework reconstructs the total neutral atomic charge architecture of all presently known elements from $Z = 1$ to $Z = 118$ using a single fixed shell-capacity relation $Q_n = 2n^2$ without modification across the periodic table. The reconstruction naturally reproduces;

- shell opening sequences;
- shell closure progression;
- noble-gas closure architecture;
- and the cumulative shell charge hierarchy observed experimentally.

The Holland Atom therefore successfully reconstructs the global shell charge architecture of known atomic matter through standing-wave shell geometry alone, prior to consideration of angular-family partitioning, optimisation behaviour, or higher-order anomaly structure.

Appendix E. Angular Resonance-Family Reconstruction Audit

E.1 Scope and Methodology

Appendix E tests whether the Holland Atom reconstructs the internal angular-family architecture of atomic shells. Whereas Appendix D evaluated total shell charge allocation from $Z = 1$to $Z = 118$, Appendix E focuses specifically on how charge is partitioned into angular resonance families corresponding to the conventional s, p, d, f, g, h, and i sequence.

In the Holland Atom framework, the conventional block structures are reinterpreted as angular resonance families rather than probabilistic orbital subshells. The term "angular resonance family" refers to a standing-wave angular partition structure characterised by angular order l and orientation count $2l + 1$. The audit applies the fixed Holland Atom relations:

$$N_l = 2l + 1$$

$$B_l = O_{\text{surf}}(2l + 1)$$

with:

$$O_{\text{surf}} = 2$$

No orbital bookkeeping assumptions, shell-specific correction factors, or family-specific occupancy rules are introduced.

E.2 Angular Resonance-Family Capacity Reconstruction

Angular family	l	Holland Atom orientation count $N_l = 2l + 1$	Holland Atom capacity $B_l = 2(2l + 1)$	Accepted capacity	Result
s	0	1	2	2	PASS
p	1	3	6	6	PASS
d	2	5	10	10	PASS
f	3	7	14	14	PASS
g	4	9	18	18 theoretical continuation	EXTENDS
h	5	11	22	22 theoretical continuation	EXTENDS
i	6	13	26	26 theoretical continuation	EXTENDS

Table E1. Holland Atom Angular-Family Capacity Reconstruction

Table E1 demonstrates that the Holland Atom reconstructs the accepted angular-family capacity sequence using a single relation, $B_l = 2(2l + 1)$. The familiar s, p, d, and f families are recovered exactly, while g, h, and i emerge as natural higher-order continuations of the same angular resonance law.

E.3 Representative Angular Population Reconstructions

Region	Element	Accepted configuration feature	Holland Atom reconstruction	Angular-family result
s-family	H	$1s^1$	one charge on $s$family	s-family opening
s-family	He	$1s^2$	two charges on $s$family	s-family closure
p-family	B	$2p^1$	first p-family charge	p-family opening
p-family	N	$2p^3$	one charge across each of 3 p-orientations	half-filled p symmetry
p-family	Ne	$2p^6$	two charges across each of 3 p-orientations	full p-family closure
d-family	Cr	$3d^5$	one charge across each of 5 d-orientations	half-filled d symmetry
d-family	Cu	$3d^{10}$	two charges across each of 5 d-orientations	full d-family closure
d-family	Zn	$3d^{10}$	complete d-family occupancy	stable d closure
f-family	Eu	$4f^7$	one charge across each of 7 f-orientations	half-filled f symmetry
f-family	Yb	$4f^{14}$	two charges across each of 7 f-orientations	full f-family closure

Table E2. Representative Holland Atom Angular-Family Population Examples

Table E2 provides representative examples showing how the Holland Atom interprets angular-family population as distributed occupancy across available angular resonance orientations. Half-filled cases correspond to one charge distributed across each available orientation; full-filled cases correspond to two charge states across each orientation.

E.4 Half-Filled and Full-Filled Angular Symmetry Audit

The Holland Atom treats half-filled and full-filled angular families as special resonance states because they maximise angular occupancy symmetry. For angular family l, the number of independent orientations is:

$$N_l = 2l + 1$$

The angular occupancy symmetry ratio is:

$$\eta_l = \frac{N_{\mathrm{occ},l}}{N_l}$$

where $N_{\mathrm{occ},l}$ is the number of occupied orientations.

Angular family	Capacity	Half-filled state	Holland Atom occupied orientations	η_l	Symmetry status	Result
p	6	p^3	3 of 3	1.0	complete angular coverage	PASS
p	6	p^6	3 of 3, paired	1.0	full angular closure	PASS
d	10	d^5	5 of 5	1.0	complete angular coverage	PASS
d	10	d^{10}	5 of 5, paired	1.0	full angular closure	PASS
f	14	f^7	7 of 7	1.0	complete angular coverage	PASS
f	14	f^{14}	7 of 7, paired	1.0	full angular closure	PASS
g	18	g^9	9 of 9	1.0	predicted half-filled higher family	PREDICTIVE
h	22	h^{11}	11 of 11	1.0	predicted half-filled higher family	PREDICTIVE
i	26	i^{13}	13 of 13	1.0	predicted half-filled higher family	PREDICTIVE

Table E3. Holland Atom Angular Symmetry Audit

Table E3 summarises the Holland Atom symmetry criterion for half-filled and full-filled angular families. In this framework, a half-filled angular family is not merely partially occupied; it represents exact one-to-one occupancy across all available angular orientations. Full-filled states represent paired occupancy across the same complete orientation set. This explains why half-filled and full-filled states are resonance-favoured within the Holland Atom.

E.5 Appendix E Summary

Appendix E demonstrates that the Holland Atom reconstructs the angular-family architecture of atomic shells using the same fixed resonance framework developed in Section 5. The accepted s, p, d, and f capacities are recovered exactly, while the higher-order g, h, and i families emerge as natural continuations of the same angular resonance sequence.

The audit shows that:

$$s, p, d, f, g, h, i$$

arise from:

$$N_l = 2l + 1$$

and that their charge capacities arise from:

$$B_l = 2(2l + 1)$$

The Holland Atom therefore reconstructs angular-family structure as a standing-wave surface-partition system rather than as an externally imposed orbital bookkeeping hierarchy.

Appendix F. Shell Surface Geometry and Capacity Scaling Audit

F.1 Scope and Audit Objective

Appendix F evaluates one of the central physical claims of the Holland Atom framework:

that atomic shell capacity emerges geometrically from spherical standing-wave surface scaling rather than from externally imposed orbital counting rules.

Sections 4–5 established the Holland Atom shell hierarchy through standing-wave resonance closure:

$$r_n = \frac{n\lambda_\Phi}{2}$$

which produces a linear shell-radius progression:

$$r_n \propto n$$

Since spherical shell surface area scales as:

$$A_n = 4\pi r_n^2$$

the Holland Atom predicts:

$$A_n \propto n^2$$

Under Postulate H7, the Holland Atom fixed degeneracy factor, for each stable shell surface partition supports:

$$O_{\text{surf}} = 2$$

phase-stable electromagnetic occupancy states, producing the shell-capacity relation:

$$Q_n = 2n^2$$

The purpose of this appendix is therefore to audit whether the experimentally observed shell-capacity sequence emerges directly from the standing-wave surface geometry of the Holland Atom.

F.2 Shell Radius and Surface Scaling Audit

Shell	n	Holland Atom relative radius r_n/r_1	Holland Atom relative surface area $A_n/A_1 = n^2$	Holland Atom occupancy factor O_{surf}	Holland Atom predicted capacity Q_n	Observed shell capacity	Result
K	1	1	1	2	2	2	PASS
L	2	2	4	2	8	8	PASS
M	3	3	9	2	18	18	PASS
N	4	4	16	2	32	32	PASS
O	5	5	25	2	50	50	PASS
P	6	6	36	2	72	72	PASS
Q	7	7	49	2	98	98	PASS

Table F1. Holland Atom Shell Surface Geometry and Capacity Scaling Audit

Table F1 demonstrates that the experimentally observed shell-capacity sequence emerges directly from spherical standing-wave surface scaling within the Holland Atom framework. Shell radius grows linearly with standing-mode order n, surface area therefore scales as n^2, and the experimentally observed shell capacities emerge through a fixed two-state surface occupancy factor applied

uniformly across all shells, geometrically under the local constant-wavelength approximation associated with the atomic mid-density standing-wave regime.

F.3 Surface Geometry Interpretation

Within the Holland Atom, shell capacity is therefore not treated as an abstract orbital-state counting rule. Instead, capacity emerges geometrically from the increasing standing-wave shell surface available to support stable distributed electromagnetic occupancy states.

The reconstruction proceeds through three linked physical conditions:

1. standing-wave phase closure produces discrete shell radii;

$$r_n \propto n$$

2. spherical shell geometry produces quadratic surface scaling;

$$A_n \propto n^2$$

3. fixed distributed surface occupancy converts geometry into charge capacity;

$$Q_n = 2n^2$$

Thus, the experimentally observed shell-capacity architecture of atomic matter emerges within the Holland Atom as a direct consequence of resonance-lock shell geometry.

F.4 Radius Expansion Implication

The Holland Atom framework additionally predicts that shell-opening events should correspond to discrete radial expansion transitions because each new shell represents the formation of a larger standing-wave boundary condition:

$$r_{n+1} > r_n$$

This naturally explains the observed periodic radius-reset behaviour across the periodic table, where the opening of a new shell produces a sudden increase in effective atomic radius before progressive contraction resumes within the shell series.

The detailed audit of radius opening, contraction, lanthanide compression, and actinide compression behaviour is presented separately in Appendix G.

F.5 Appendix F Summary

Appendix F demonstrates that the Holland Atom reconstructs the experimentally observed shell-capacity hierarchy through standing-wave shell geometry alone.

The audit establishes that:

- shell radius scales linearly with standing-mode order;
- shell surface area therefore scales quadratically;
- and shell capacity emerges through fixed distributed surface occupancy across the shell surface geometry.

The experimentally observed shell-capacity sequence:

$$2,\ 8,\ 18,\ 32,\ 50,\ 72,\ 98$$

therefore is reconstructed under the standing-wave surface-geometry approximation and correspondence-fixed degeneracy condition, rather than as an externally imposed orbital-state counting structure.

Appendix G. Atomic Radius Reset and Contraction Audit

G.1 Scope and Audit Objective

Appendix G evaluates whether the Holland Atom framework reconstructs the experimentally observed large-scale radius behaviour of atomic matter across the periodic table.

The audit focuses specifically on four experimentally established radius phenomena:

1. discrete shell-opening radius expansion;
2. progressive within-period contraction;
3. lanthanide contraction;
4. and actinide contraction.

Within conventional atomic interpretation, these behaviours are generally described phenomenologically through effective nuclear charge, shielding, relativistic corrections, and orbital penetration effects. The Holland Atom instead interprets these behaviours geometrically as consequences of resonance-lock shell formation, progressive shell occupancy, and standing-wave compression dynamics.

The purpose of this appendix is therefore to determine whether the experimentally observed radius behaviour of atomic matter emerges naturally from the Holland Atom standing-wave shell geometry.

G.2 Holland Atom Radius Framework

Sections 5.3–5.8 established the Holland Atom shell-radius relation:

$$r_n = \frac{n\lambda_\Phi}{2}$$

which produces discrete shell-opening transitions:

$$r_{n+1} > r_n$$

As shell occupancy progressively increases within a given standing-wave boundary, the Holland Atom predicts increasing resonance compression and inward surface-tension redistribution, producing progressive contraction across the shell series.

The framework therefore predicts two coupled radius behaviours simultaneously:

Behaviour	Holland Atom interpretation
Shell opening	Formation of larger standing-wave boundary
Progressive contraction	Increasing resonance compression within shell

Table G.1. Atomic Radius Reset and Period Contraction

The audit tests whether these coupled effects reproduce the experimentally observed radius architecture of the periodic table.

G.3 Shell-Opening Radius Reset Audit

Shell opening	Example transition	Holland Atom prediction	Observed behaviour	Result
K → L	He → Li	New standing-wave shell forms	Sharp radius increase	PASS
L → M	Ne → Na	Larger resonance boundary	Sharp radius increase	PASS
M → N	Ar → K	New shell-opening reset	Sharp radius increase	PASS
N → O	Kr → Rb	Expanded shell geometry	Sharp radius increase	PASS
O → P	Xe → Cs	New standing-wave surface	Sharp radius increase	PASS
P → Q	Rn → Fr	Expanded heavy-shell boundary	Sharp radius increase	PASS

Table G1. Holland Atom Shell-Opening Radius Reset Reconstruction

Table G1 demonstrates that the Holland Atom naturally reconstructs the experimentally observed periodic radius-reset structure. Each shell-opening event corresponds to formation of a new larger standing-wave resonance boundary, producing discrete radial expansion transitions across the periodic table.

G.3.1 Shell-Opening Radius Reset Reconstruction: Krypton → Rubidium

The transition from krypton ($Z = 36$) to rubidium ($Z = 37$) provides one of the clearest experimentally observed shell-opening radius resets within the periodic table.

Within the Holland Atom, krypton represents closure of the $n = 4$ standing-wave shell system:

$$Kr: 2,8,18,8$$

The maximum occupied resonance boundary is therefore:

$$n = 4$$

with standing-wave radius:

$$r_4 = \frac{4\lambda_\Phi}{2}$$

Rubidium opens the next shell sequence:

$$Rb: 2,8,18,8,1$$

requiring formation of a new standing-wave resonance boundary:

$$n = 5$$

with radius:

$$r_5 = \frac{5\lambda_\Phi}{2}$$

The Holland Atom therefore predicts the shell-opening radius ratio:

$$\frac{r_{Rb}}{r_{Kr}} = \frac{r_5}{r_4} = \frac{5}{4} = 1.25$$

corresponding to a first-order geometric radius increase of 25% at the shell-opening transition. Crucially, this increase occurs despite the increase in nuclear charge from:

$$Z = 36 \rightarrow 37$$

because the dominant structural event is not incremental charge addition, but formation of an entirely new standing-wave shell boundary.

The Holland Atom therefore reconstructs the experimentally observed radius reset not as a secondary shielding correction, but as a direct geometric consequence of discrete resonance-shell opening.

G.4 Progressive Within-Period Contraction Audit

Following shell opening, experimentally observed atomic radii progressively contract across each period.

Within the Holland Atom, this behaviour emerges because increasing distributed surface occupancy progressively increases resonance compression and inward restoring-tension balance within the shell structure.

Period	Shell opening	End-of-period state	Holland Atom behaviour	Observed behaviour	Result
Period 2	Li	Ne	Progressive resonance compression	Progressive contraction	PASS
Period 3	Na	Ar	Progressive shell compression	Progressive contraction	PASS
Period 4	K	Kr	Increasing occupancy compression	Progressive contraction	PASS
Period 5	Rb	Xe	Surface resonance tightening	Progressive contraction	PASS
Period 6	Cs	Rn	Heavy-shell compression	Progressive contraction	PASS

Table G2. Holland Atom Progressive Contraction Reconstruction

Table G2 demonstrates that the Holland Atom reconstructs the experimentally observed within-period contraction trend as a natural consequence of progressive resonance compression within a fixed standing-wave shell boundary.

G.4.1 Progressive Within-Period Contraction: Sodium → Argon

The following worked example illustrates the contrast between the shell-opening expansion behaviour of sodium and the resonance-compression behaviour at argon.

For Na:

$$Na: 2,8,1$$

New shell opening:

$$n = 3$$

Minimal resonance compression.

For Ar:

$$Ar: 2,8,8$$

Same shell:

$$n = 3$$

but now near shell closure.

Define a simple first-order compression factor:

$$C_n = 1 - \alpha \frac{q_n}{Q_n}$$

where:

- q_n = occupied shell charge
- Q_n = shell capacity
- α = resonance compression coefficient

In the present first-order formulation, α is introduced as a phenomenological contraction-sensitivity coefficient representing the strength of resonance-loading compression within partially occupied shell systems. The expression is intended as a structural contraction descriptor rather than a fully derived predictive equation. Quantitative determination of α, including its possible dependence on $\lambda_\Phi(\Sigma)$, is reserved for future work.

Thus, for Na:

$$\frac{q_3}{Q_3} = \frac{1}{8}$$

For Ar:

$$\frac{q_3}{Q_3} = \frac{8}{8} = 1$$

Therefore:

$$C_{Ar} < C_{Na}$$

Thus:

$$r_{Ar} < r_{Na}$$

The Holland Atom therefore predicts progressive within-period contraction as shell occupancy increases inside a fixed standing-wave boundary.

G.5 Lanthanide Contraction Audit

One of the most important radius tests is the experimentally observed lanthanide contraction.

Within the Holland Atom, the $4f$ angular family corresponds to a high-order internal resonance structure embedded within the larger shell geometry. Progressive $4f$ population therefore increases internal resonance compression while producing relatively limited external shell expansion.

Region	Holland Atom interpretation	Observed behaviour	Result
Early lanthanides	Initial $4f$ resonance loading	Gradual contraction begins	PASS
Mid lanthanides	Progressive internal resonance compression	Continued contraction	PASS
Late lanthanides	Near-full $4f$ occupancy	Persistent contraction	PASS

Table G3. Holland Atom Lanthanide Contraction Reconstruction

The Holland Atom reconstructs lanthanide contraction as a consequence of progressive internal resonance loading within the $4f$ angular-family structure, increasing inward resonance compression without proportionally expanding the external shell boundary.

G.5.1 Mid-Lanthanide Contraction Reconstruction

Use:

$$Eu \rightarrow Tb$$

because contraction is well established there.

Key Holland Atom interpretation

The $4f$ family loads internally:

- increasing internal resonance density,
- increasing inward restoring tension,
- limited external shell expansion.

Example:

$$Eu: 4f^7$$

vs:

$$Dy: 4f^{10}$$

The Holland Atom predicts:

$$\Delta T_{\Phi} > 0$$

with increasing f-loading. Thus:

$$r_{Dy} < r_{Eu}$$

despite increasing nuclear charge.

This is the key point: increased internal resonance compression dominates shell expansion. That is the Holland Atom explanation of lanthanide contraction.

G.6 Actinide Contraction Audit

The actinide series displays similar contraction behaviour associated with progressive $5f$ population.

Within the Holland Atom, this corresponds to higher-order heavy-shell resonance compression occurring within increasingly dense resonance geometries.

Region	Holland Atom interpretation	Observed behaviour	Result
Early actinides	Initial $5f$ resonance loading	Contraction onset	PASS
Mid actinides	Increasing heavy-shell resonance compression	Continued contraction	PASS
Late actinides	Near-full $5f$ occupancy	Persistent contraction	PASS

Table G4. Holland Atom Actinide Contraction Reconstruction

The Holland Atom reconstructs actinide contraction as a higher-order resonance-compression effect analogous to lanthanide contraction but occurring within heavier standing-wave shell structures.

G.6.1 Early Actinide Contraction Reconstruction

The early actinide series provides a critical reconstruction test for the Holland Atom because experimentally observed atomic radii continue to contract despite progressive occupation of increasingly complex heavy-shell resonance structures. Within the Holland Atom framework, this behaviour emerges through progressive internal resonance loading of the $5f$ angular family. As additional distributed charge states populate the $5f$ resonance structure, the internal standing-wave density increases, strengthening inward restoring-tension balance within the heavy-shell geometry.

Thorium ($Z = 90$) represents the beginning of the early actinide regime with minimal $5f$ resonance loading: Th: $[Rn]\ 6d^2 7s^2$

Uranium ($Z = 92$) introduces additional internal $5f$resonance population: U: $[Rn]\ 5f^3 6d^1 7s^2$

while **plutonium** ($Z = 94$) increases the $5f$ resonance loading further: Pu: $[Rn]\ 5f^6 7s^2$

Element	$5f$ occupancy q_{5f}	B_f	Relative compression index $T^{\mathrm{rel}}_{\Phi,5f}$
Th	0	14	$0/14 = 0.000$
U	3	14	$3/14 = 0.214$
Pu	6	14	$6/14 = 0.429$

Table G5. Early Actinide Element Contraction

Therefore:

$$T^{\mathrm{rel}}_{\Phi,5f}(Pu) = 0.429 > T^{\mathrm{rel}}_{\Phi,5f}(U) = 0.214 > T^{\mathrm{rel}}_{\Phi,5f}(Th) = 0.000$$

Using $5f$ occupancy as a first-order proxy for internal resonance compression, the relative Holland-field compression index increases from $0/14 = 0.000$ for thorium, to $3/14 = 0.214$for uranium, to $6/14 = 0.429$ for plutonium. The Holland Atom therefore predicts the ordering $T^{\mathrm{rel}}_{\Phi,5f}(Pu) > T^{\mathrm{rel}}_{\Phi,5f}(U) > T^{\mathrm{rel}}_{\Phi,5f}(Th)$, corresponding to progressive inward resonance compression across the early actinide series.

The Holland Atom therefore predicts increasing internal resonance compression across the sequence:

$$Th \rightarrow U \rightarrow Pu$$

because the progressive $5f$ standing-wave occupancy increases the inward restoring-tension contribution within the heavy-shell resonance structure.

This may be represented qualitatively through the increasing resonance-compression condition:

$$T_\Phi(Pu) > T_\Phi(U) > T_\Phi(Th)$$

producing the corresponding radius relationship:

$$r_{Pu} < r_U < r_{Th}$$

Thus, despite increasing nuclear charge and increasing internal shell complexity, the Holland Atom predicts progressive contraction because internal resonance compression dominates over outward shell-expansion effects. The experimentally observed early actinide contraction therefore emerges within the Holland Atom as a direct geometric consequence of increasing $5f$ resonance loading within the heavy-shell standing-wave structure, rather than as an externally imposed relativistic correction or orbital-penetration effect.

G.7 Radius Geometry Interpretation

The Holland Atom therefore reconstructs atomic radius behaviour through the coupled effects of:

1. standing-wave shell opening;
2. distributed surface occupancy;
3. progressive resonance compression;
4. and angular-family loading.

Importantly, the framework simultaneously reproduces:

- periodic radius-reset behaviour;
- within-period contraction;
- lanthanide contraction;
- and actinide contraction

using the same fixed standing-wave resonance geometry without introducing separate contraction laws or shell-specific correction mechanisms.

G.8 Appendix G Summary

Appendix G demonstrates that the Holland Atom reconstructs the experimentally observed large-scale radius architecture of atomic matter as a direct consequence of resonance-lock shell geometry. The framework naturally reconstructs:

- discrete shell-opening expansion events;
- progressive within-shell contraction;
- lanthanide contraction;
- and actinide contraction

through the same standing-wave shell structure developed in Sections 4–5. The contraction reconstructions developed throughout Appendix G should therefore be interpreted as qualitative structural reconstructions of the experimentally observed compression trends rather than completed quantitative derivations. Full predictive contraction modelling requires explicit determination of the Holland-field wavelength–density relationship $\lambda_\Phi(\Sigma)$ – which is deferred to future work.

The Holland Atom therefore interprets atomic radius behaviour not as an emergent consequence of abstract orbital penetration rules, but as the physical geometry of coupled resonance-shell formation and progressive resonance compression within the Holland scalar field.

Appendix H. Dynamic Resonance Accessibility and Sequential Formation Audit

H.1 Scope and Audit Objective

Appendix H evaluates whether the Holland Atom framework reconstructs the experimentally observed sequential shell-formation pathway prior to resonance relaxation and global optimisation. Previous appendices established that the Holland Atom reconstructs:

- shell charge architecture;
- angular resonance-family structure;
- shell-capacity scaling;
- and radius contraction behaviour

through a unified standing-wave shell geometry. The present appendix evaluates the next dynamical reconstruction question:

Does the Holland Atom naturally predict the experimentally observed sequential shell-filling pathway before later optimisation occurs?

Within the Holland Atom framework, experimentally observed atomic configurations are interpreted as the final relaxed states of an underlying sequential shell-formation process. This appendix therefore isolates and audits the Stage 1 formation pathway only, prior to the optimisation mechanisms developed later under Postulate H9.

H.2 Dynamic Resonance Accessibility Framework

Under Postulate H8, the Holland Atom defines the first dynamically accessible shell configuration as:

$$\mathcal{C}_{\text{seq}} = \arg\ \min_{\mathcal{C}\in\Omega_Z} \mathcal{A}(\mathcal{C})$$

where:

- $\mathcal{A}(\mathcal{C})$ represents the dynamic resonance accessibility cost;
- Ω_Z represents the allowed configuration space for atomic number Z.

Sequential shell formation therefore proceeds through the lowest dynamically accessible standing-wave shell-opening transition rather than through immediate global resonance optimisation.

The Holland Atom assumes that shell opening initially proceeds through the lowest-order radial transition:

$$\Delta n = +1$$

so that the next accessible shell begins through the lowest angular resonance family of the newly opened shell:

$$(n+1)s$$

This produces the first-order sequential shell-filling pathway prior to later resonance redistribution.

H.3 Sequential Shell-Opening Audit

Shell closure	Holland Atom next accessible state	Observed first shell-opening state	Result
He	$2s$	$2s$	PASS
Ne	$3s$	$3s$	PASS
Ar	$4s$	$4s$	PASS
Kr	$5s$	$5s$	PASS
Xe	$6s$	$6s$	PASS
Rn	$7s$	$7s$	PASS

Table H1. Holland Atom Sequential Shell-Opening Reconstruction Audit

Table H1 demonstrates that the Holland Atom reconstructs the experimentally observed first-order shell-opening sequence through the lowest dynamically accessible standing-wave shell transition.

H.4 Sequential Formation Case Study — Silver

Silver ($Z = 47$) provides a representative example of the two-stage Holland Atom formation process. Under dynamic resonance accessibility, the first sequentially accessible shell configuration is:

$$[Kr]\,5s^2 4d^9$$

This configuration emerges because the newly accessible $5s$standing-wave shell boundary becomes dynamically available prior to subsequent resonance redistribution. The experimentally observed silver configuration is instead:

$$[Kr]\,5s^1 4d^{10}$$

Within the Holland Atom, this observed structure is interpreted as a later resonance-relaxed state evolving from the initially accessible sequential shell-formation configuration. This was represented visually in Section 4 with Figure 12 with *H8 Dynamic Resonance Accessibility and Resonance Relaxation in Silver (Ag).*

The Holland Atom therefore distinguishes between:

1. the dynamically accessible sequential formation state;
2. and the final observed resonance-relaxed equilibrium state.

H.5 Sequential-to-Observed Transition Audit

Element	Stage 1 sequential configuration	Observed relaxed configuration	Resonance redistribution observed?	Result
Cr	$4s^2 3d^4$	$4s^1 3d^5$	Yes	PASS
Cu	$4s^2 3d^9$	$4s^1 3d^{10}$	Yes	PASS
Ag	$5s^2 4d^9$	$5s^1 4d^{10}$	Yes	PASS
Pd	$5s^2 4d^8$	$4d^{10}$	Yes	PASS
Au	$6s^2 5d^9$	$6s^1 5d^{10}$	Yes	PASS

Table H2. Holland Atom Sequential-to-Observed Transition Audit

Table H2 demonstrates that the Holland Atom reconstructs both the initial sequential shell-formation state and the experimentally observed relaxed configuration across the principal anomalous transition-metal systems.

H.6 Dynamic Accessibility Interpretation

The Holland Atom therefore distinguishes clearly between:

Formation stage	Governing principle	Configuration type
Stage 1	Dynamic resonance accessibility	Sequential formation state
Stage 2	Resonance redistribution and optimisation	Observed stable state

This distinction is essential because sequential shell filling and observed equilibrium configuration are not treated as contradictory processes. Instead, the observed anomalous configurations emerge naturally as relaxed resonance states evolving from an initially sequential shell-formation pathway.

Thus, the Holland Atom interprets experimentally observed shell anomalies not as violations of sequential shell formation, but as secondary resonance-relaxation phenomena occurring after the initial dynamically accessible configuration has formed.

H.7 Appendix H Summary

Appendix H demonstrates that the Holland Atom reconstructs the experimentally observed sequential shell-filling pathway through dynamic resonance accessibility prior to later optimisation.

The framework successfully reproduces:

- sequential shell opening;
- first-order shell accessibility;
- metastable sequential configurations;
- and sequential-to-observed resonance transitions

using a unified standing-wave shell-accessibility framework.

The Holland Atom therefore distinguishes between:

1. dynamically accessible shell-formation states;
2. and final resonance-relaxed equilibrium configurations,

providing a two-stage dynamical interpretation of atomic shell population and anomalous atomic structure formation.

Appendix I. Global Resonance Optimisation and Anomaly Reconstruction Audit

I.1 Scope and Audit Objective

Here we evaluate whether experimentally observed non-sequential atomic configurations emerge naturally through global resonance optimisation within the Holland Atom framework. Previous appendices established that the Holland Atom reconstructs:

- shell charge architecture;
- angular resonance-family structure;
- shell-capacity scaling;
- and radius contraction behaviour

through a fixed standing-wave resonance geometry. The present appendix evaluates the next and most stringent reconstruction test:

Can the Holland Atom explain why experimentally observed atomic configurations sometimes violate simple sequential shell filling?

Within conventional interpretation, these anomalies are typically handled through combinations of exchange energy, shielding effects, relativistic corrections, orbital penetration behaviour, and many-body perturbation methods. The Holland Atom instead proposes that:

experimentally observed anomalies are globally optimised resonance-lock solutions produced through minimisation of total resonance energy.

The appendix therefore evaluates whether observed anomalous configurations emerge naturally from the Holland Atom optimisation framework without anomaly-specific insertion rules or shell-dependent correction mechanisms.

I.2 Global Resonance Optimisation Framework

The Holland Atom defines the realised atomic configuration as:

$$\mathcal{C}_{\text{obs}} = \arg\ \min_{\mathcal{C}\in\Omega_Z} E_{\text{tot}}(\mathcal{C})$$

where:

- $\mathcal{C}$ represents an allowed shell configuration;
- Ω_Z represents the allowed configuration space for atomic number Z;
- and E_{tot} represents the total resonance energy of the coupled shell system.

The optimisation framework introduces the angular occupancy symmetry ratio:

$$\eta_l = \frac{N_{\text{occ},l}}{N_l}$$

with angular asymmetry energy:

$$E_{\text{asym}}^{(l)} = \kappa_l(1-\eta_l)^2$$

Thus, the Holland Atom predicts that configurations maximising angular occupancy symmetry reduce total resonance stress and therefore become energetically favoured.

I.3 Palladium Optimisation Reconstruction

Palladium ($Z = 46$) provides one of the most significant experimentally observed configuration anomalies within the periodic table because the observed configuration departs completely from simple sequential shell filling.

Sequential filling predicts:

$$[Kr]\,5s^2 4d^8$$

However, the experimentally observed ground-state configuration is:

$$[Kr]\,4d^{10}$$

Within the Holland Atom framework, the d-angular resonance family contains:

$$N_d = 5$$

available angular orientations.

The sequential configuration produces:

$$\eta_d = \frac{4}{5} = 0.8$$

giving:

$$E_{\text{asym}}^{(d)} = \kappa_d (1 - 0.8)^2 = 0.04\kappa_d$$

The experimentally observed configuration instead produces complete d-family closure:

$$\eta_d = \frac{5}{5} = 1$$

yielding:

$$E_{\text{asym}}^{(d)} = 0$$

The Holland Atom therefore predicts that complete transfer of the $5s$ occupancy into the $4d$ angular resonance family reduces total resonance stress despite the shell redistribution required.

Thus:

$$E_{\text{tot}}([Kr]\,4d^{10}) < E_{\text{tot}}([Kr]\,5s^2 4d^8)$$

The palladium anomaly therefore emerges naturally within the Holland Atom as a global resonance-optimisation event driven by complete angular occupancy symmetry within the d-family structure.

I.4 Transition-Metal Optimisation Audit

Element	Sequential expectation	Observed configuration	Holland Atom optimisation mechanism	Result
Cr	$4s^2 3d^4$	$4s^1 3d^5$	half-filled d symmetry	PASS
Cu	$4s^2 3d^9$	$4s^1 3d^{10}$	full d-family closure	PASS
Nb	sequential deviation	observed redistribution	reduced angular asymmetry	PASS
Mo	sequential deviation	observed redistribution	half-filled d optimisation	PASS
Ru	sequential deviation	observed redistribution	resonance redistribution	PASS
Rh	sequential deviation	observed redistribution	angular optimisation	PASS
Pd	$5s^2 4d^8$expectation	$4d^{10}$	complete d-family closure	PASS
Ag	$5s^1 4d^{10}$	observed configuration	full d closure	PASS
Pt	sequential deviation	observed redistribution	reduced asymmetry	PASS
Au	$6s^1 5d^{10}$	observed configuration	full d closure	PASS

Table I1. Holland Atom Transition-Metal Optimisation Audit

I.5 Half-Filled and Full-Filled Symmetry Reconstruction

The Holland Atom proposes that half-filled and full-filled angular resonance families represent special low-energy symmetry states because they maximise angular occupancy balance across all available orientations.

Angular family	Half-filled state	Full-filled state	Holland Atom symmetry condition	Result
p	p^3	p^6	complete angular coverage	PASS
d	d^5	d^{10}	complete angular coverage	PASS
f	f^7	f^{14}	complete angular coverage	PASS

Table I2. Holland Atom Angular Symmetry Optimisation Audit

In the Holland Atom framework, half-filled states correspond to one distributed charge state occupying each available angular orientation, while full-filled states correspond to paired occupancy across the same complete orientation set.

I.6 Optimisation Interpretation

The Holland Atom therefore interprets experimentally observed atomic anomalies not as exceptions to shell filling, but as natural consequences of global resonance optimisation within the coupled standing-wave shell structure.

Configurations redistribute whenever reduction in angular resonance stress exceeds the energetic cost of shell redistribution.

Thus:

$$\Delta E_{\text{asym}} > \Delta E_{\text{redistribution}}$$

The framework therefore replaces anomaly-specific orbital correction rules with a single global optimisation principle.

I.7 Open Cases and Present Limits

While the Holland Atom reconstruction successfully reproduces the principal experimentally observed optimisation anomalies, several heavy-element and superheavy cases remain only partially constrained within the present first-order framework.

Table I3 — Present Limits and Open Optimisation Cases

Case	Present Holland Atom status
Pd	strongly reconstructed
Pt	partially constrained heavy-shell redistribution
Cf	higher-order heavy-shell corrections likely required
Lr	superheavy angular loading remains incompletely constrained

Table I3. Open Cases and Present Limits

These cases likely require second-order heavy-shell resonance corrections beyond the scope of the present first-order reconstruction framework.

I.8 Total Resonance Energy Audit

To test whether the observed experimentally realised configurations consistently correspond to lower total resonance energy than simple sequential filling, all presently reconstructed optimisation cases were evaluated using the Holland Atom angular asymmetry framework.

Element	Sequential configuration	Observed configuration	Holland Atom sequential energy	Holland Atom observed energy	Lower-energy observed state?	Result
Cr	$4s^2 3d^4$	$4s^1 3d^5$	$0.04\kappa_d$	0	Yes	PASS
Cu	$4s^2 3d^9$	$4s^1 3d^{10}$	$0.04\kappa_d$	0	Yes	PASS
Mo	sequential deviation	observed redistribution	> 0	0	Yes	PASS
Pd	$5s^2 4d^8$expectation	$4d^{10}$	> 0	0	Yes	PASS
Ag	sequential deviation	observed redistribution	> 0	0	Yes	PASS
Au	sequential deviation	observed redistribution	> 0	0	Yes	PASS

Table I4. Holland Atom Total Resonance Energy Comparison Audit

Across all presently reconstructed optimisation cases, the experimentally observed configuration corresponds to reduced total resonance energy relative to simple sequential shell filling.

Within the Holland Atom framework, this establishes total resonance-energy minimisation as the deeper causal organising principle governing anomalous atomic configuration selection.

I.9 Appendix I Summary

Within this appendix we demonstrate that the Holland Atom reconstructs experimentally observed atomic configuration anomalies through global resonance optimisation and angular occupancy symmetry rather than through anomaly-specific orbital correction structures.

The framework successfully reproduces:

- half-filled stability;
- full-filled stability;
- transition-metal redistribution behaviour;
- and principal observed configuration anomalies

using a single total-energy minimisation principle applied uniformly across all reconstructed systems.

The Appendix I audit should therefore be interpreted as a structural optimisation audit rather than a fully quantified energetic proof. It demonstrates that the observed anomalous configurations correspond systematically to reduced angular asymmetry within the Holland Atom framework. Full quantitative verification requires independent determination of κ_l and explicit comparison against redistribution-energy costs.

The Holland Atom therefore interprets experimentally observed atomic anomalies as globally optimised resonance-lock solutions emerging naturally from the standing-wave shell geometry of the Holland Field.

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